FireTwins (PyFDS)

Client demonstrations - step-by-step user manual

Example 1: room fire with door, window and heat detector
Example 2: burning couch (FDS example case)

Software	Version used for this manual
PyFDS model editor and PyFDS Results	0.3.0 (repository E:\firetwins)
FDS (Fire Dynamics Simulator, NIST)	FDS-6.10.1 release
Smokeview (NIST)	SMV-6.10.1 release
Python / PySide6 / PyVista / VTK	3.10 / 6.12.0 / 0.49.0 / 9.3.1
Screen used for every screenshot	1360 x 768, 100 % scaling, windows maximized

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1 Before you start

FireTwins demonstrations use three programs. All screenshots in this manual were taken from the running programs while the two cases were built, run and post-processed; nothing is a mock-up.

- **PyFDS (model editor).** Builds and checks the FDS model, writes the FDS input file and starts FDS.
- **PyFDS Results.** A Smokeview-style 3D viewer for FDS results that comes with PyFDS.
- **Smokeview.** NIST's own viewer, installed with FDS (fds_smv\SMV6\smokeview.exe); PyFDS can open a run in it and drive it with Smokeview scripts.

1.1 Launching the programs

The installer is not yet available. Until it is, the programs are started from the PyFDS folder in the Python environment that has PyFDS installed (here: the conda environment 'agent'):

Program	Command (run in E:\firetwins)
PyFDS model editor	python main.py (or: pyfds-gui)
PyFDS Results	python results_main.py (or: pyfds-results)
Smokeview	fds_smv\SMV6\smokeview.exe <case>.smv (or double-click the .smv file)

FDS and Smokeview 6.10.1 are in E:\firetwins\fds_smv; PyFDS finds them there. Both programs open maximized. A log of each session is written to user_data\logs\ (pyfds.log, pyfds_results.log); send it with any problem report.

1.2 The demonstration files

Folder	Contents
client_demos\example_1	Room fire: example_1.pyfds (project), example_1.fds (FDS input), run\ (FDS output), results\ (pictures and movie), example_1.pyfdsv (saved viewer setup)
client_demos\example_2	Couch fire: couch_original.fds (the unchanged FDS example), couch.pyfds, couch.fds, run\, results\, couch.pyfdsv
documentation	This manual, the screenshots (screenshots\example_1, \example_2), the screen recordings (videos\) and validation_report.md
examples	The earlier versions of both examples, kept unchanged as the reference

2 Example 1 - Room fire with door, window and heat detector

A Introduction

Purpose. A propane burner fire in a small room with an open door and window, with a heat detector and a thermocouple tree above the fire. The case is built from an empty project, so it shows the model editor's dialogs one by one.

Scenario. Room 4.8 x 3.3 x 2.4 m (inside 25 mm gypsum-board walls) in a 6 x 4.5 x 3 m computational domain open to the outside on its sides and top. Door 0.9 x 2.1 m (south wall), window 1.2 x 1.05 m with 0.9 m sill (east wall). A 0.6 x 0.6 m burner in the middle of the floor releases 500 kW/m2 on its top face, i.e. 180 kW, reached along a t-squared curve in 15 s, and burns propane (soot yield 0.024, CO yield 0.005). Simulated time 60 s.

Engineering objectives. Follow the hot gas layer and smoke as they form under the ceiling and leave through the openings; find the time the heat detector (RTI 5 (m s)^0.5, 74 C) activates; read the temperature profile above the fire; check visibility at head height.

- Building a model from scratch: mesh, library construction, Draw Room, burner, reaction, vents
- Devices (heat detector, thermocouples) and output slices (temperature, visibility)
- Model validation, the Run dialog and the live Run Monitor
- Post-processing in PyFDS Results (slices, 3D smoke, devices, time histories, pictures, movie) and in Smokeview

Prerequisites. PyFDS, FDS 6.10.1 and Smokeview 6.10.1 (section 1), about 2 GB of free disk space, and time for the run: it took 40 minutes of wall clock time on the test computer (one MPI process). For a live demonstration, open the finished run instead (section E).

Screen recordings. documentation\videos\example_1_build_and_run.mp4 records the whole of sections B to D on screen (the FDS run as a time-lapse, one frame per 10 s); example_1_post_processing.mp4 records section E.

B Opening the program

Start the PyFDS model editor (section 1.1). It opens maximized with an empty project. The main areas are: the menu bar and toolbar at the top; the drawing tools on the left edge; the Project Explorer (every object of the model, grouped by kind) with the Properties panel under it; the 3D view in the centre with its icon strip (rendering mode, standard views, grid, snapping, screenshot, show/hide); and the status bar.

To demonstrate a finished case instead of building it, use File > Open and choose client_demos\example_1\example_1.pyfds, then continue at section D.

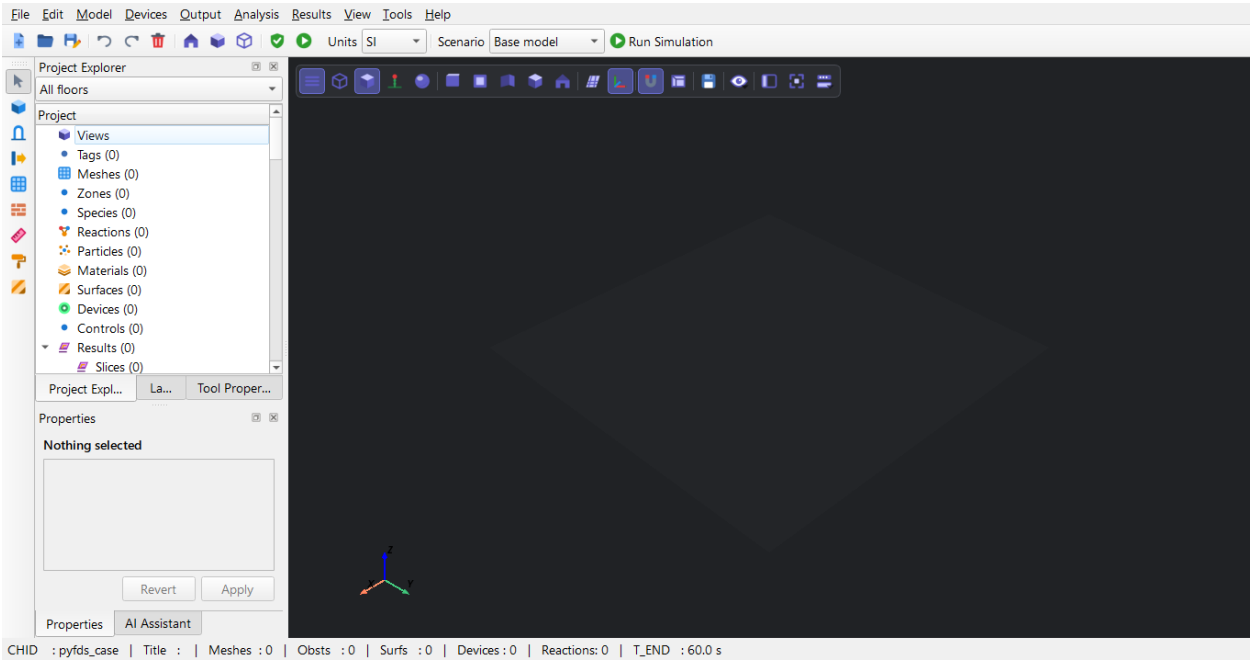


Figure 1. PyFDS opens maximized with an empty project: menus and toolbars at the top, the Project Explorer and Properties on the left, the 3D view in the centre.

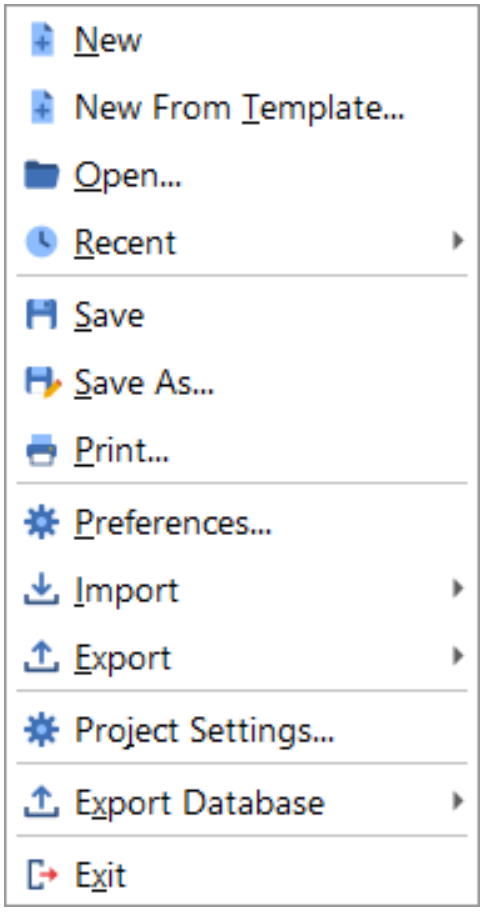


Figure 2. File menu: New, Open, Save, Import and Export.

C Building and configuring the model

Each dialog below is shown completed, with the values used for this case. Press OK to apply a dialog; every change can be undone with Edit > Undo (Ctrl+Z). Values are in SI units (m, s, kW, C).

Step 2 - Simulation parameters. Only the end time is changed (T_END = 60 s); the other tabs show the defaults that are used (they match FDS's own defaults).

Step 1 - Project settings. CHID is the base name of every file FDS writes.



Figure 3. File > Project Settings: CHID (the name of every output file) and the case title.

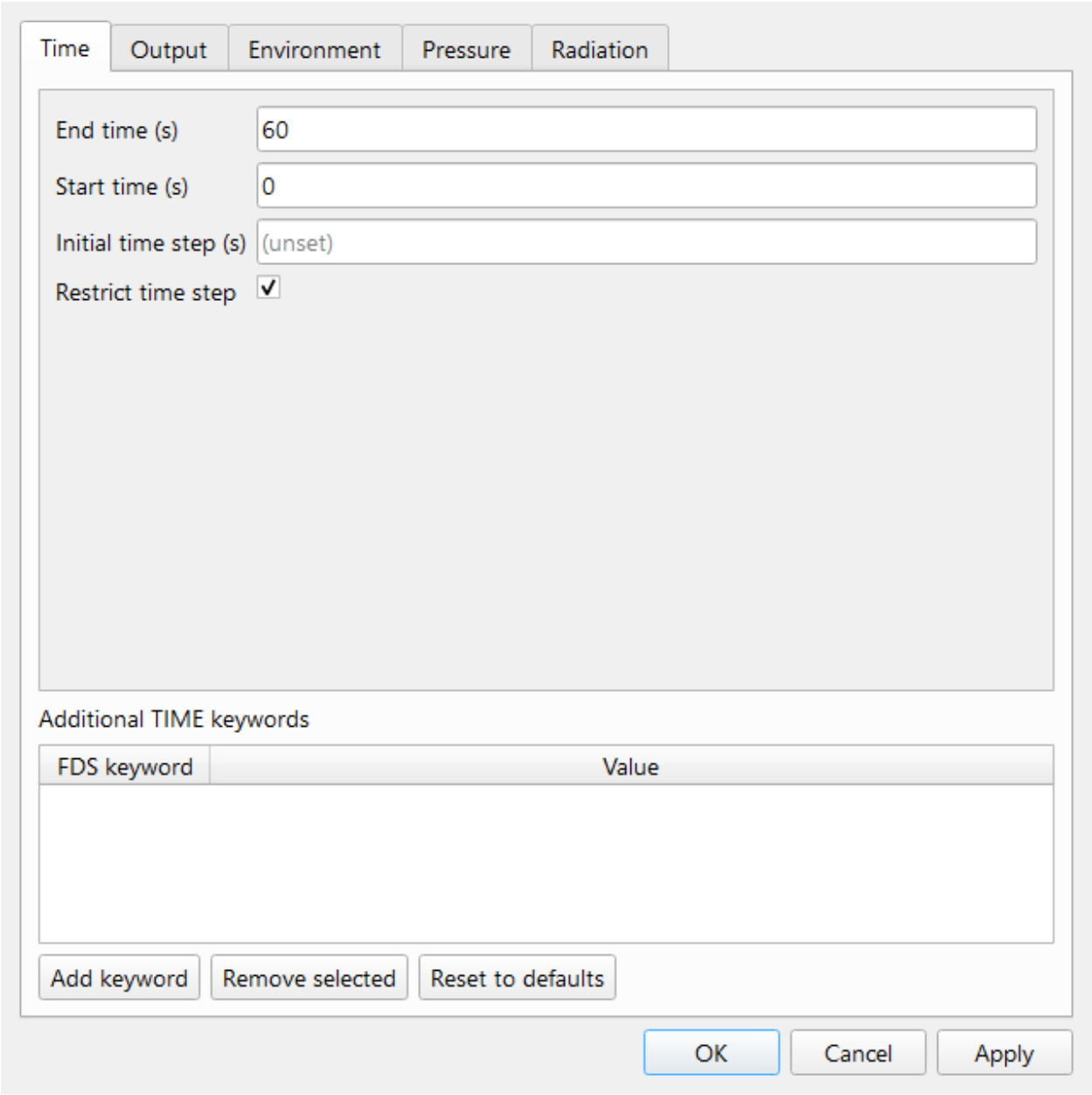


Figure 4. Model > Simulation Parameters, Time tab: the simulation ends at T_END = 60 s.

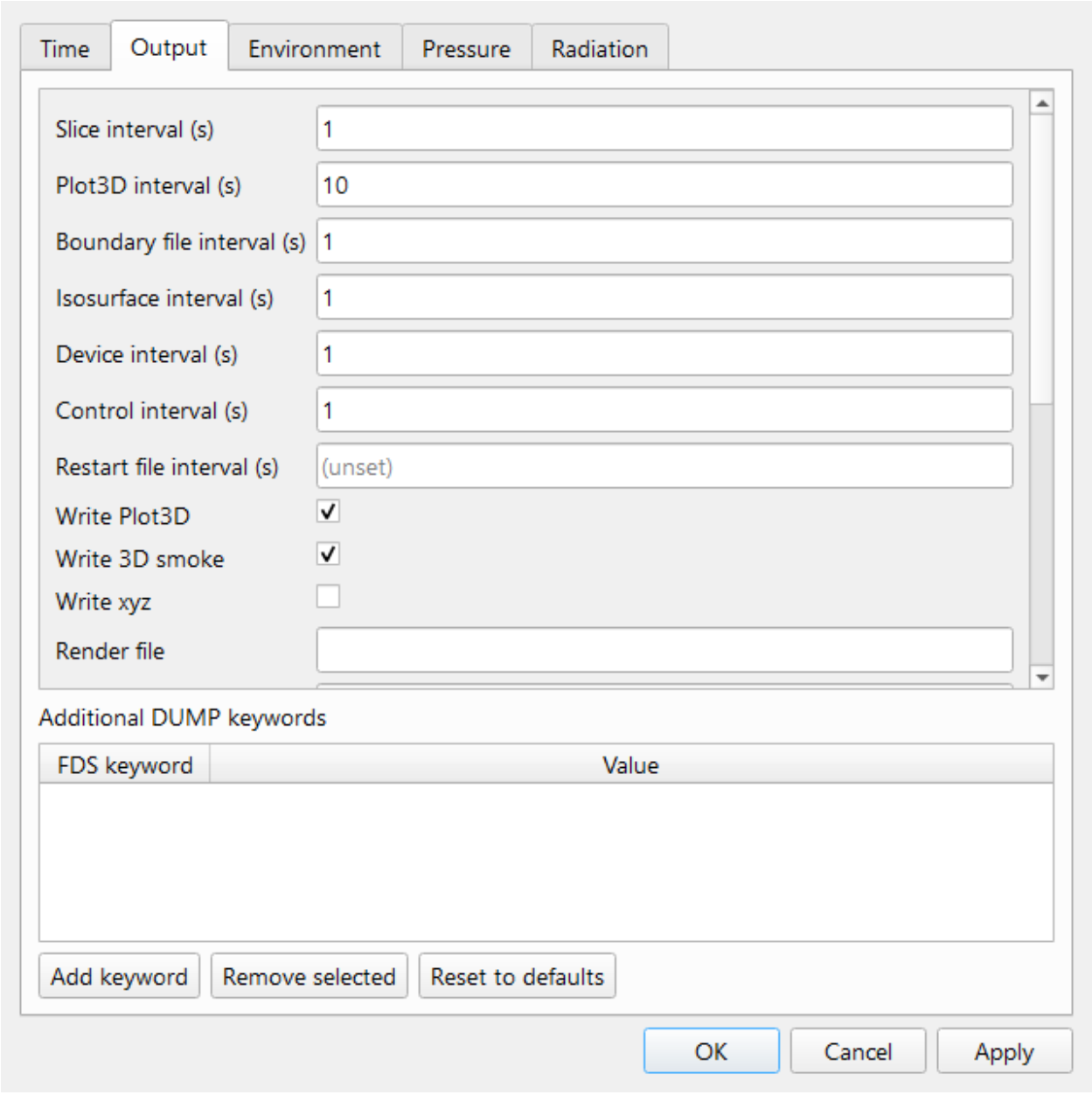


Figure 5. Model > Simulation Parameters: Output tab

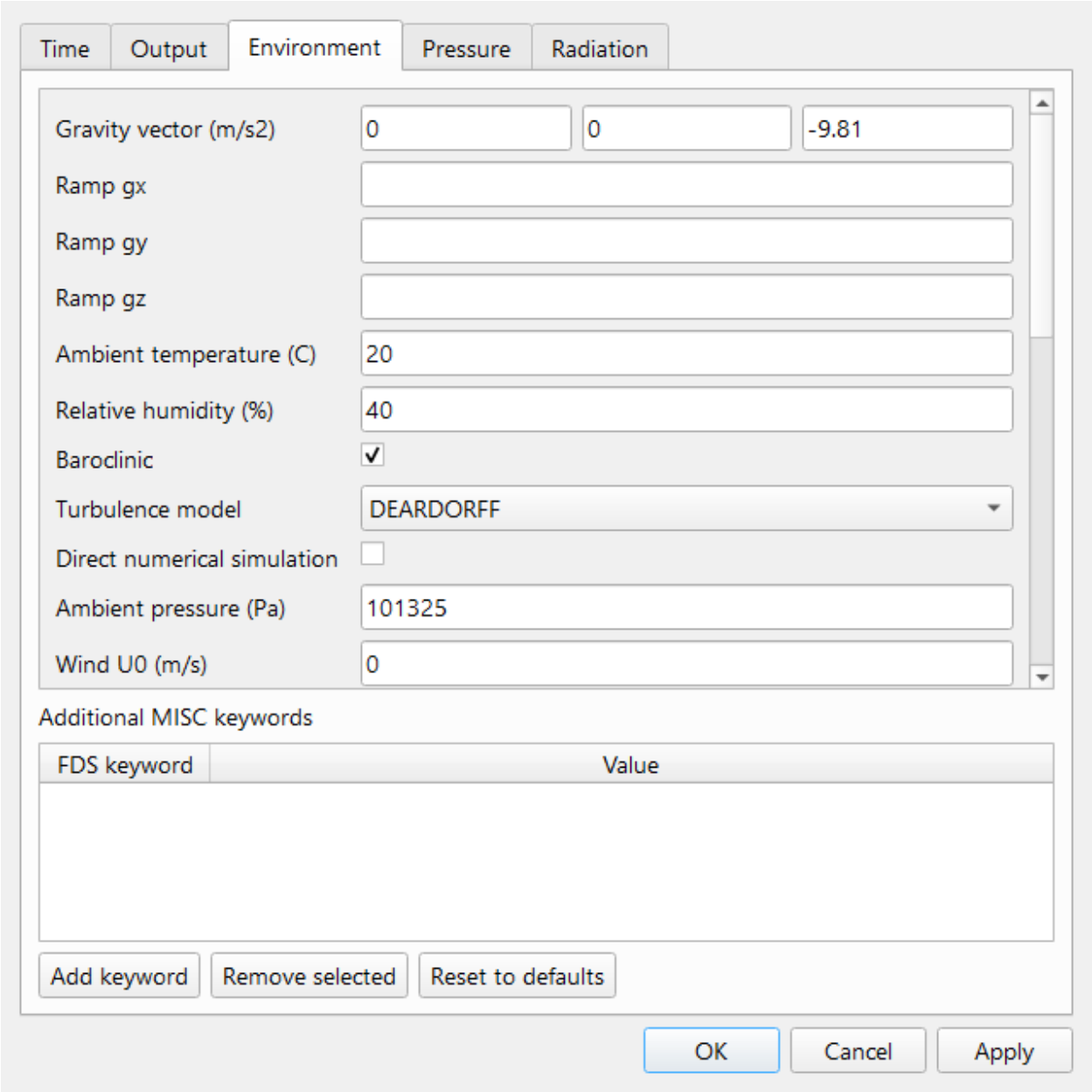


Figure 6. Model > Simulation Parameters: Environment tab

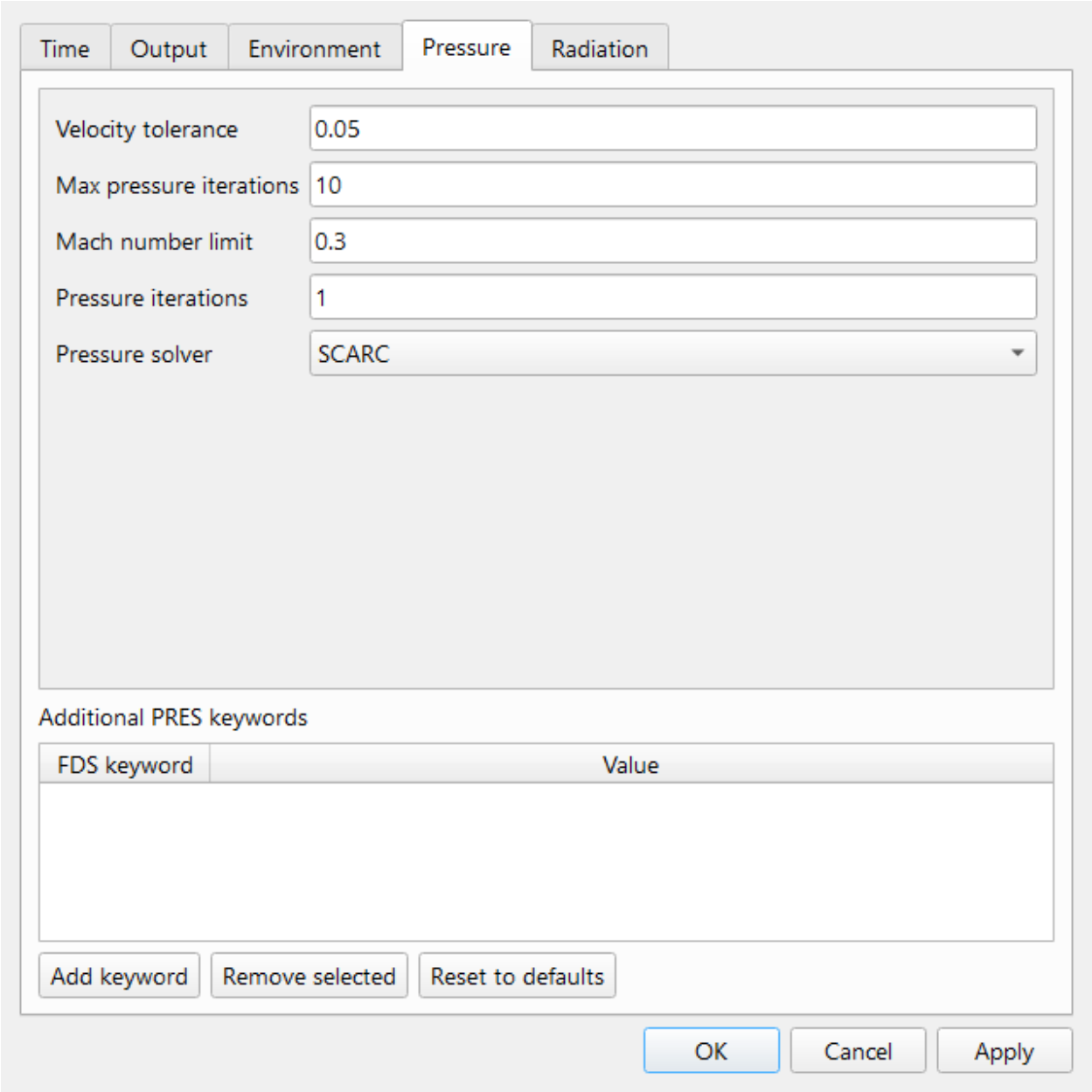


Figure 7. Model > Simulation Parameters: Pressure tab

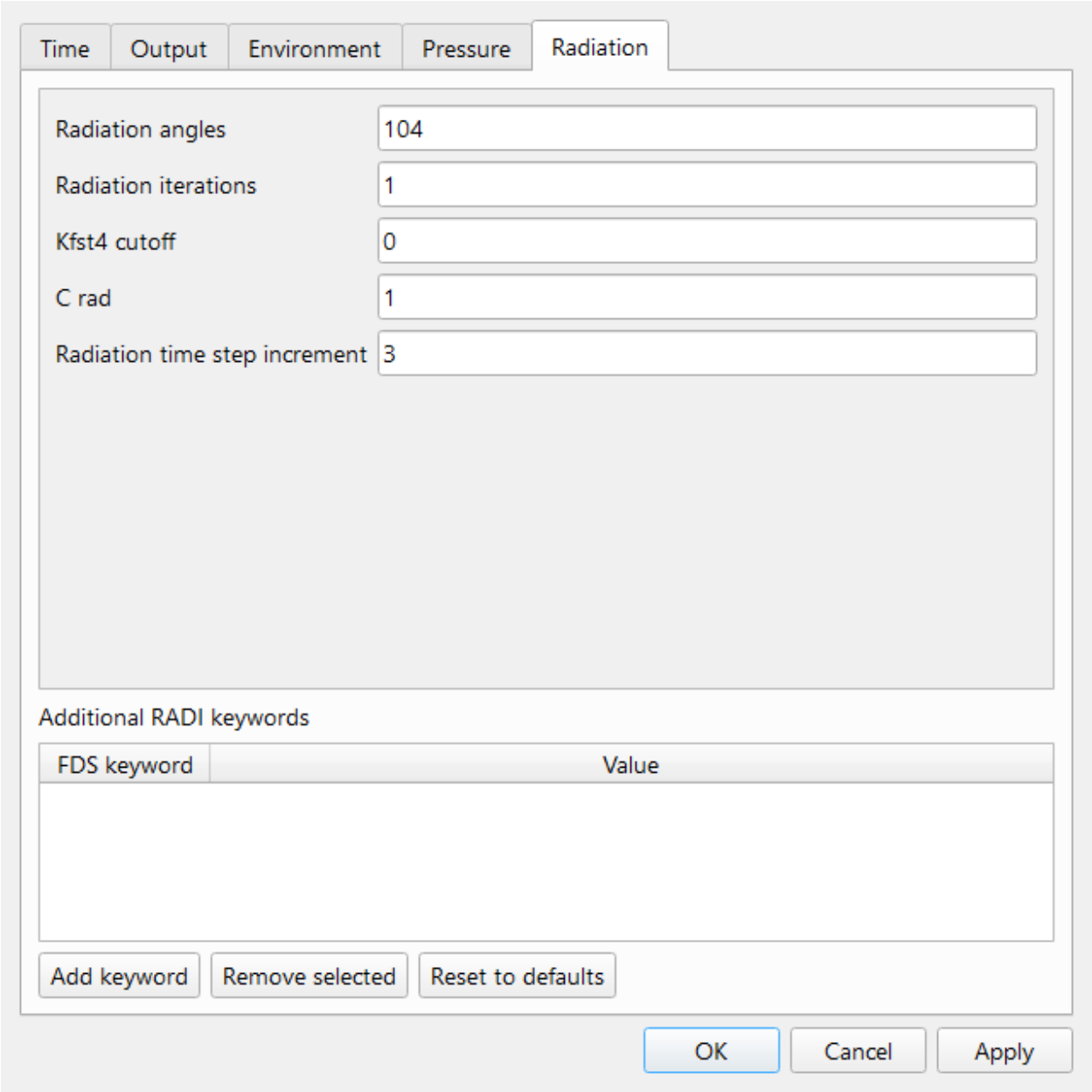


Figure 8. Model > Simulation Parameters: Radiation tab

Step 3 - Mesh. The cell size is 6 m / 60 = 0.1 m in every direction. The cell size has to resolve the fire: FDS guidance uses D*/dx, the characteristic fire diameter over the cell size. For 180 kW, D* = 0.48 m; 0.15 m cells give only 3.2, which PyFDS's validation rejects (below 4), while 0.1 m cells give 4.8.

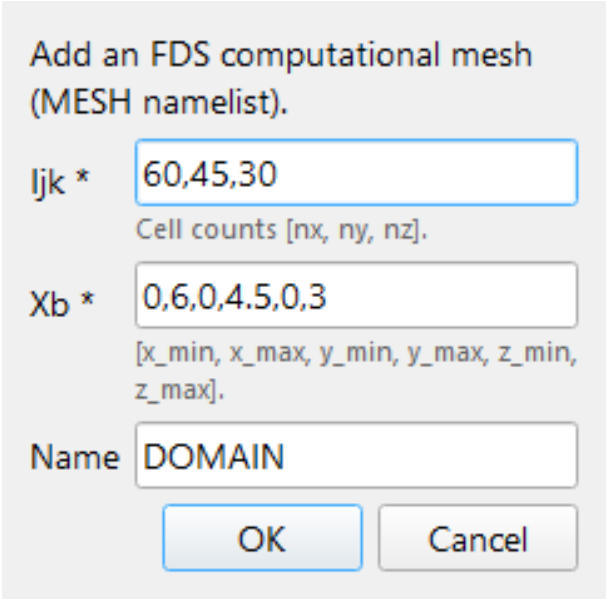


Figure 9. Model > Add Mesh: a 6 x 4.5 x 3 m domain in 60 x 45 x 30 cells (0.1 m cells, 81,000 in all). 0.1 m is the coarsest cell that resolves this 180 kW fire (validation: D*/dx at least 4).

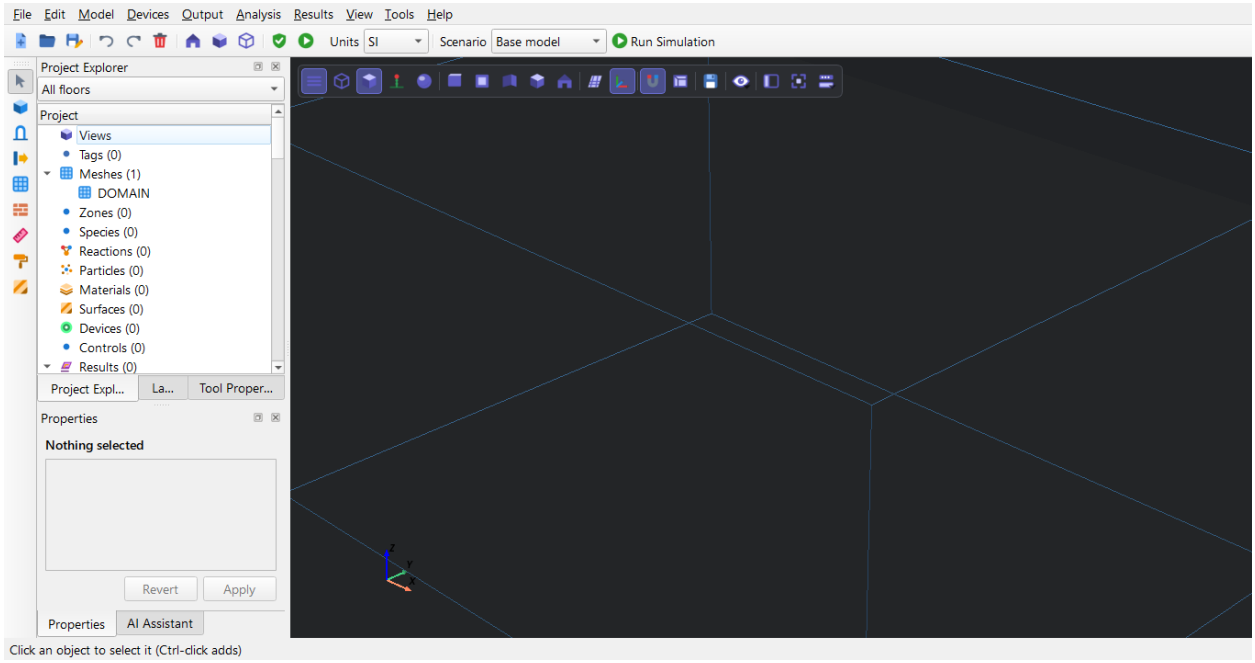


Figure 10. The mesh appears in the 3D view and under Meshes in the Project Explorer.

Step 4 - Wall construction from the library.

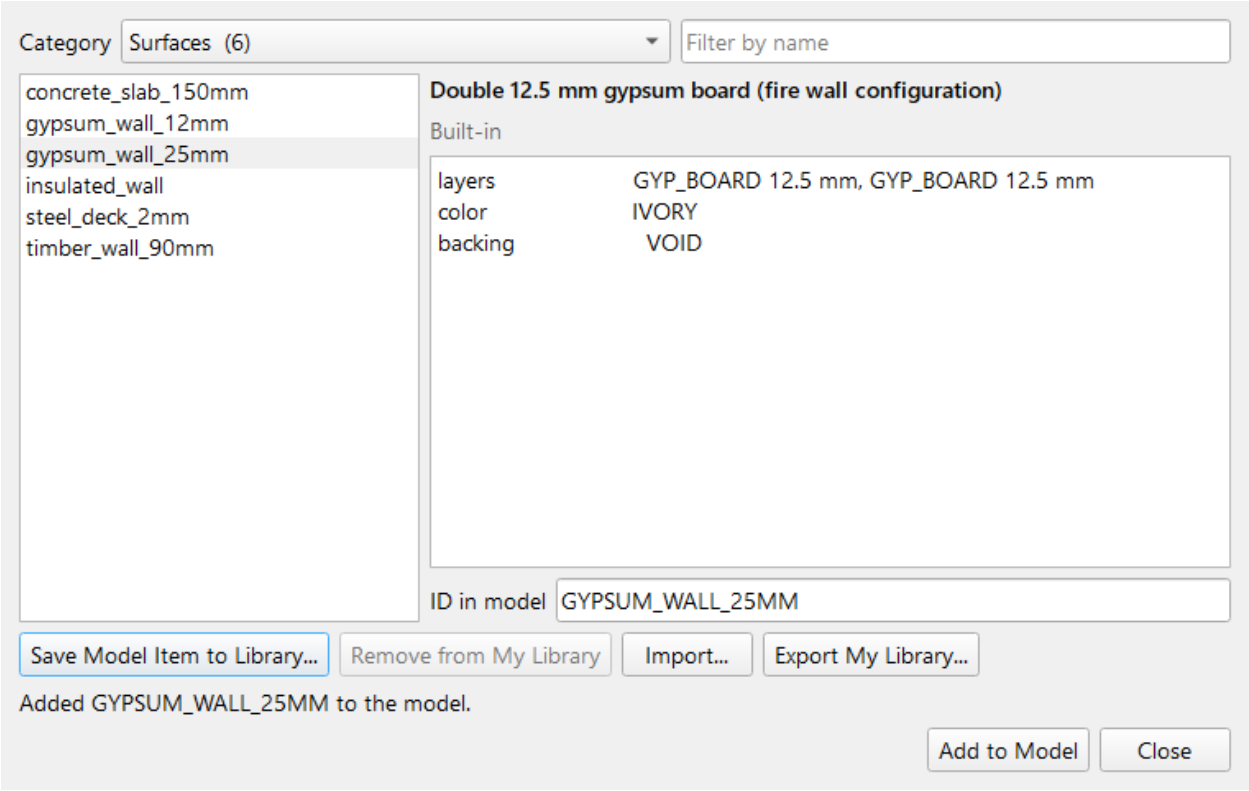


Figure 11. Model > Library: the 25 mm gypsum wall construction (two 12.5 mm boards) is added to the project.

Step 5 - Draw the room with its two openings.

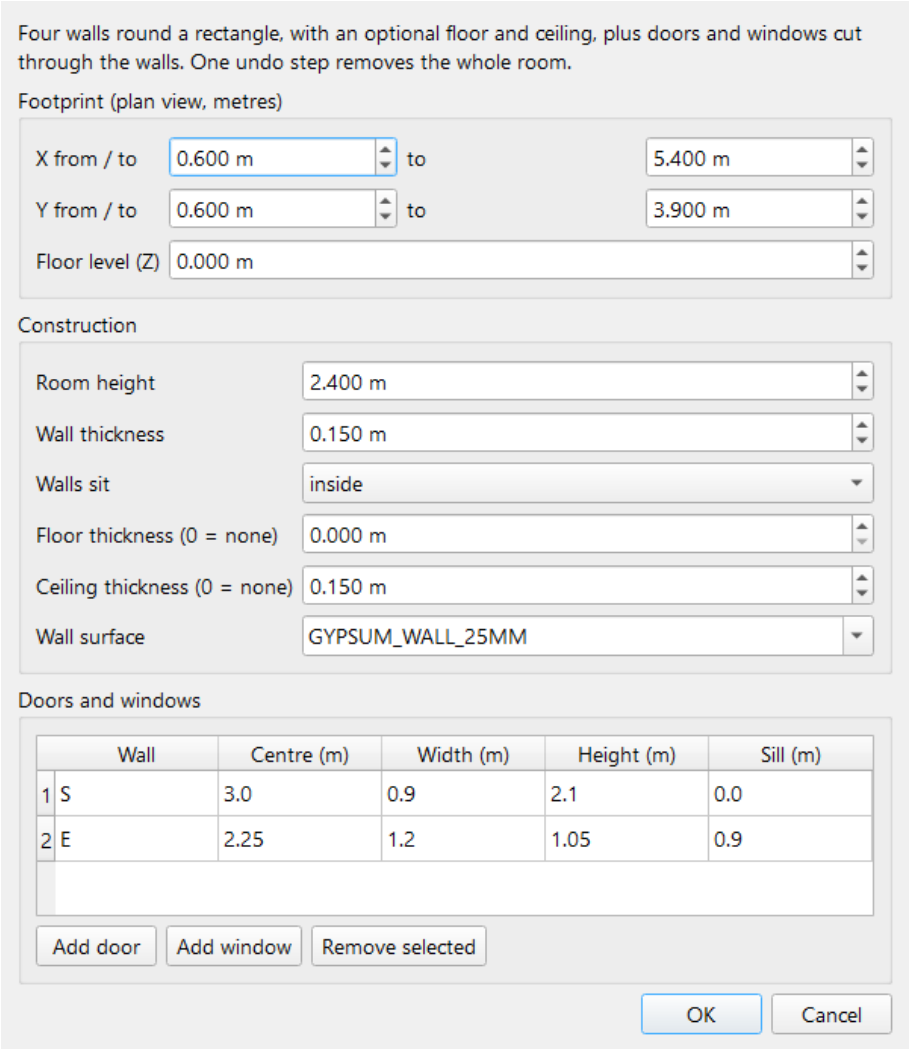


Figure 12. Draw > Draw Room: walls, ceiling and two openings - a 0.9 x 2.1 m door in the south wall and a 1.2 x 1.05 m window (sill 0.9 m) in the east wall.

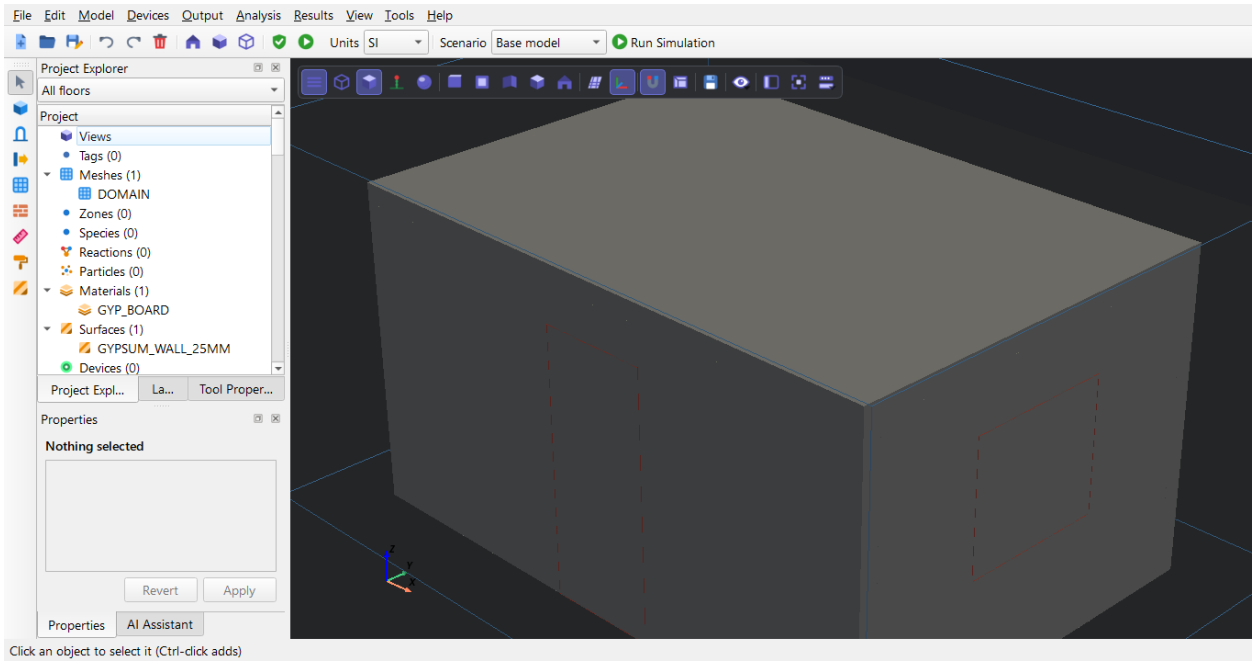


Figure 13. The room (4.8 x 3.3 x 2.4 m inside the mesh) with its door and window.

Step 6 - The fire: a burner surface, the burner body, the burning face and the reaction.

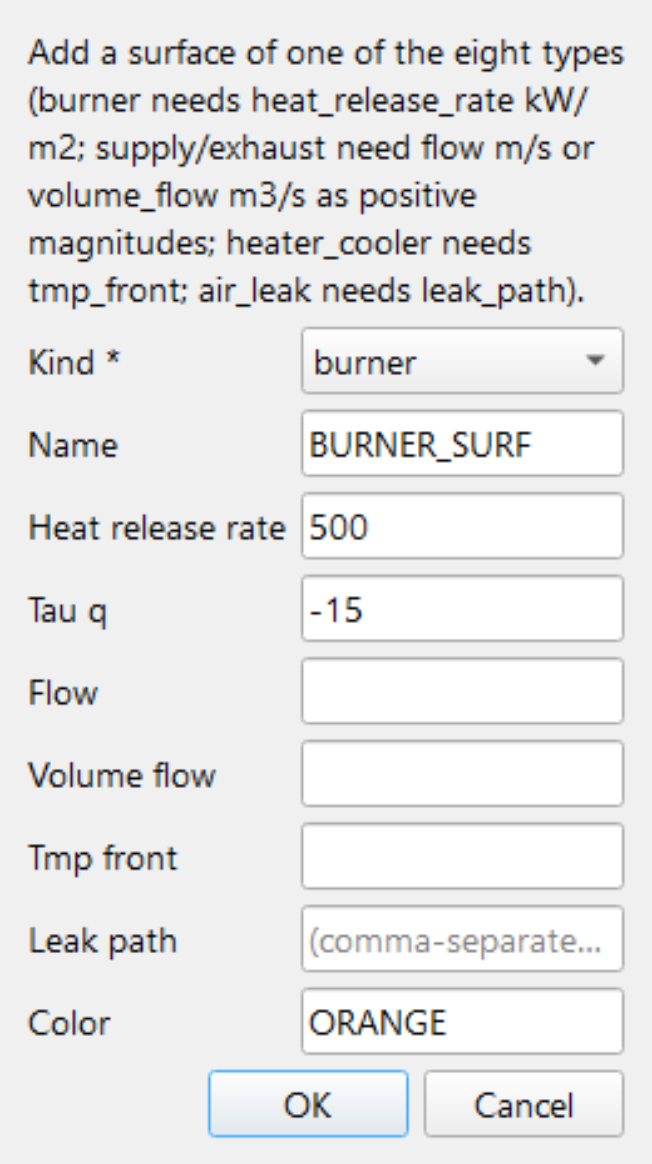


Figure 14. Model > Add Typed Surface, burner: 500 kW/m2, reached along a t-squared curve in 15 s (TAU_Q = -15).

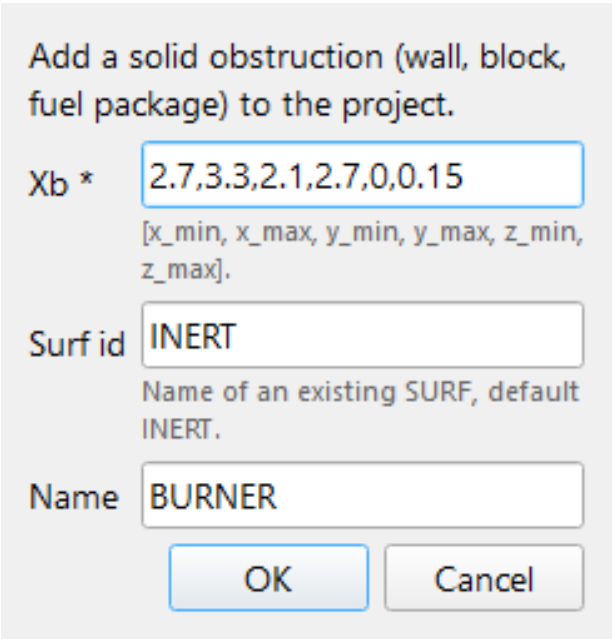


Figure 15. Model > Add Obstruction: the burner body, a 0.6 x 0.6 x 0.15 m inert block in the middle of the room.

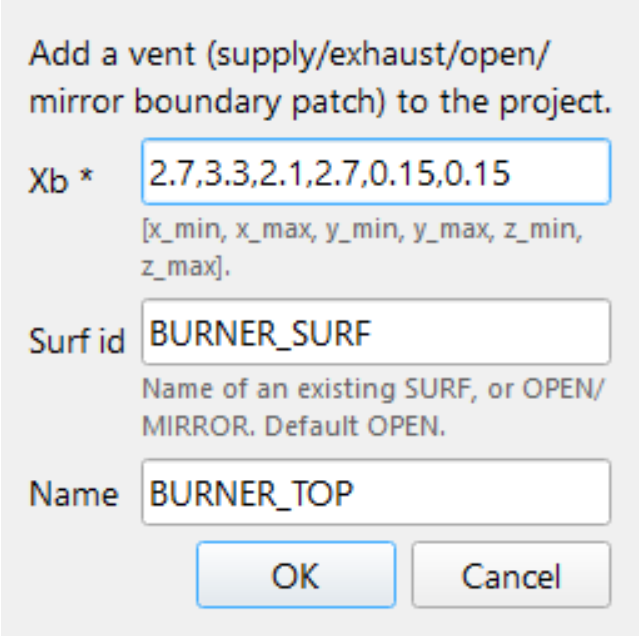


Figure 16. Model > Add Vent: the burning surface is a vent on the top face of the block only (0.36 m2 x 500 kW/m2 = 180 kW). Giving the whole block the burner surface would also burn its four sides and double the fire.

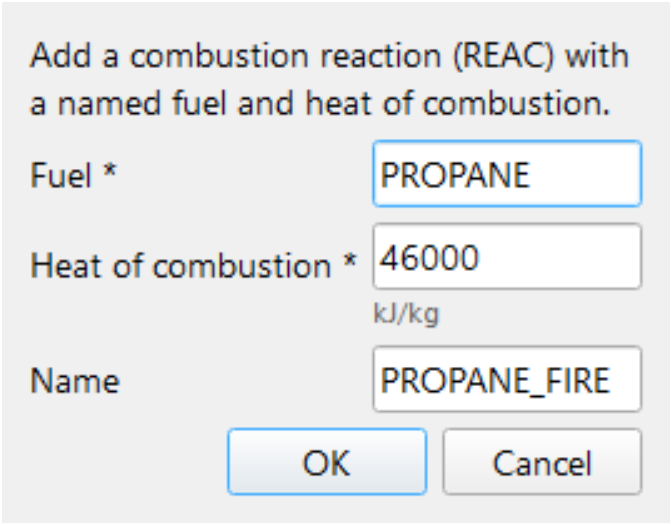


Figure 17. Model > Add Reaction: propane, heat of combustion 46,000 kJ/kg.

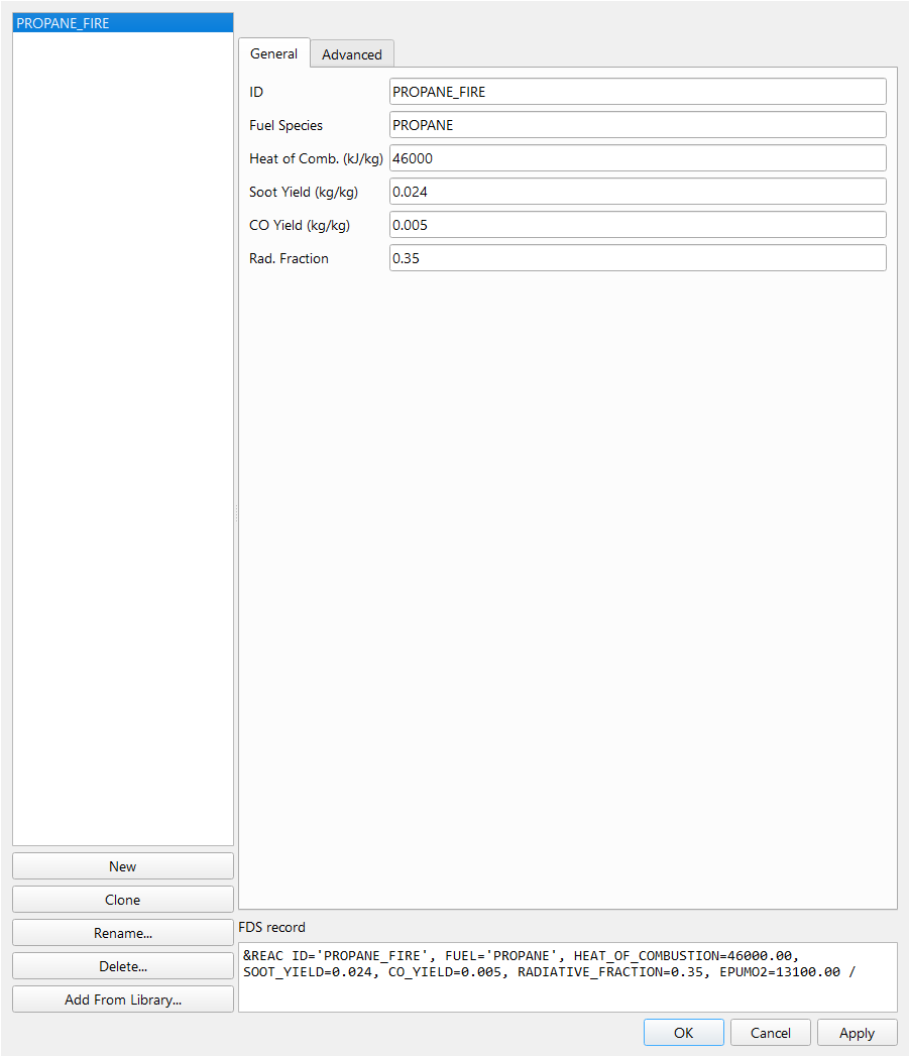


Figure 18. Model > Edit Reactions: soot yield 0.024 kg/kg and CO yield 0.005 kg/kg for propane (well-ventilated values, SFPE Handbook, Tewarson). With the default yield of 0 the fire would make no smoke.

Step 7 - Outer boundaries. Model > Open Mesh Boundaries makes all six faces of the mesh OPEN. The room stands on the ground, so the floor vent is deleted again.

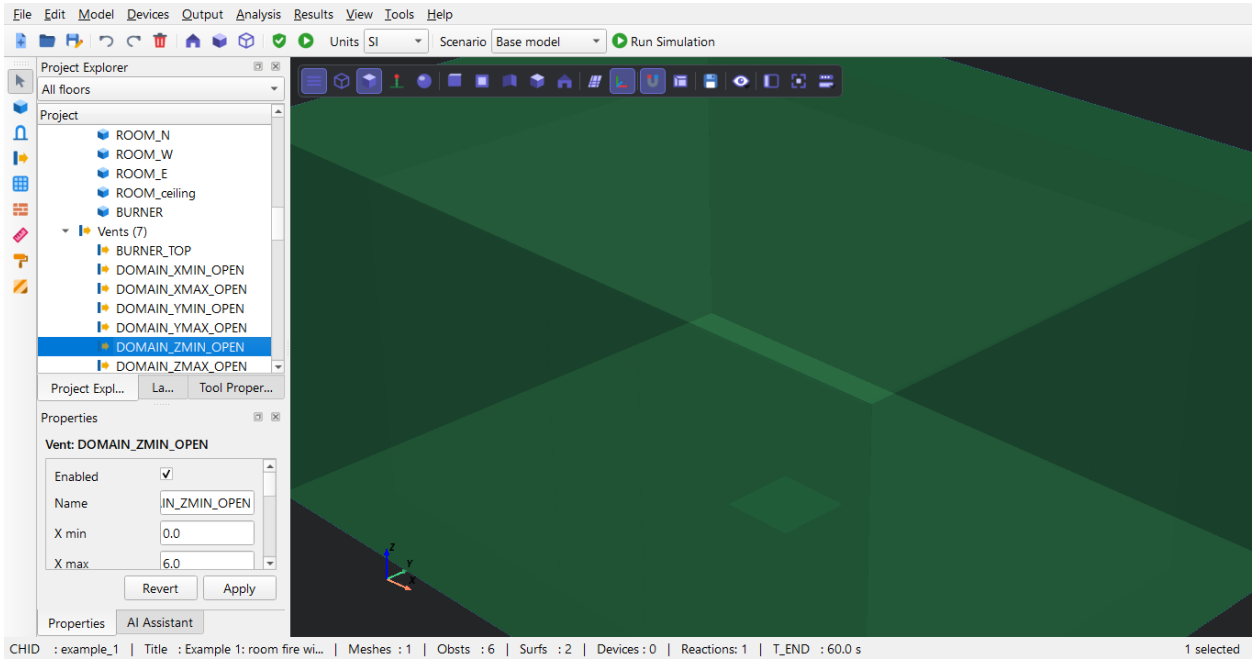


Figure 19. Open Mesh Boundaries opens all six faces of the mesh. The floor of a room standing on the ground must stay closed: select DOMAIN_ZMIN_OPEN in the Project Explorer and press Delete.

Step 8 - Instruments and outputs.

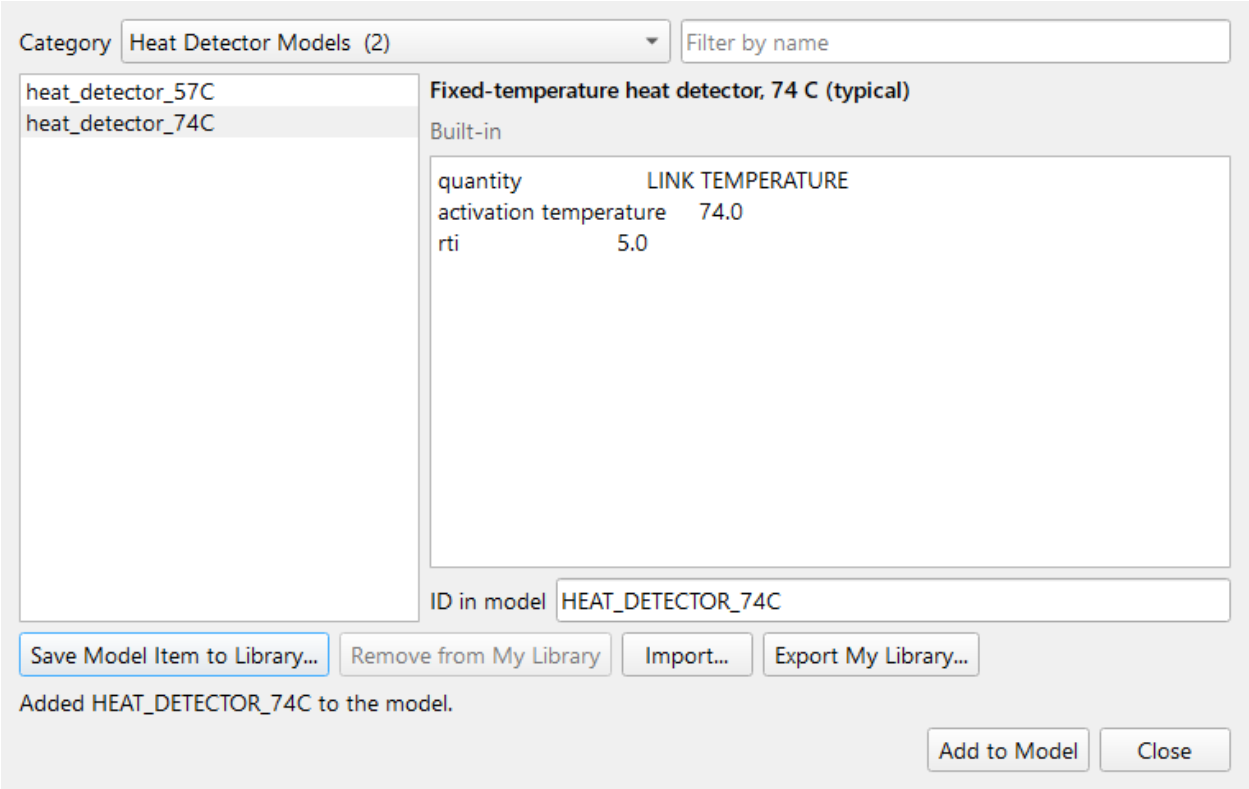


Figure 20. Model > Library: a heat detector property (RTI 5, activation 74 C).

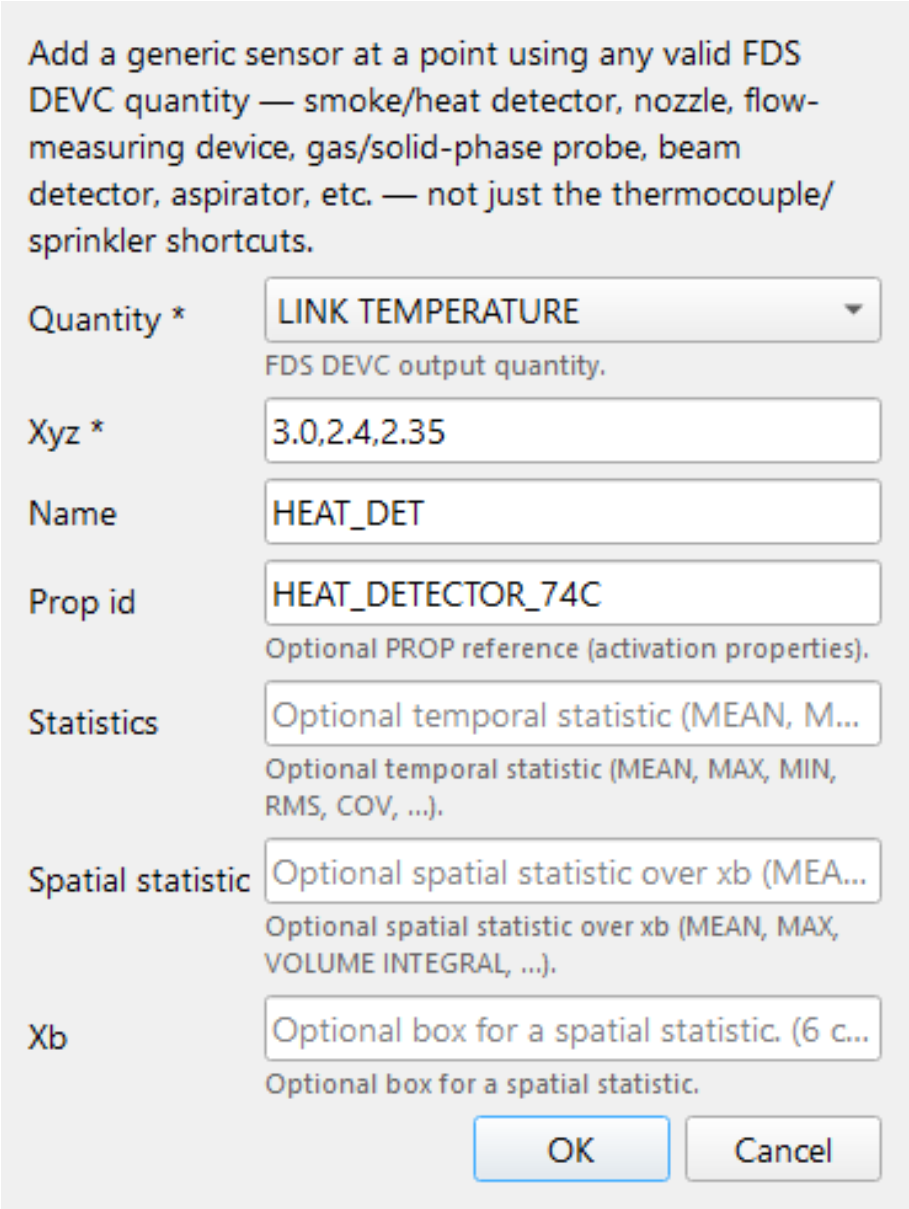


Figure 21. Model > Add Device: the heat detector 5 cm below the ceiling above the burner.

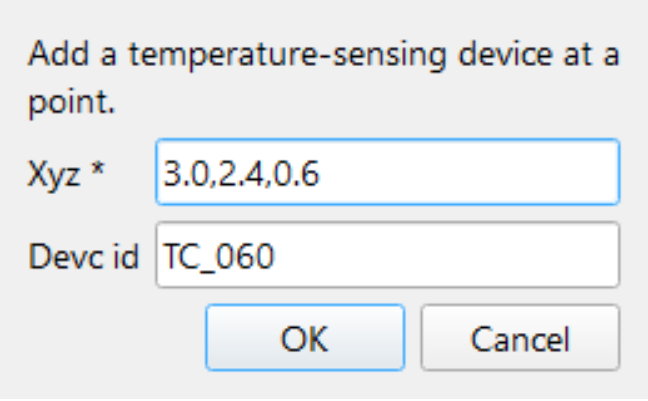


Figure 22. Model > Add Thermocouple: a vertical tree of four thermocouples at 0.6, 1.2, 1.8 and 2.3 m above the burner centre (the same dialog is used four times).

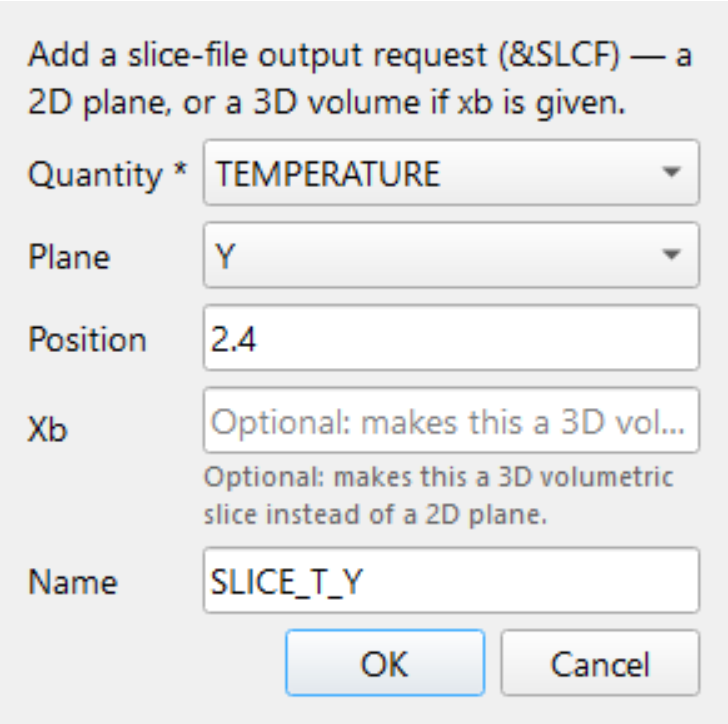


Figure 23. Model > Add Slice: a temperature slice through the burner (Y = 2.4 m).

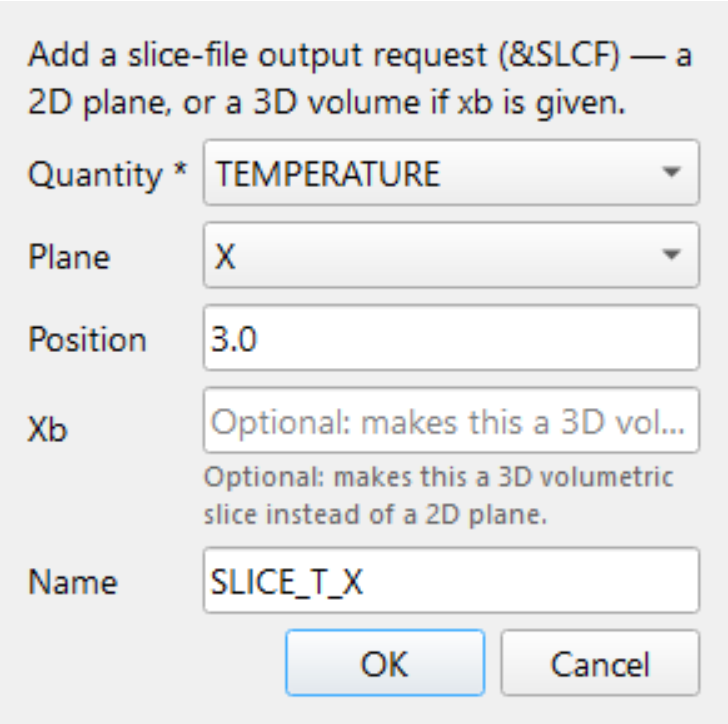


Figure 24. Model > Add Slice: a second temperature slice across the room (X = 3.0 m).

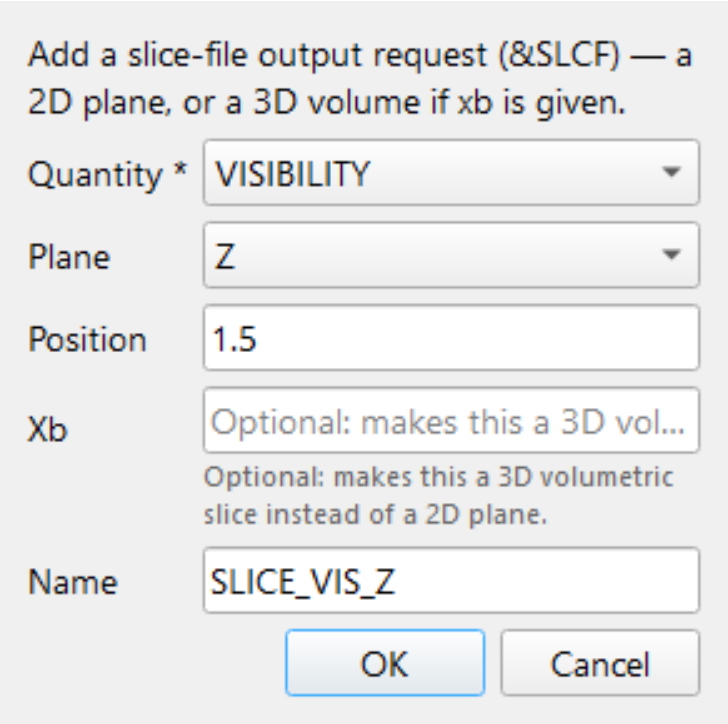


Figure 25. Model > Add Slice: a visibility slice at head height (Z = 1.5 m).

Step 9 - Check the finished model.

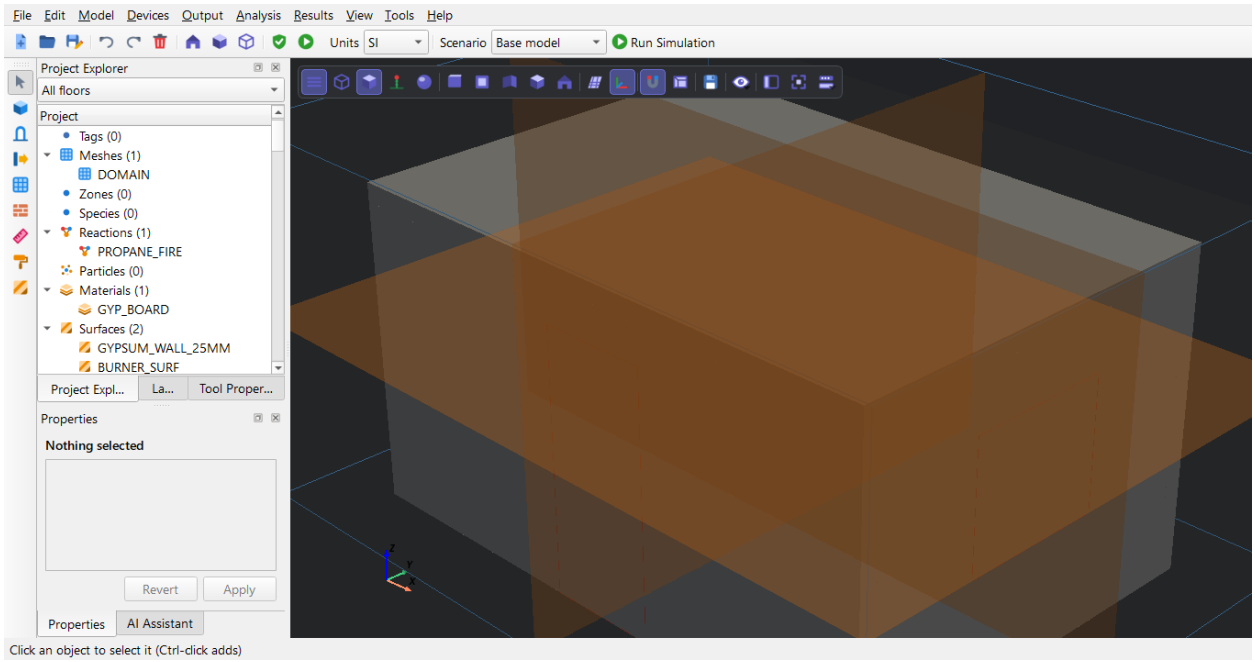


Figure 26. The complete model (open-boundary vents hidden): room, door, window, burner, detector and thermocouple tree, and the three slice planes.

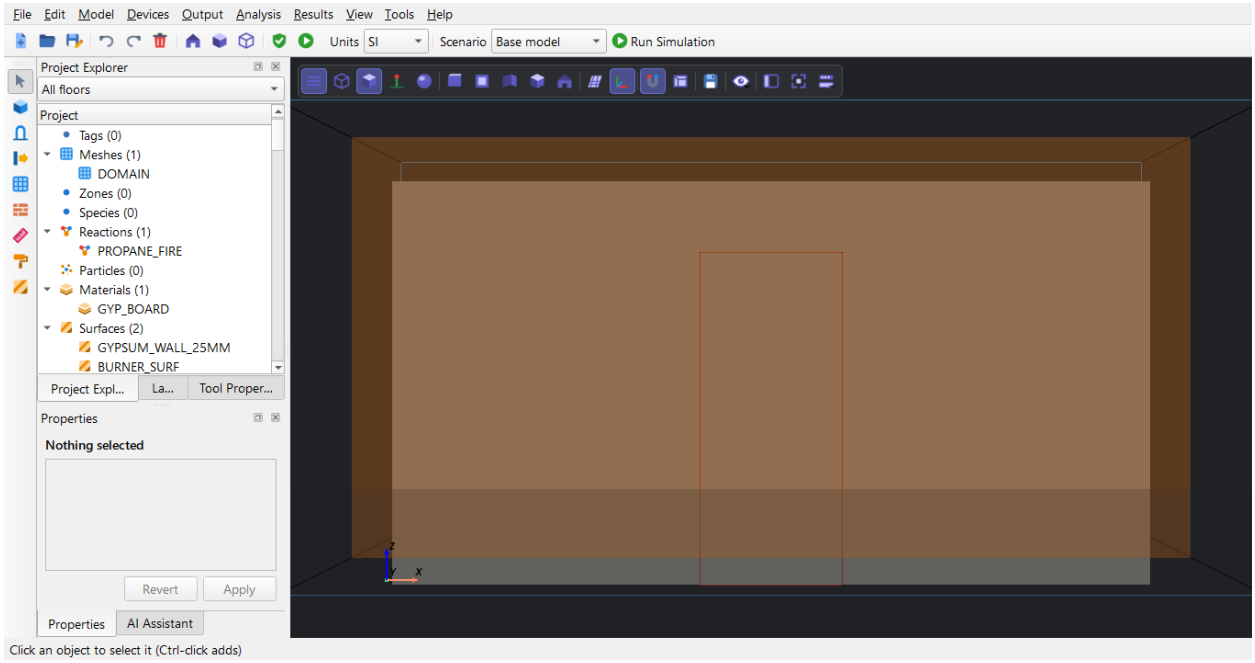


Figure 27. Front view (View > Front): the door in the south wall and the burner behind it.

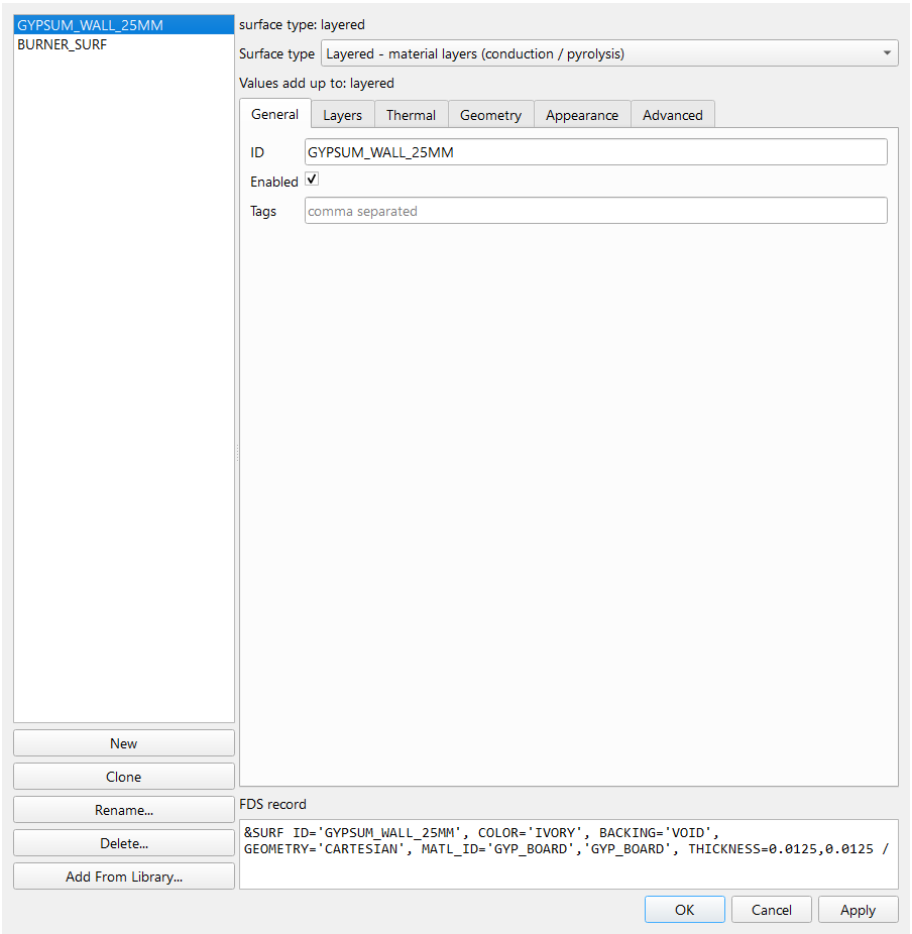


Figure 28. Model > Edit Surfaces: every object of the kind in one list, its settings on the right and the FDS text it will write at the bottom.

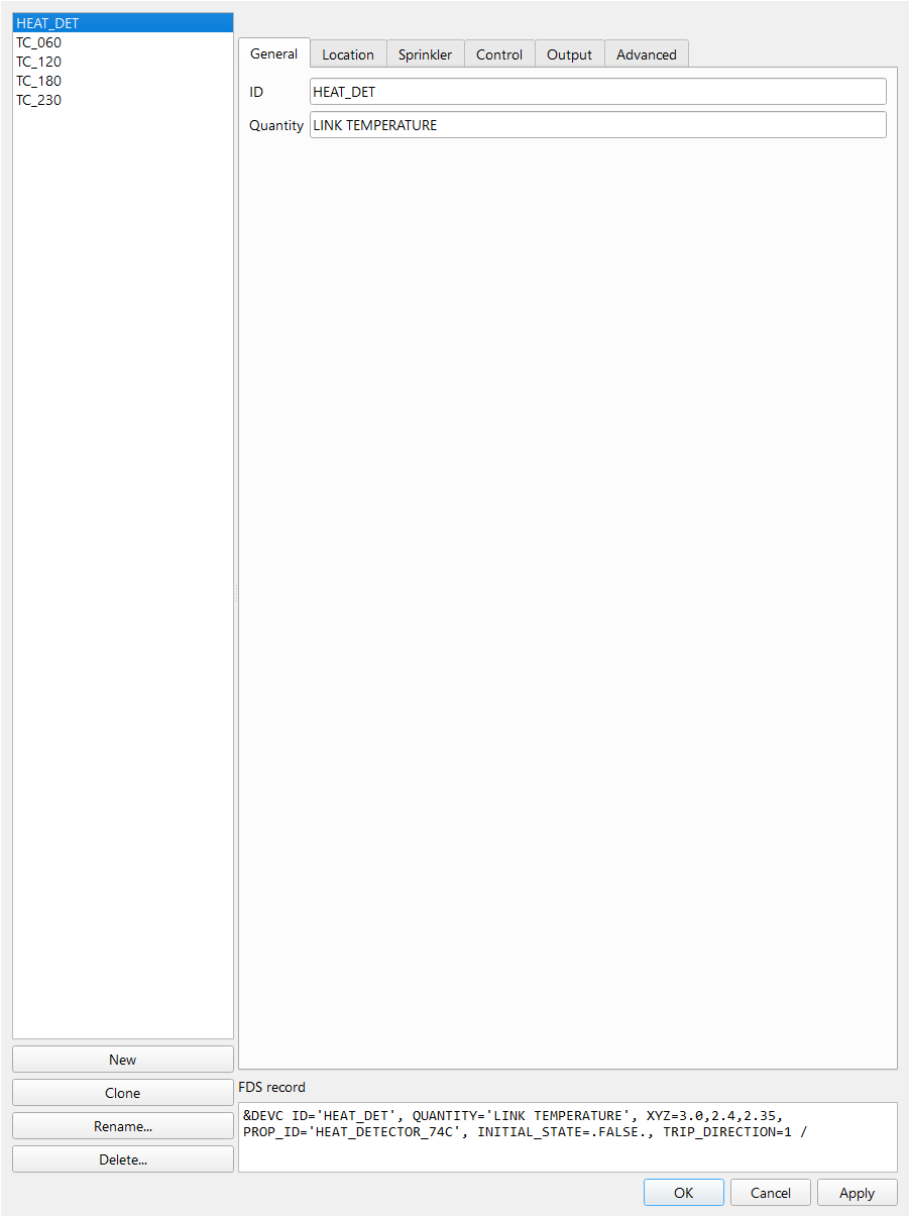


Figure 29. Model > Edit Devices: every object of the kind in one list, its settings on the right and the FDS text it will write at the bottom.

Save the project with File > Save As (example_1.pyfds). File > Export > FDS writes the FDS input file example_1.fds, which is also written automatically when the run starts.

D Validation and running the simulation

Model > Validate checks the model before it is run: meshes and their resolution, objects outside the mesh, references to surfaces, materials and properties, ramps, controls, HVAC and the time and output settings.

Result for this case: 0 errors, 1 warning(s). Warning: Mesh 'DOMAIN': D*/dx ≈ 4.8 — borderline resolution. Recommended ratio is ≥ 10.0. Errors block a run; warnings do not, but should be understood. A resolution warning (D*/dx between 4 and 10) is expected for a demonstration mesh: the plume is resolved, but coarsely (see section F).

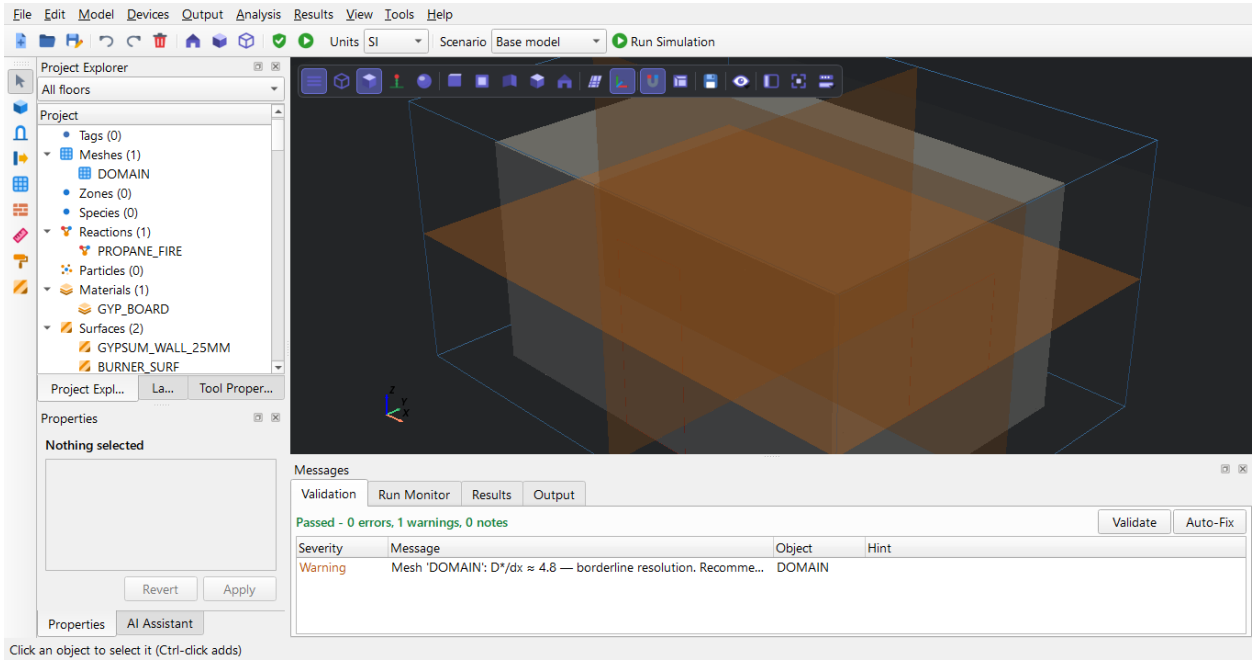


Figure 30. Model > Validate: the findings appear in the Validation tab at the bottom. Double-click a finding to select the object it is about.

Run > Run Simulation opens the Run dialog. For a one-mesh model it proposes 'This computer' with one process. Choose the folder for the input and results (Browse...), check the FDS version (PyFDS uses the bundled FDS 6.10.1 when none is configured; Locate FDS... picks another), read the 'Before you run' summary, and press Run. 'Check the model first' stops the run if the model has errors.

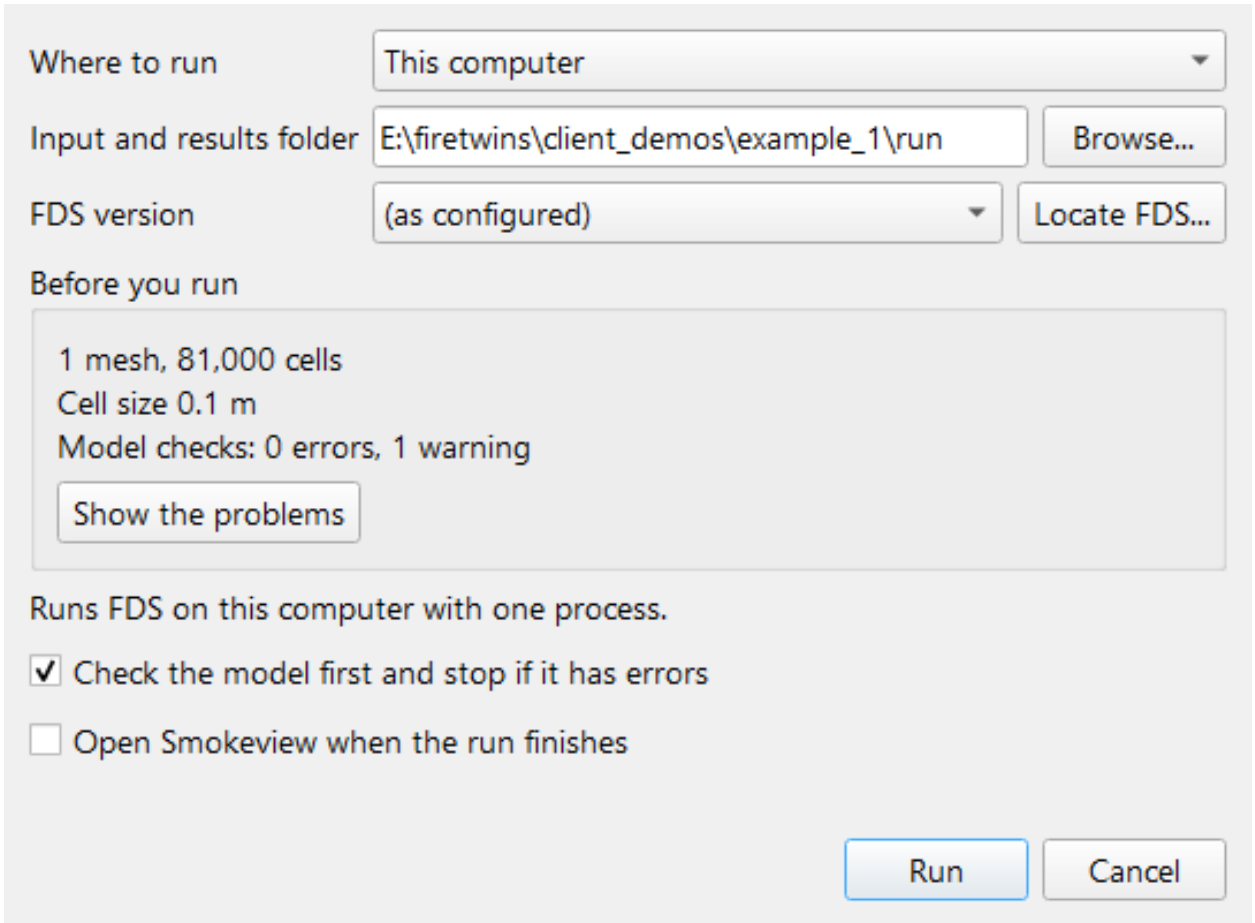


Figure 31. Run > Run Simulation: where to run (this computer, one process - the model has one mesh), the folder for the input and results, the FDS version, and the 'Before you run' summary of the model and its checks.

The Messages panel opens with the Run Monitor: progress (simulated time / end time), the heat release rate curve as FDS computes it, and the last lines of the FDS log (.out file), refreshed twice a second. The status bar shows the progress too. Run > Stop Simulation asks FDS to finish cleanly and write a restart file; Run > Cancel Simulation ends it at once.

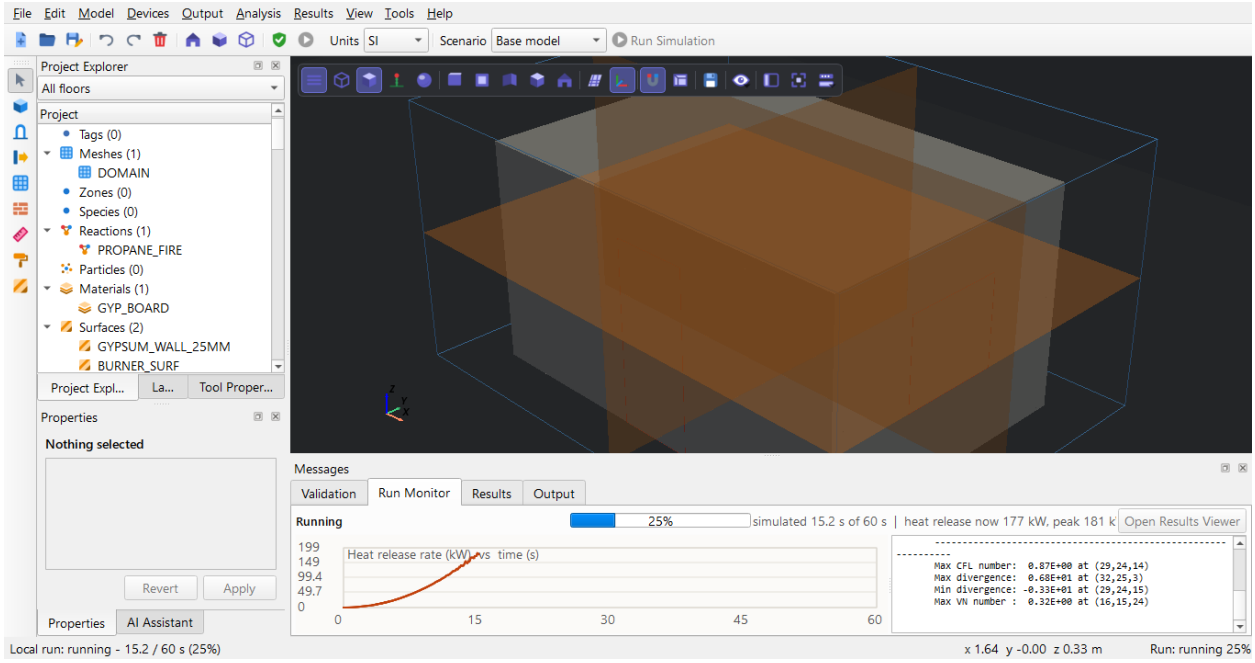


Figure 32. Run Monitor at 25 %: progress, heat release rate curve and the last lines of the FDS log.

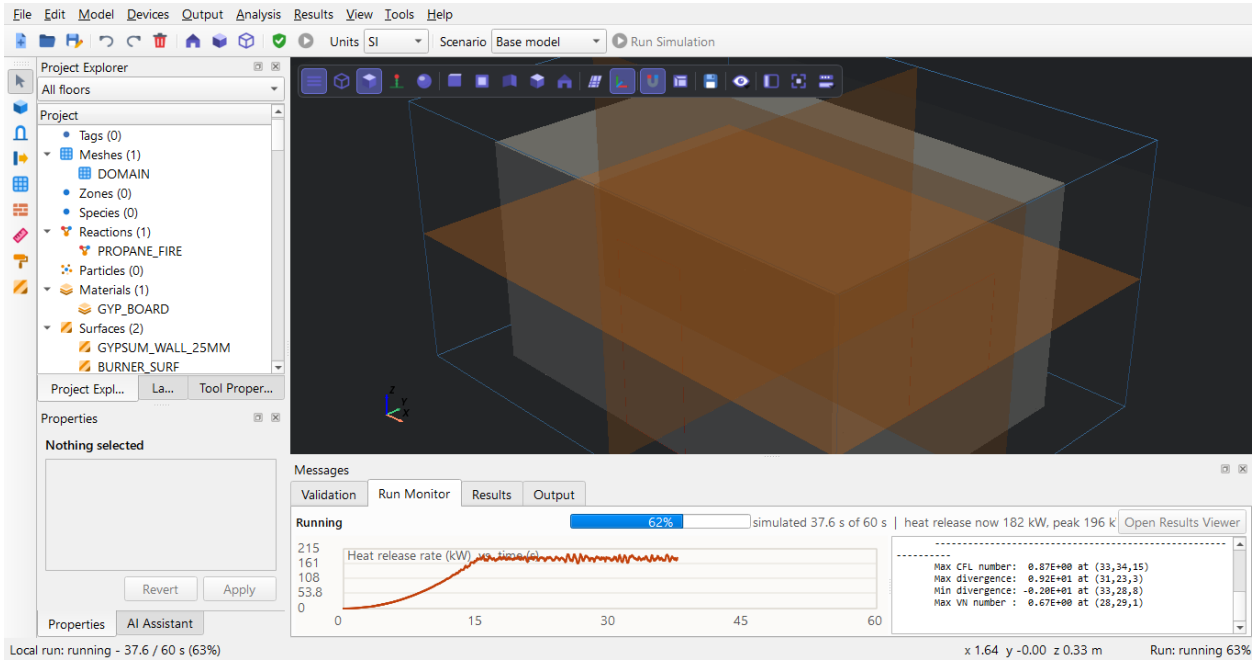


Figure 33. Run Monitor at 60 %: progress, heat release rate curve and the last lines of the FDS log.

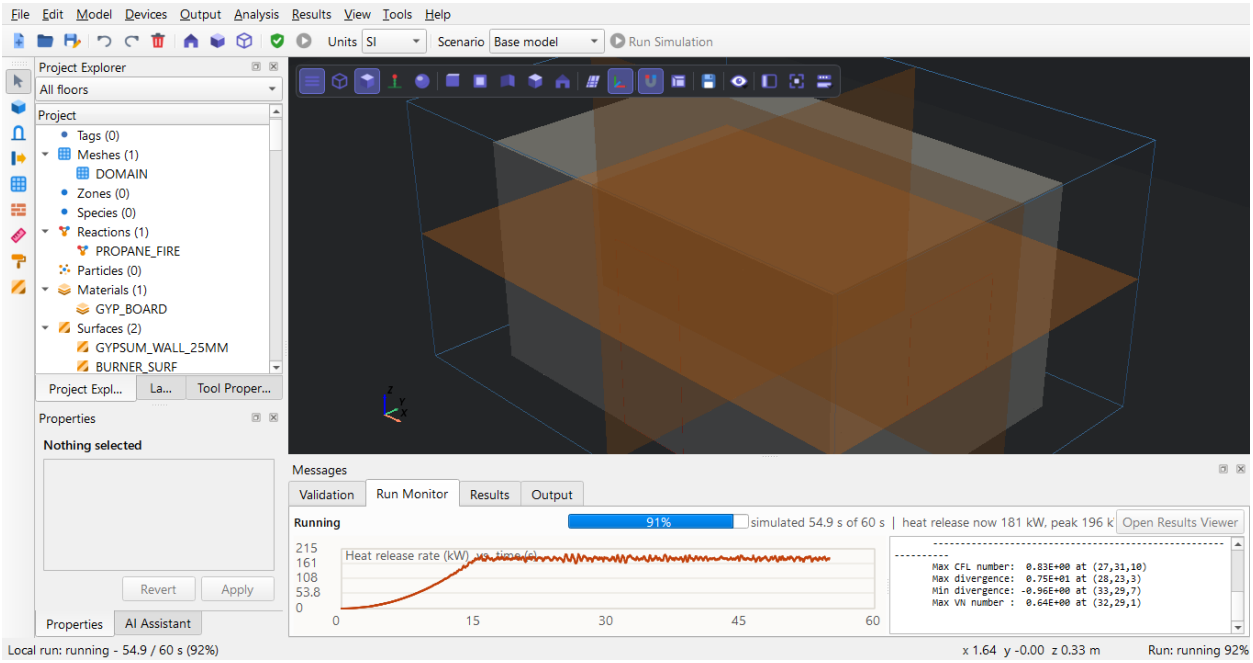


Figure 34. Run Monitor at 90 %: progress, heat release rate curve and the last lines of the FDS log.

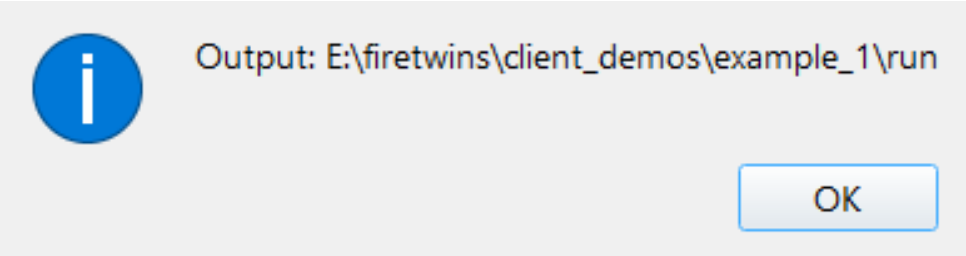


Figure 35. Message: Simulation complete

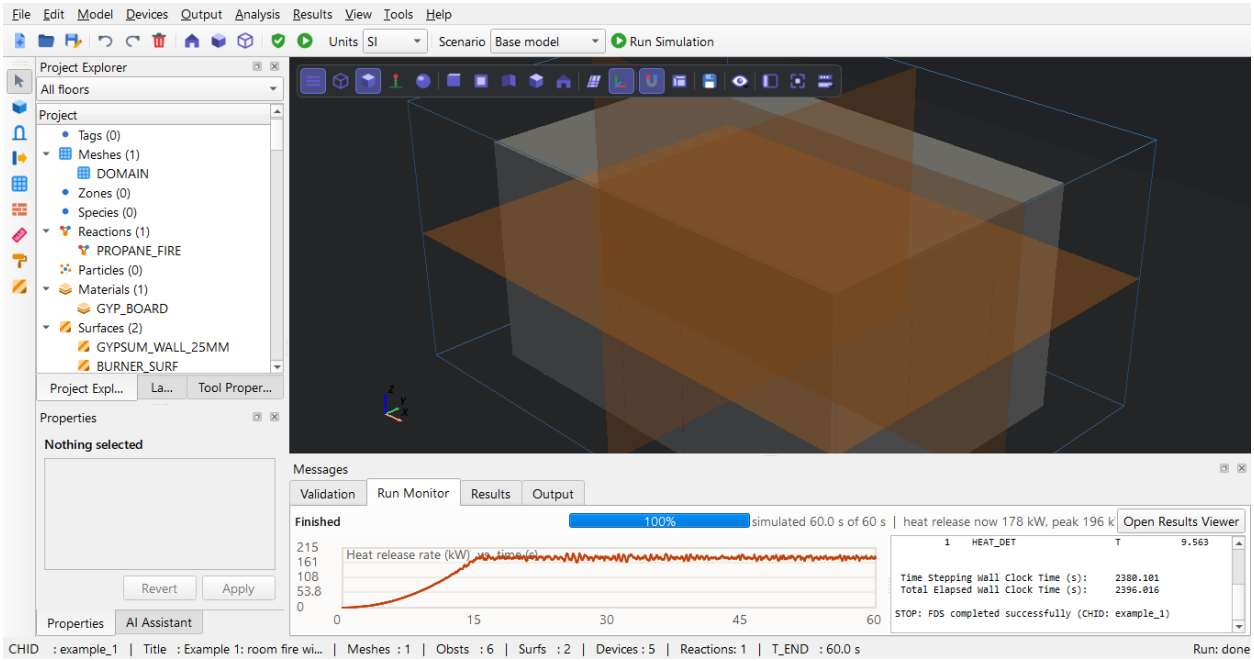


Figure 36. Run finished: 'STOP: FDS completed successfully' in the log, Open Results Viewer enabled.

Outcome on the test computer: FDS reported 'STOP: FDS completed successfully'; wall clock time 2396.016 s for 60 s of simulated time.

E Post-processing

E.1 PyFDS Results

Open the viewer from the editor with Results > Open PyFDS Results (or start it on its own, section 1.1) and load the run with File > Load Results: choose client_demos\example_1\run\example_1.smv. The Navigation View lists everything the run wrote: geometry, devices, views, 3D smoke, Plot3D, slices. Click a result to show it; use the time bar at the bottom (Play, step, drag the slider) to move through the run.

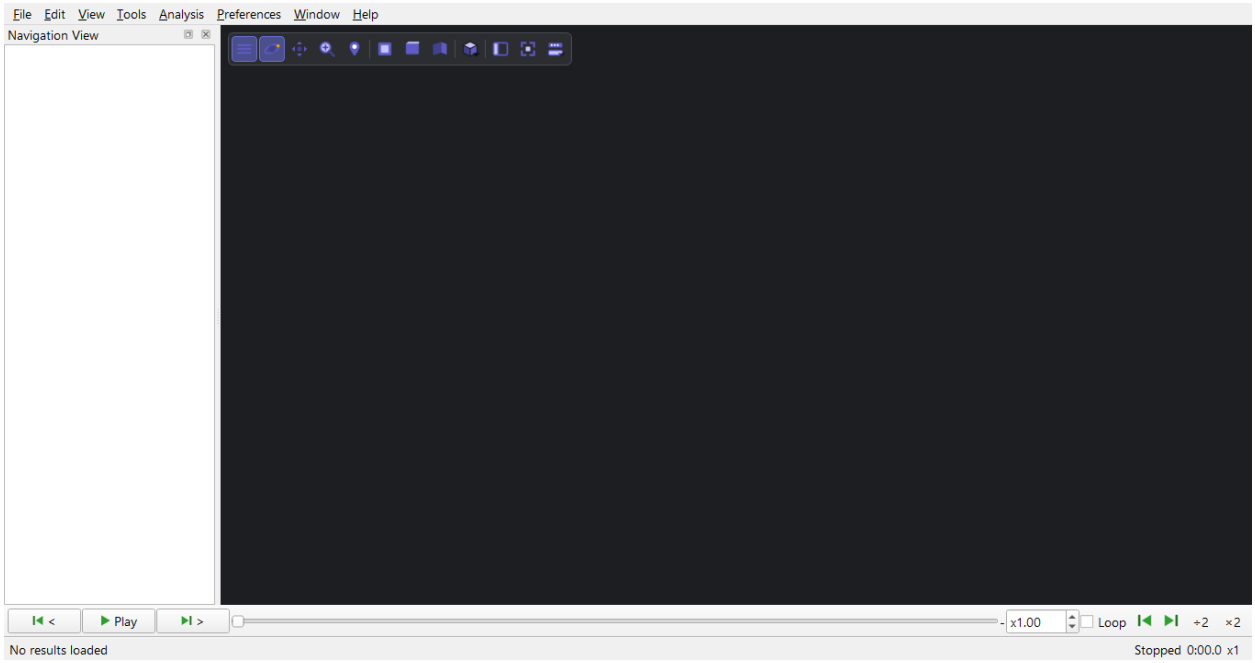


Figure 37. PyFDS Results (Results > Open PyFDS Results in the editor, or the PyFDS Results shortcut) before a run is loaded.

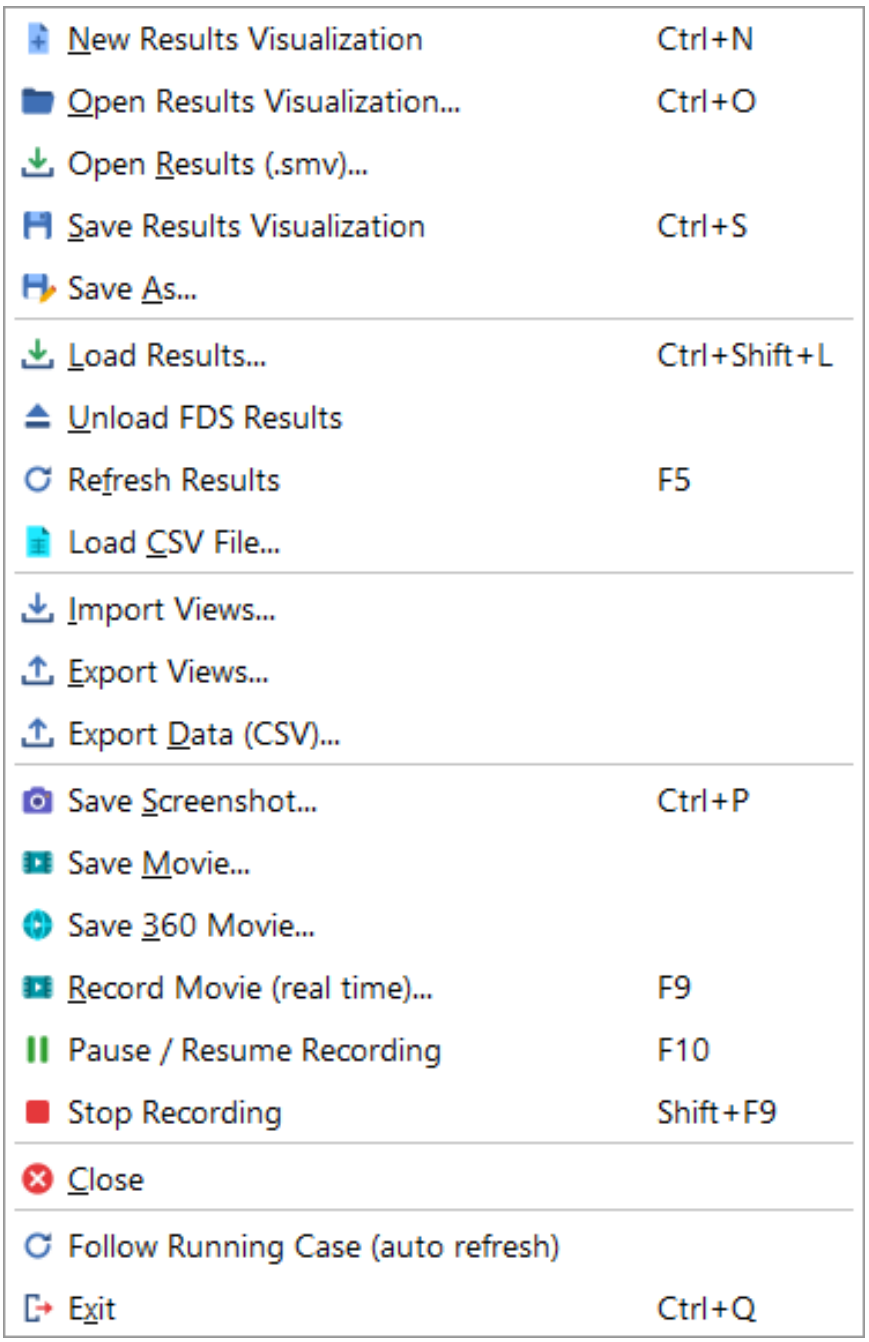


Figure 38. PyFDS Results, File menu: Load Results opens the .smv file of a run.

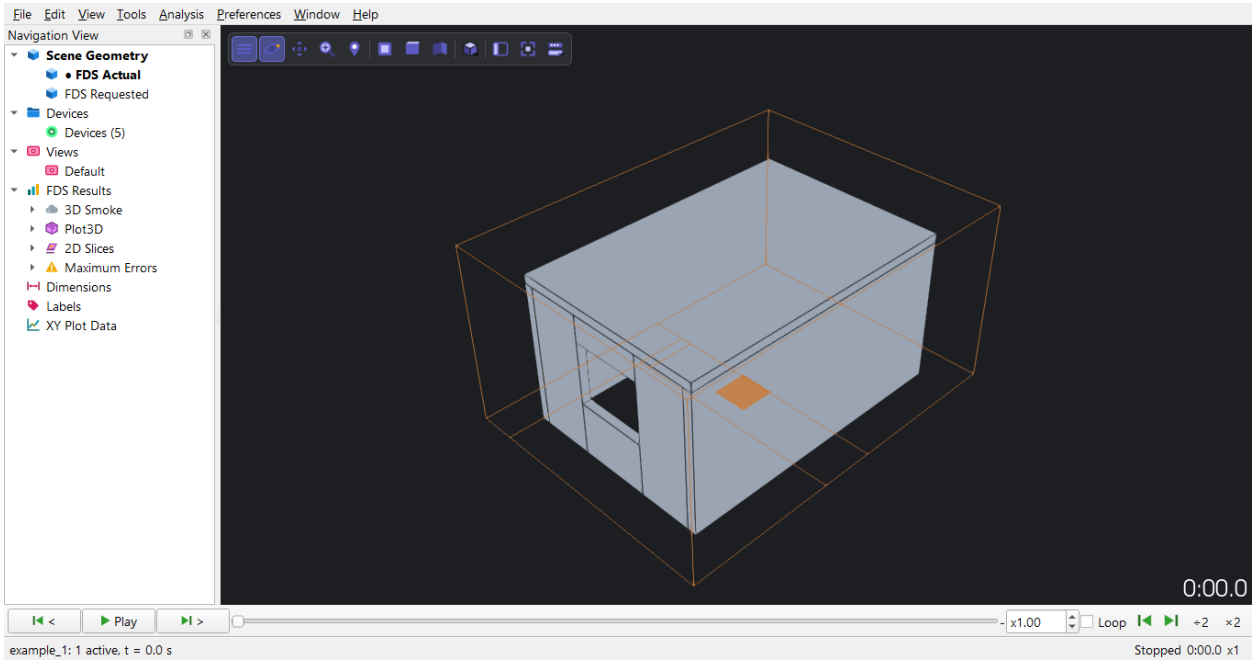


Figure 39. File > Load Results with example_1.smv: the geometry is shown and the Navigation View on the left lists every result of the run.

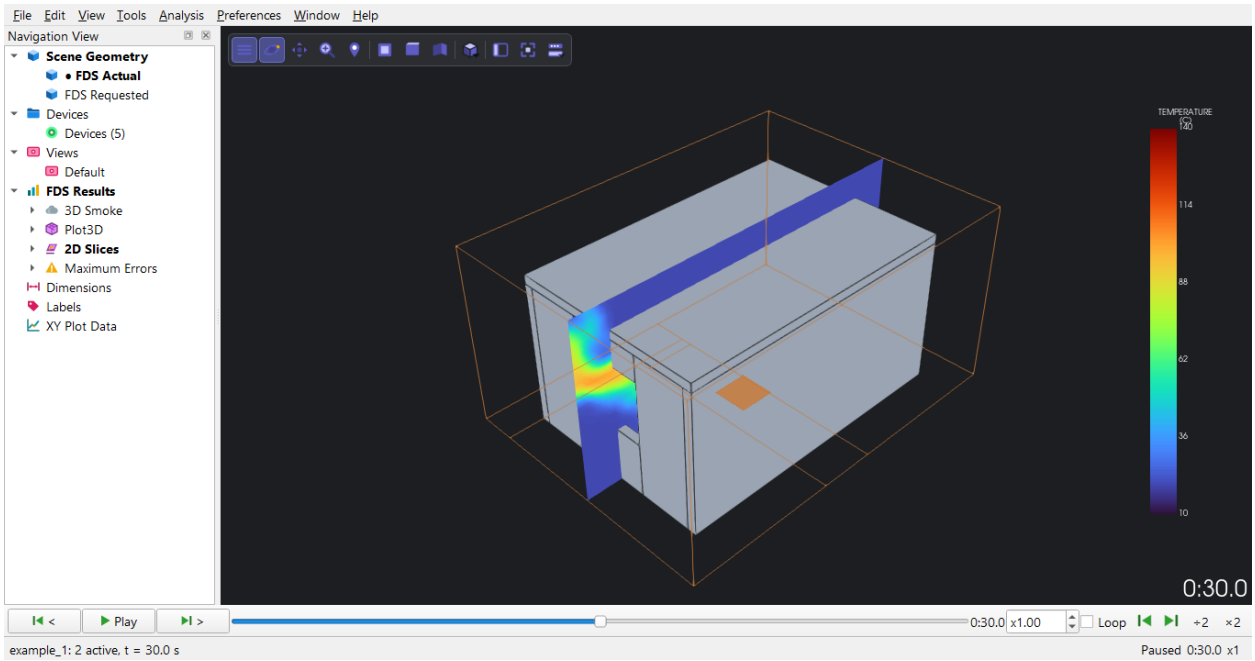


Figure 40. Temperature slice through the burner (Y = 2.4 m) at 30 s: the plume rises to the ceiling and the hot layer spreads under it.

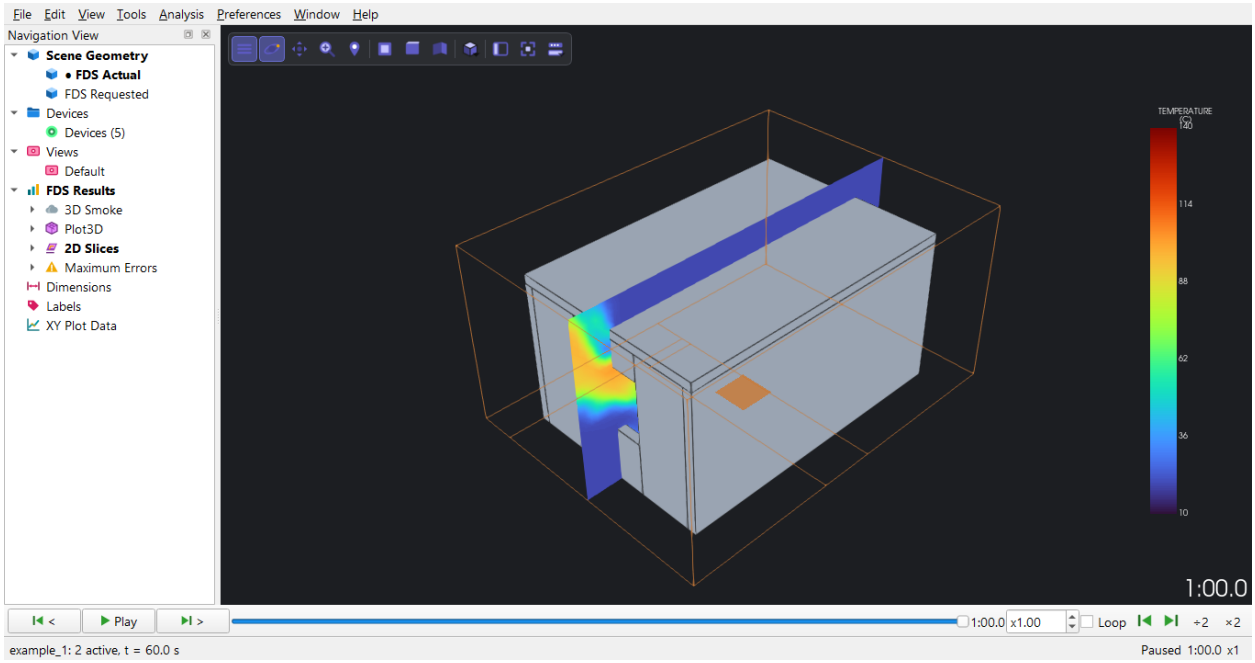


Figure 41. The same slice at the end of the run (60 s).

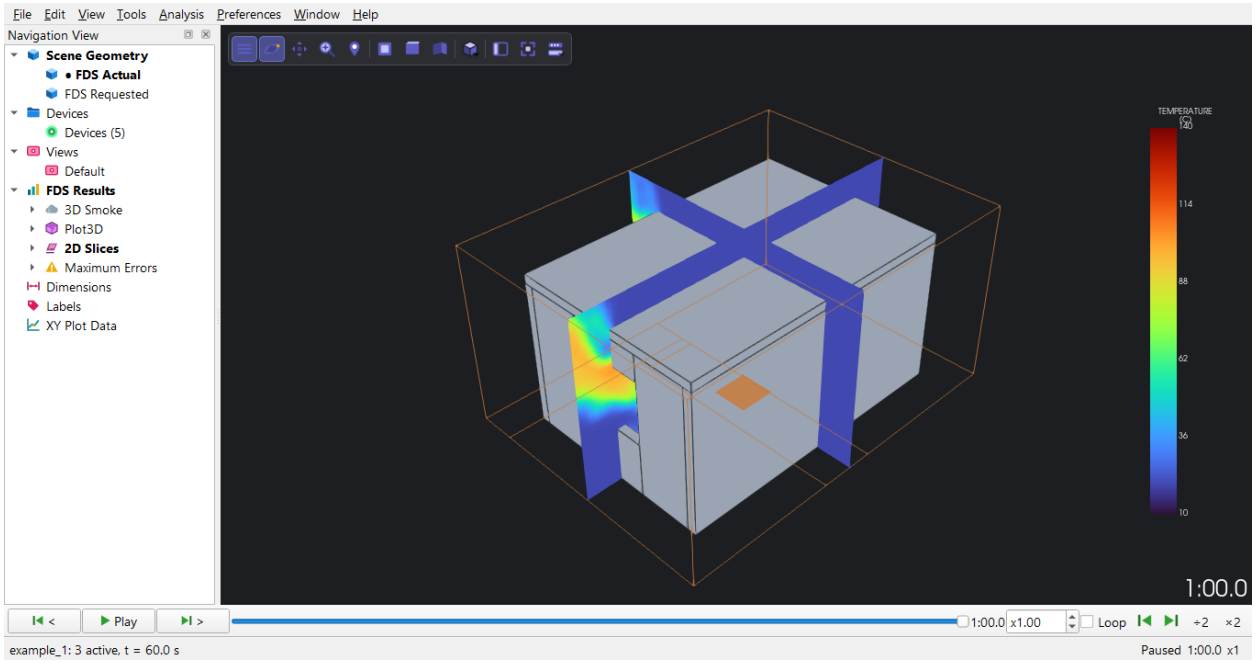


Figure 42. Two temperature slices at 60 s: along the room (Y = 2.4 m) and across it (X = 3.0 m).

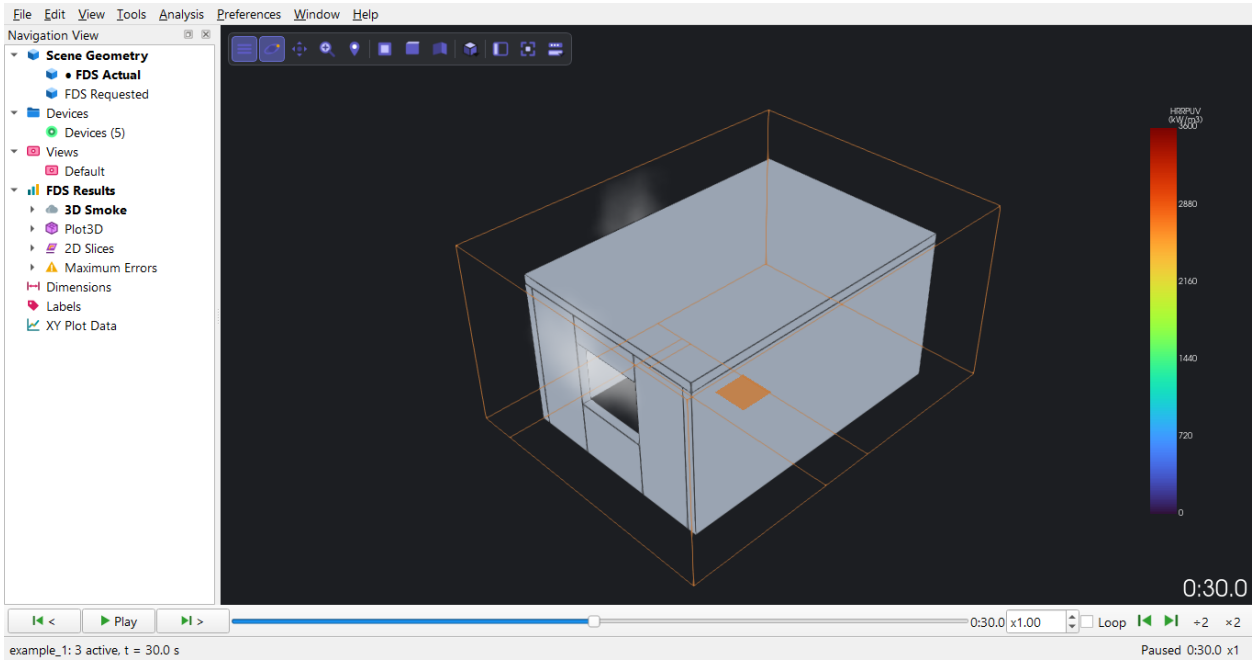


Figure 43. 3D smoke (soot density) and the flame (heat release rate per unit volume) at 30 s.

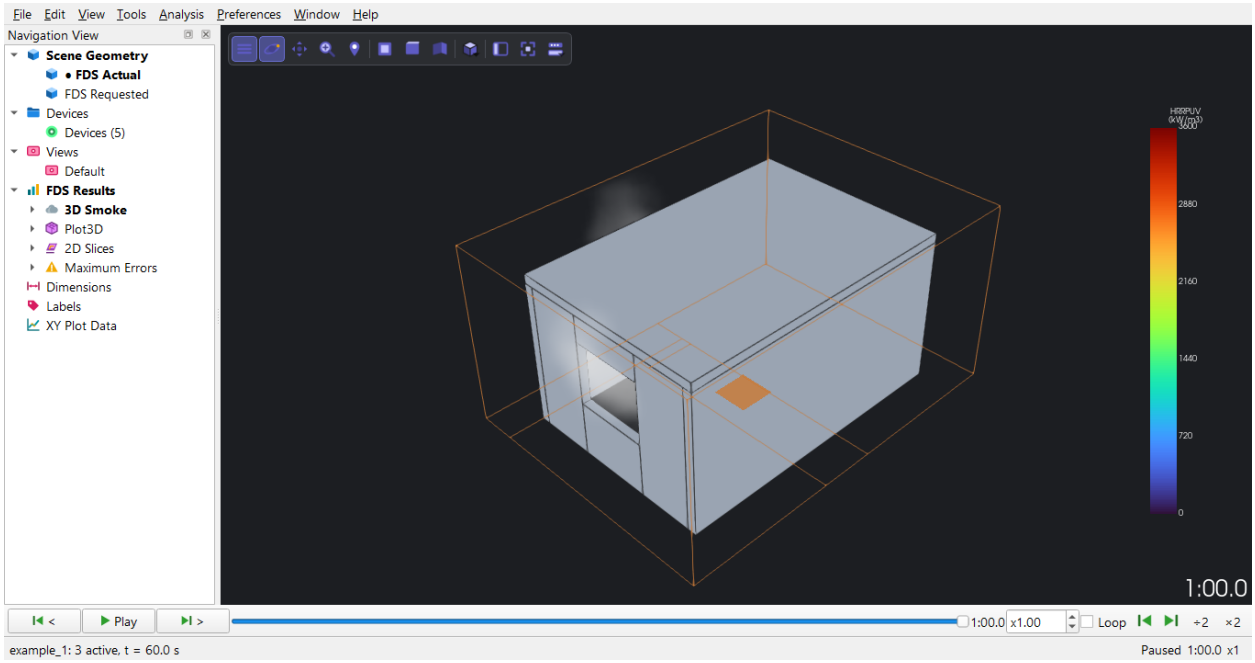


Figure 44. 3D smoke at 60 s: smoke leaves through the top of the door and the window.

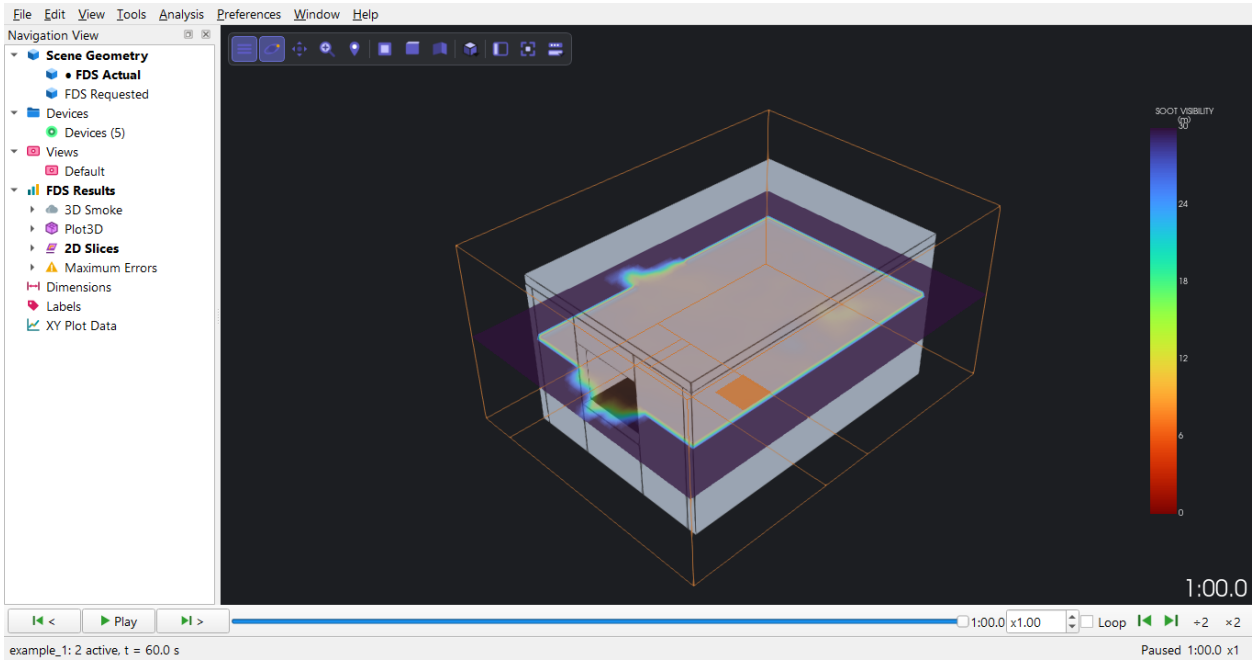


Figure 45. Visibility at head height (Z = 1.5 m) at 60 s (metres; 30 m is the FDS upper limit, clear air).

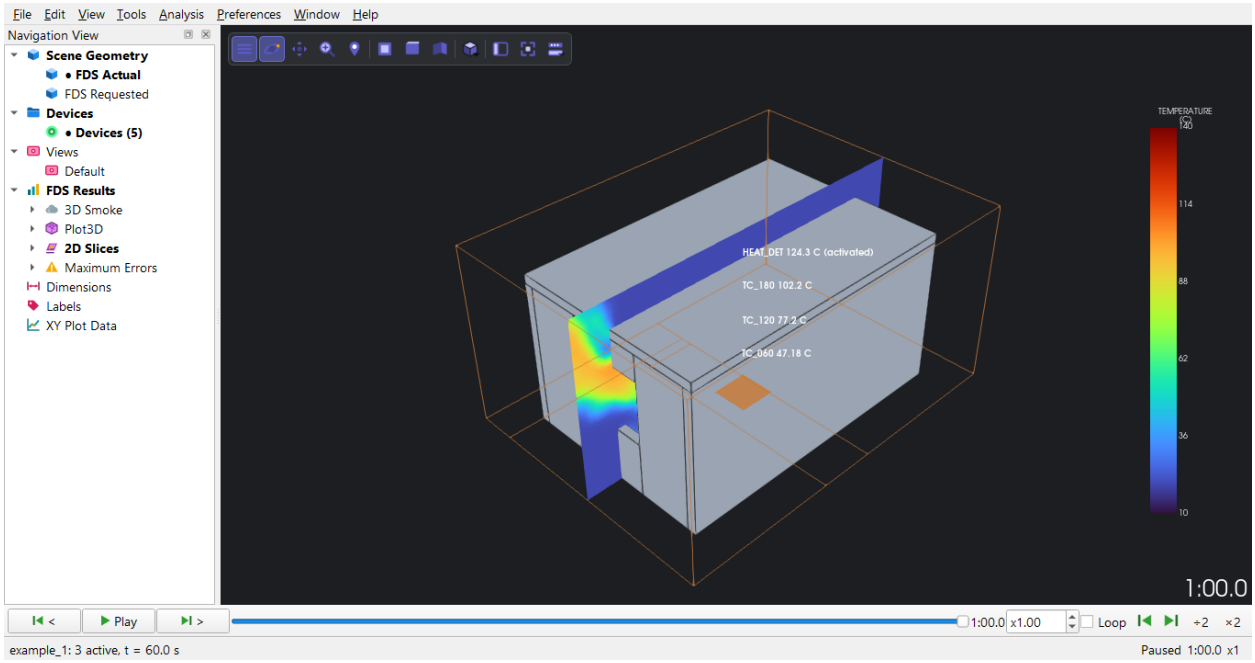


Figure 46. Devices at 60 s: the heat detector and the thermocouple tree above the burner, coloured by their current value.

Time histories: File > Load CSV with example_1_hrr.csv and example_1_devc.csv, then Analysis > Time History Plot and tick the series.

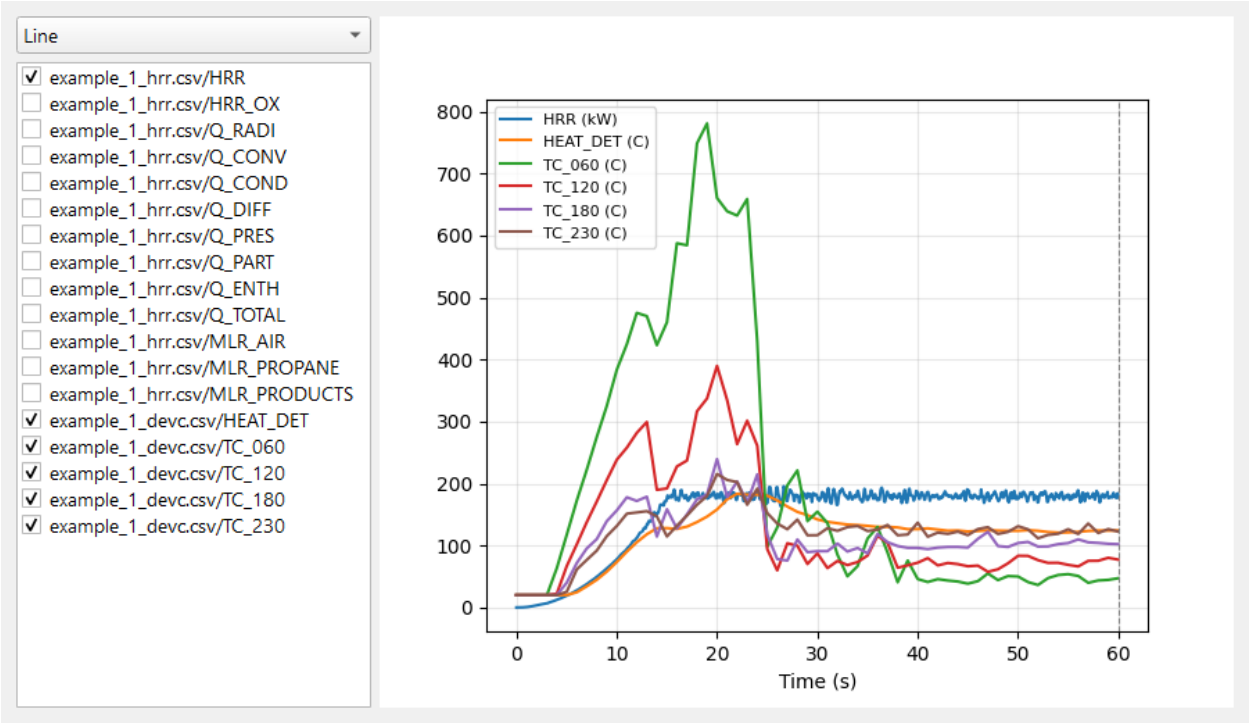


Figure 47. Analysis > Time History Plot after File > Load CSV with the heat release (_hrr.csv) and device (_devc.csv) files: tick the series to plot.

Pictures and movies: Preferences > Colorbar adjusts a colour bar; File > Save Screenshot and File > Save Movie write pictures and MP4 files. File > Save Visualization keeps the whole viewer setup (example_1.pyfdsv) so the same views open next time.

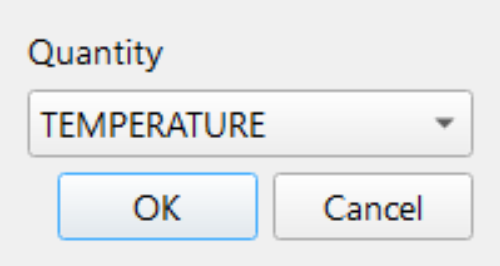


Figure 48. Preferences > Colorbar: range mode, limits, colour map and labels of a quantity's colour bar.

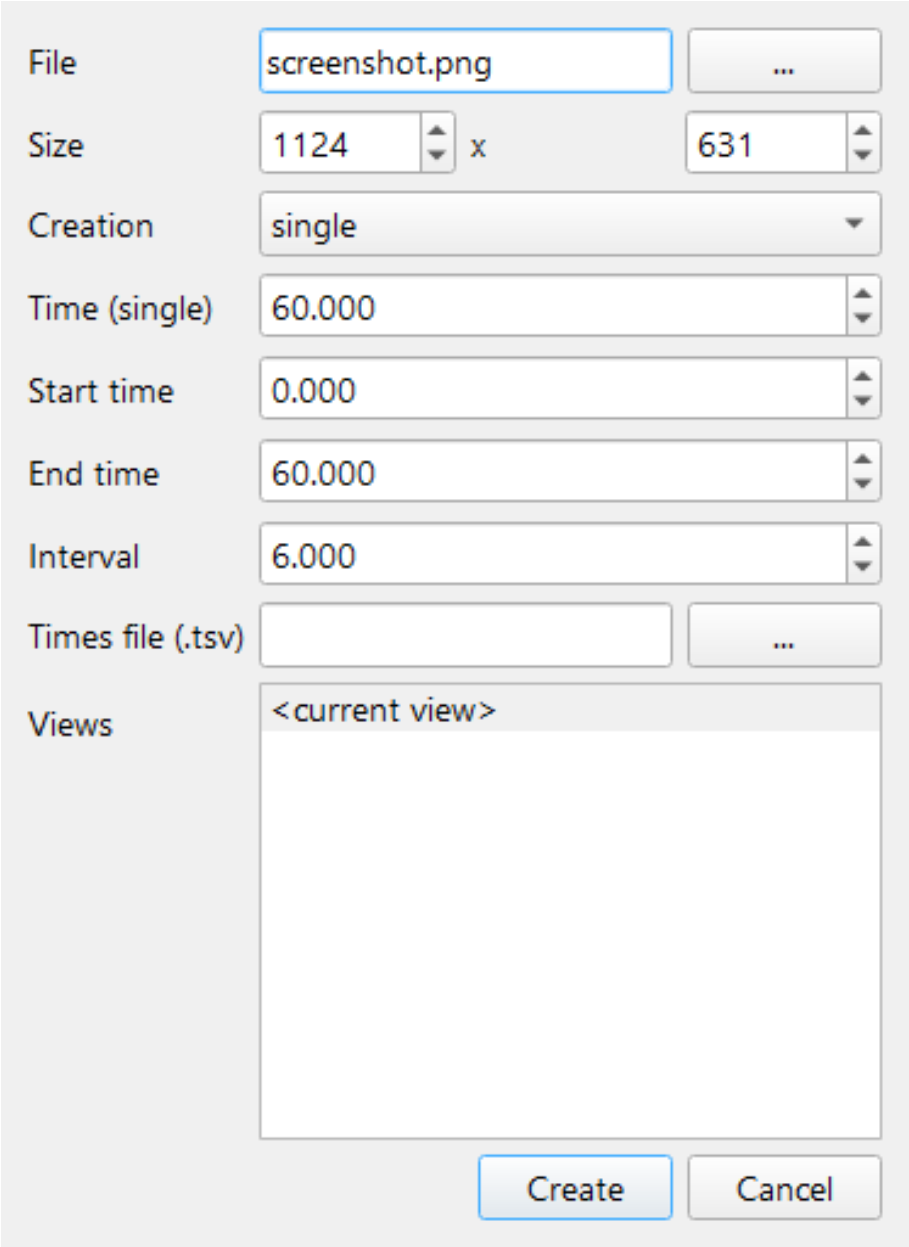


Figure 49. File > Save Screenshot: one picture, a series over time, or one per saved view, at a chosen size.

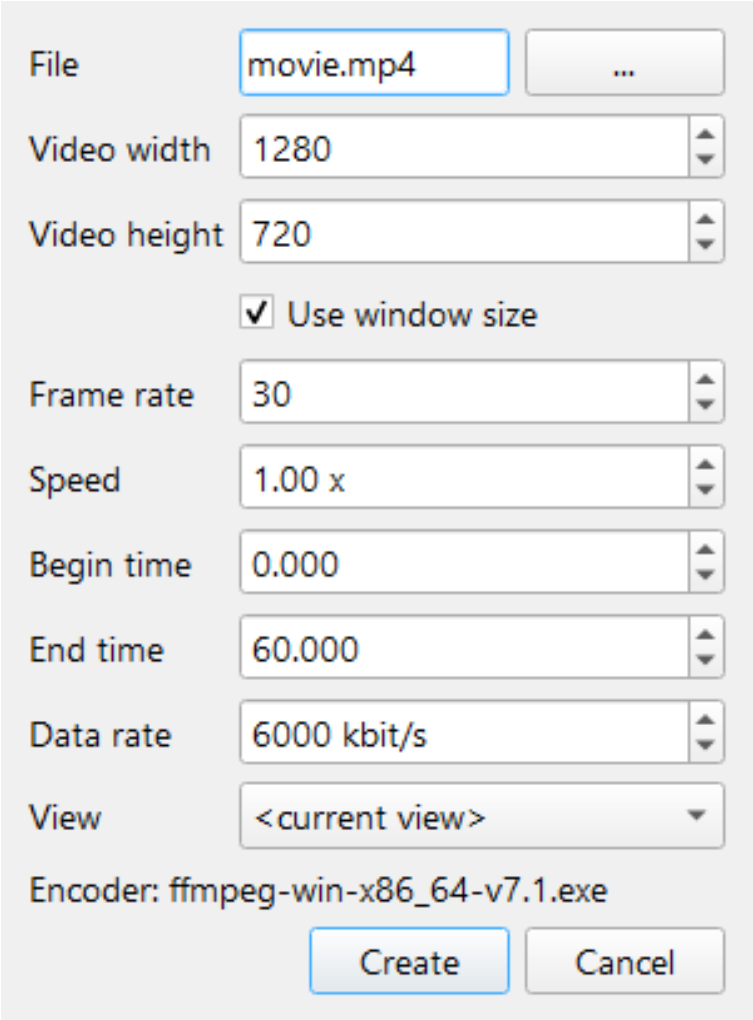


Figure 50. File > Save Movie: an MP4 of the time range at a chosen frame rate and speed.

E.2 Smokeview

The same run opens in Smokeview: double-click example_1.smv, or start fds_smv\SMV6\smokeview.exe with the file. Right-click in the Smokeview window for its menus: Load/Unload > Slice file, 3D smoke; the time bar runs the animation; File/Bookmark/Render saves pictures. The pictures below were made by a Smokeview script (client_demos\results_walkthrough.py writes it), the same commands a user can type in a .ssf file.

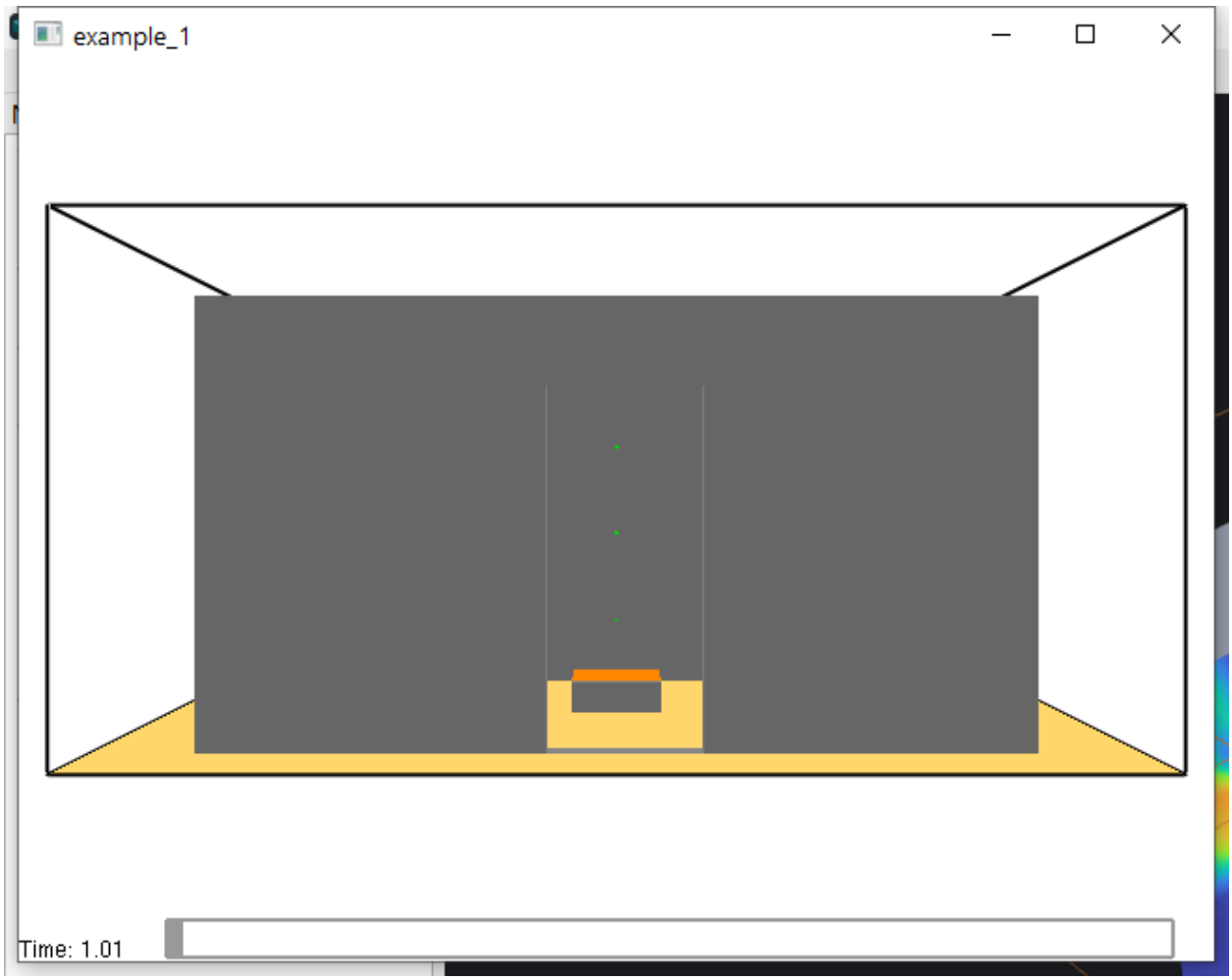


Figure 51. Smokeview 6.10.1 (Run > Open Smokeview, or double-click the .smv file) while the script runs: the room seen from the front at t = 1 s, before smoke has formed; the pictures the script rendered follow. Right-click in the window for the menus.

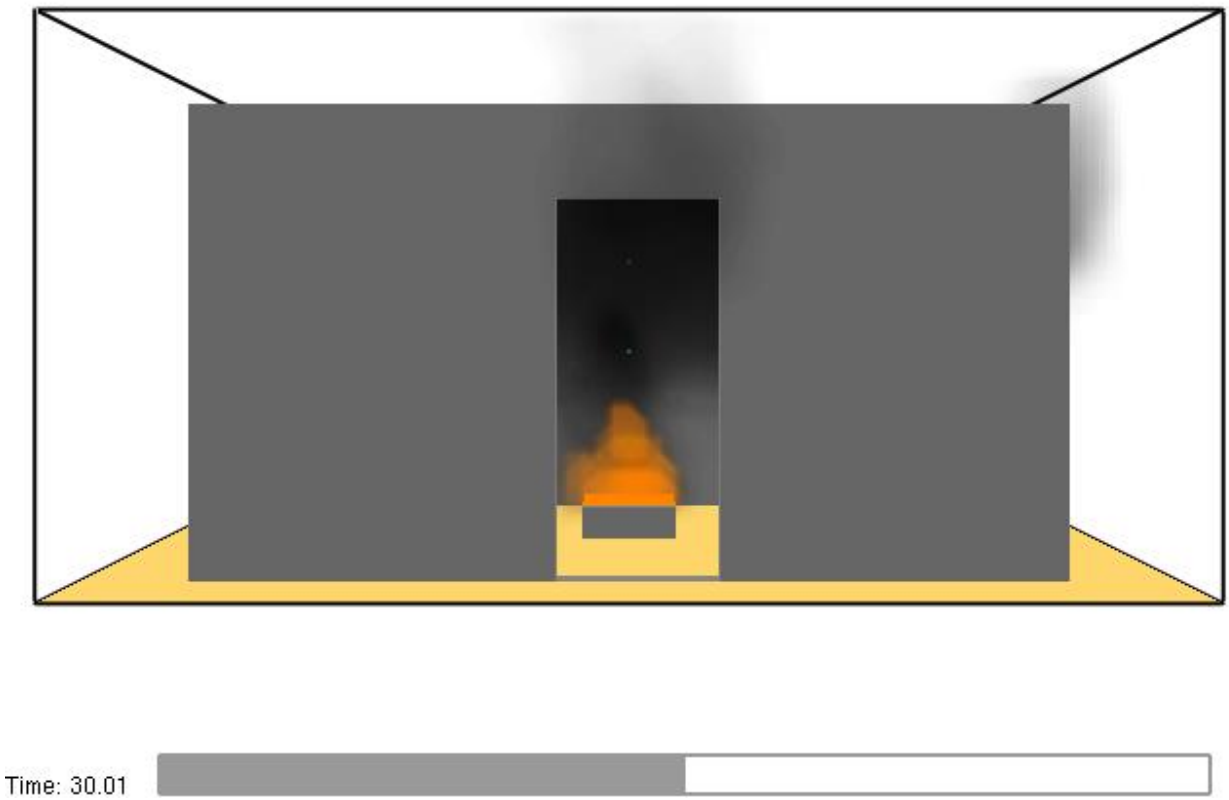


Figure 52. Rendered by Smokeview from the script: smv_smoke_t30.png

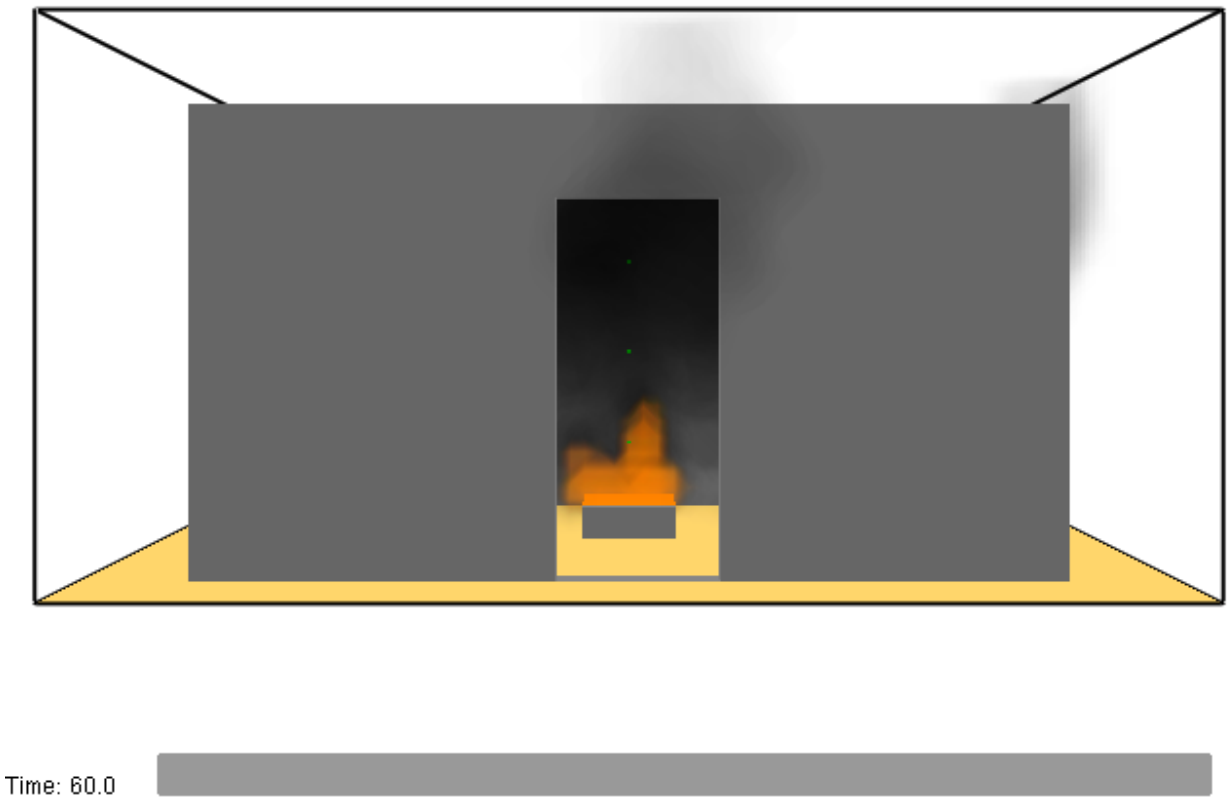


Figure 53. Rendered by Smokeview from the script: smv_smoke_t60.png

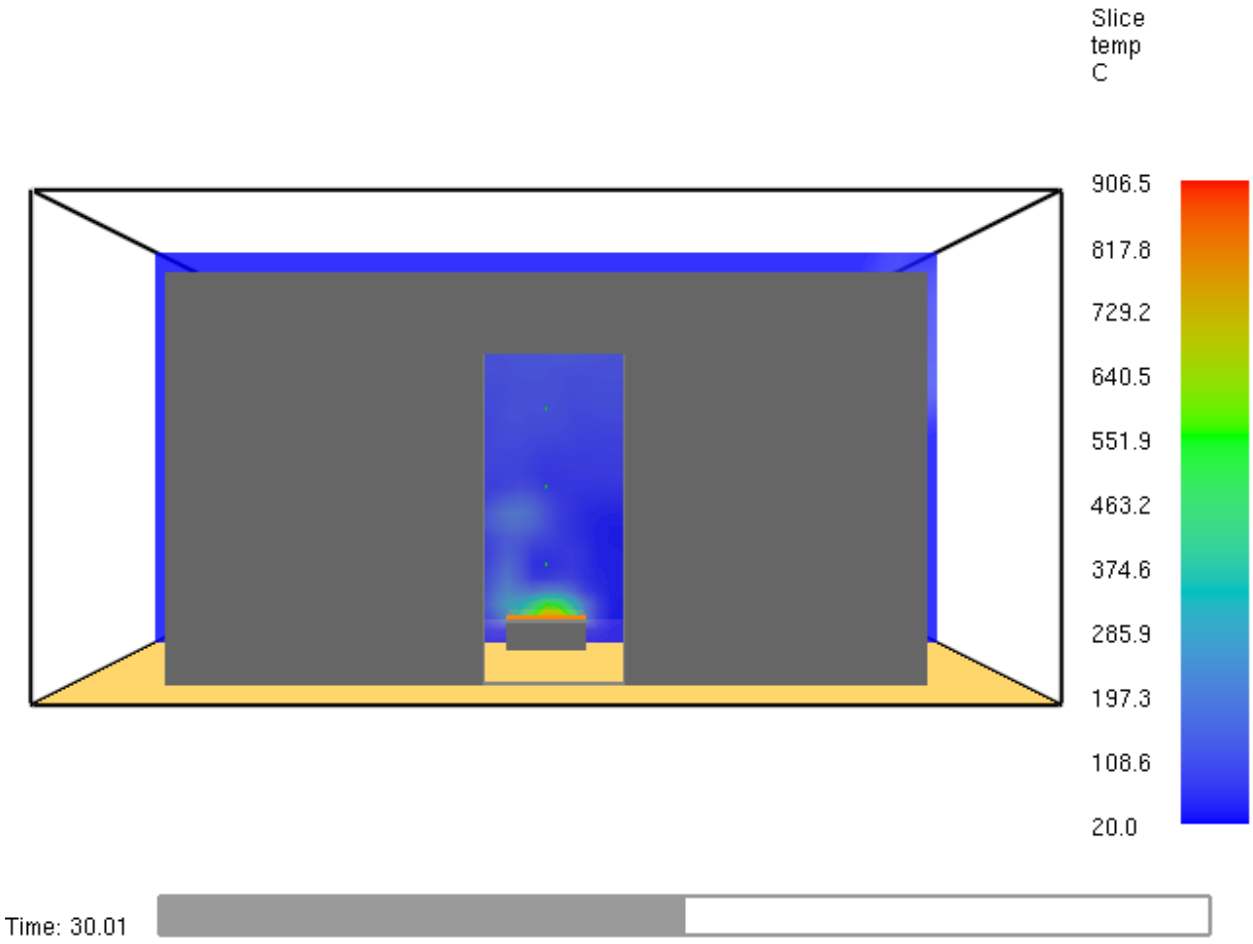


Figure 54. Rendered by Smokeview from the script: smv_temperature_slice_t30.png

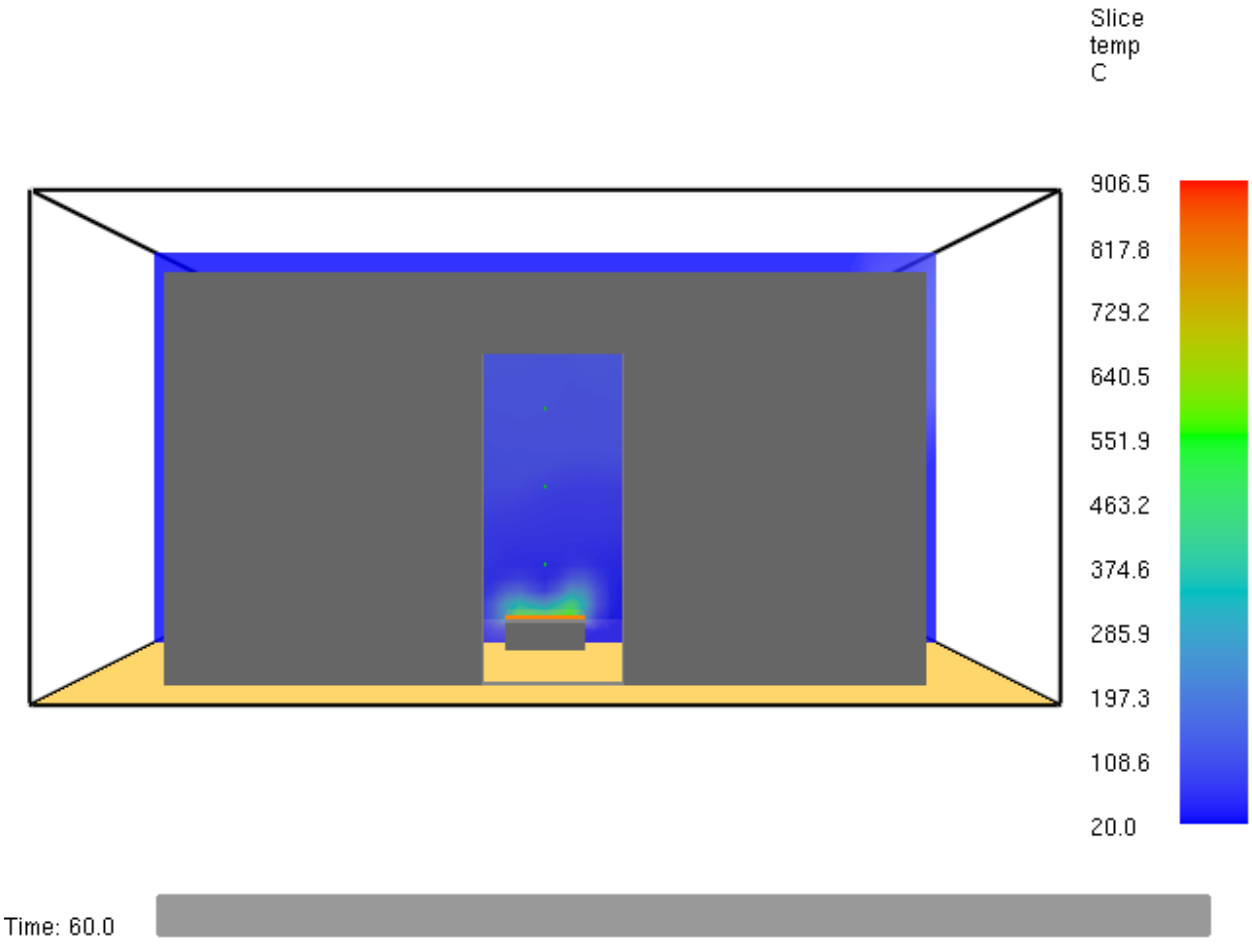


Figure 55. Rendered by Smokeview from the script: smv_temperature_slice_t60.png

F Results and interpretation

Quantity (from the run's CSV files)	Value
Heat release rate at 15 s (end of the t-squared growth)	177 kW
Mean heat release rate 20-60 s (design 180 kW)	180 kW
Heat detector (74 C, RTI 5) activation	11 s
Gas temperature above the burner at 60 s: 0.6 / 1.2 / 1.8 / 2.3 m	47 / 77 / 102 / 122 C

- The temperature slices show the plume rising from the burner and a hot layer forming under the ceiling, spilling out through the top of the door and the window - the openings are the vents of the room.
- The thermocouples sit in the plume centreline, so they read plume temperatures, not the room's average layer temperature.
- The 3D smoke shows where soot goes; its density follows from the assumed soot yield (0.024 kg/kg).
- Visibility at 1.5 m stays near its maximum where smoke has not reached head height; where it drops, smoke has come down to that level.

Limitations. This is a demonstration case. The mesh resolves the fire only coarsely (D*/dx about 4.8; 10 or more is recommended for quantitative work), the run covers 60 s, the burner is a prescribed heat release (no fire growth from materials), and the soot and CO yields are handbook values for well-ventilated propane burning. The results show what the model computes; they are not a validated fire-engineering assessment of a real room, which would need a mesh-sensitivity study and case-specific inputs.

G Troubleshooting

Symptom	Cause and remedy
Run button disabled; red text 'choose a	Set 'Input and results folder' in the Run dialog.

folder'	
'FDS was not found on this computer'	Use Locate FDS... and pick fds.exe (for example fds_smv\FDS6\bin\fds.exe).
'the model has errors; see Model Checks'	Model > Validate, fix each error (double-click it to select the object), or untick 'Check the model first'.
Run Monitor stays at 'Waiting for FDS to start'	FDS has not written the .out file yet; if it never does, check the folder is writable and see Output > Log.
'FDS stopped with an error'	Read the last lines in the Run Monitor (the ERROR line names the namelist); the full log is <CHID>.out in the run folder.
Heat release about twice the expected value	The burner surface was put on the whole obstruction: use an INERT block with a burner vent on its top face.
Validation error 'D*/dx ... Minimum acceptable ratio is 4.0'	The cells are too large for the fire: refine the mesh (0.1 m here) or check the heat release rate is the one intended.
PyFDS Results shows an empty view	Tick an item in the Navigation View (geometry, a slice ...).
Smokeview shows no data	Load it from the right-click menu (Load/Unload); check the .smv and the data files are in the same folder.

H Quick reference

1. Start PyFDS; File > Project Settings: CHID example_1
2. Model > Simulation Parameters: T_END 60 s
3. Model > Add Mesh: 60,45,30 cells over 0,6,0,4.5,0,3 (0.1 m)
4. Model > Library: gypsum_wall_25mm; Draw > Draw Room: door S 0.9 x 2.1, window E 1.2 x 1.05 sill 0.9
5. Model > Add Typed Surface: burner 500 kW/m2, TAU_Q -15; Add Obstruction (INERT) and Add Vent (BURNER_SURF) on its top; Add Reaction: propane, then Edit Reactions: soot 0.024, CO 0.005
6. Model > Open Mesh Boundaries, then delete the floor vent
7. Library: heat_detector_74C; Add Device (LINK TEMPERATURE); 4 thermocouples; 3 slices
8. Model > Validate (0 errors); File > Save As
9. Run > Run Simulation: folder, Run; watch the Run Monitor until 'Finished'
10. PyFDS Results: File > Load Results (example_1.smv), tick slices / 3D smoke / devices, use the time bar
11. Smokeview: open example_1.smv, right-click > Load/Unload

3 Example 2 - Burning couch (FDS example case)

A Introduction

Purpose. Shows that a case written for FDS by hand opens in PyFDS, can be inspected and run in parallel, and post-processed - the route a consultant's existing FDS models take.

Scenario. The case is 'Examples/Fires/couch.fds' from the FDS 6.10.1 distribution (a teaching case; its material properties are marked 'completely fabricated'). A couch of fabric over foam on a gypsum base stands in a 5 x 5 x 2.4 m room with one open vent; a hot (1000 C) static particle on the seat ignites it. Burning materials make the fire grow on its own - there is no prescribed heat release rate.

Engineering objectives. Watch the fire spread over the couch surfaces, the flame and smoke, and the heat release rate it produces.

- Opening an existing FDS input file
- Multiplied meshes (MULT) and a 4-process MPI run
- Layered, burning-away surfaces and pyrolysis materials
- Boundary-file results on the couch surfaces

Prerequisites. The original runs for 600 s. Here T_END is set to 25 s in the GUI so the demo runs in about 20 minutes on 4 processes (4 free CPU cores); nothing else in the physics is changed.

Screen recordings. documentation\videos\example_2_open_and_run.mp4 (sections B to D, the run as a time-lapse) and example_2_post_processing.mp4 (section E).

B Opening the example

Start PyFDS, then File > Open and choose client_demos\example_2\couch_original.fds (a copy of the untouched FDS example). PyFDS reads the FDS namelists into objects; anything it has no editor for is kept and written back unchanged.

What was read: 1 &MESH line with a multiplier, giving 4 meshes; 5 obstructions; surfaces UPHOLSTERY, WALL, ignitor; materials FABRIC, FOAM, GYPSUM PLASTER; T_END 600.0 s.

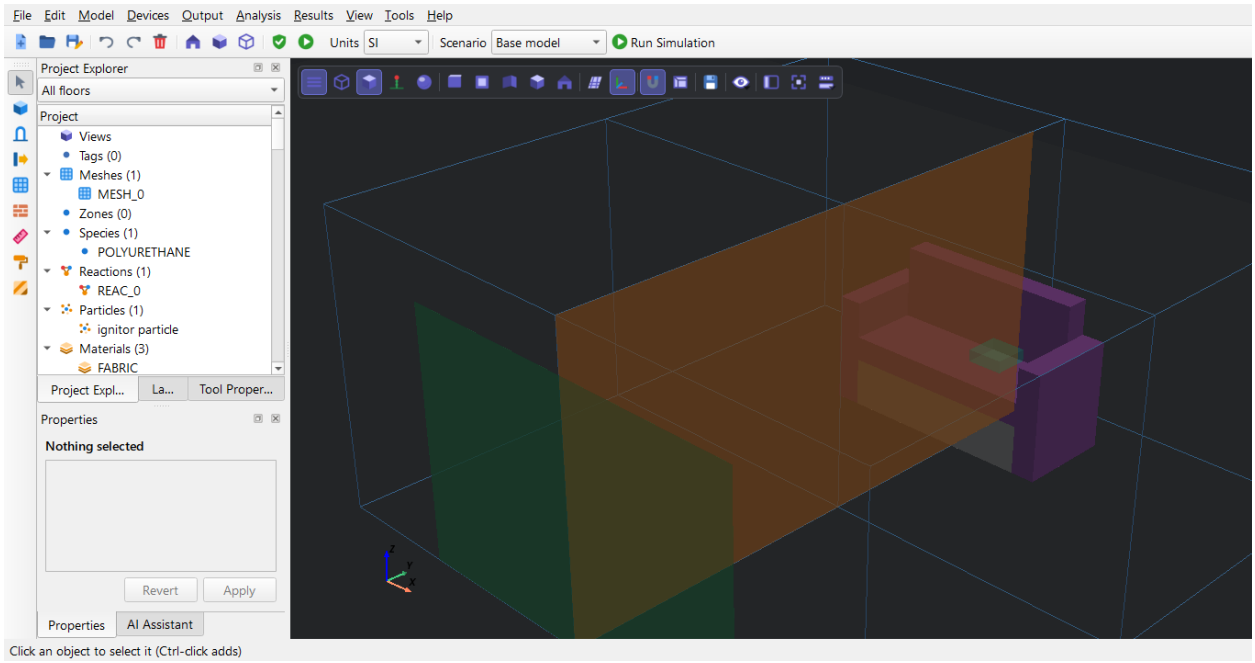


Figure 56. File > Open with couch_original.fds: the couch (purple upholstery on a gypsum base) and the four meshes made by the MULT line are listed in the Project Explorer and drawn in the 3D view.

C Inspecting and configuring the case

The editors under Model show what the file defines. Nothing is changed in them for this demonstration.

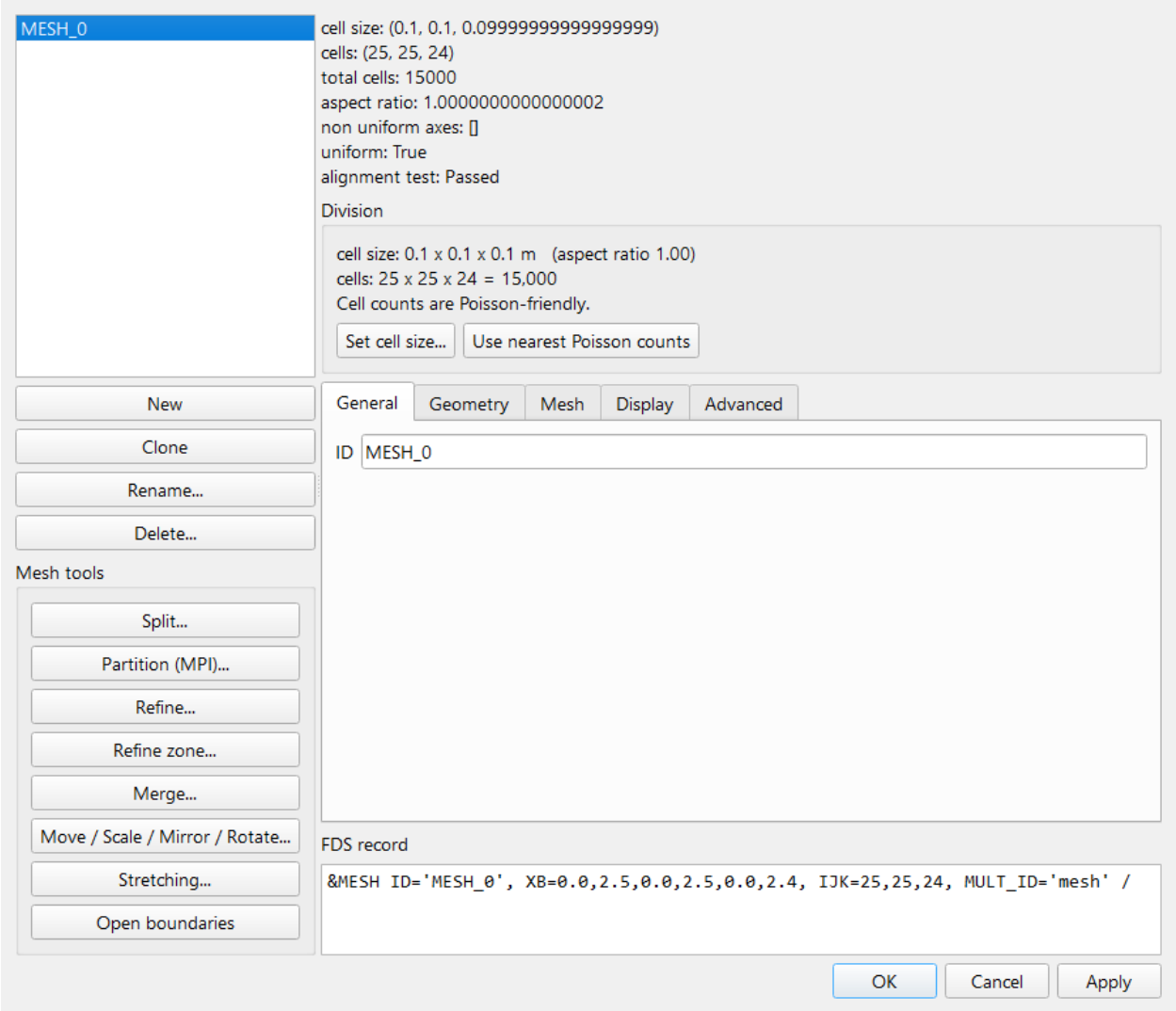


Figure 57. Model > Edit Meshes: one 2.5 x 2.5 x 2.4 m mesh of 25 x 25 x 24 cells (0.1 m); its multiplier (MULT_ID) repeats it 2 x 2 to cover 5 x 5 m.

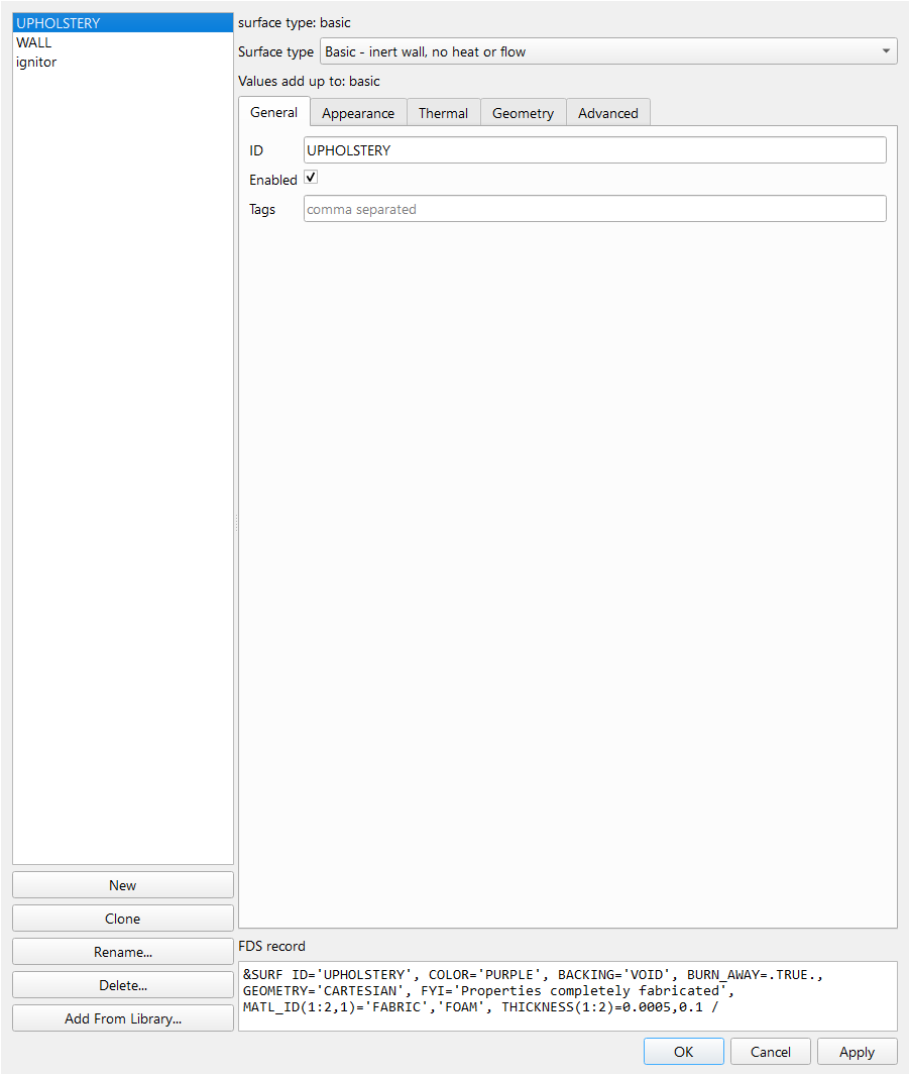


Figure 58. Model > Edit Surfaces, UPHOLSTERY: fabric (0.5 mm) over foam (100 mm), burns away; the properties are the FDS example's (marked 'fabricated').

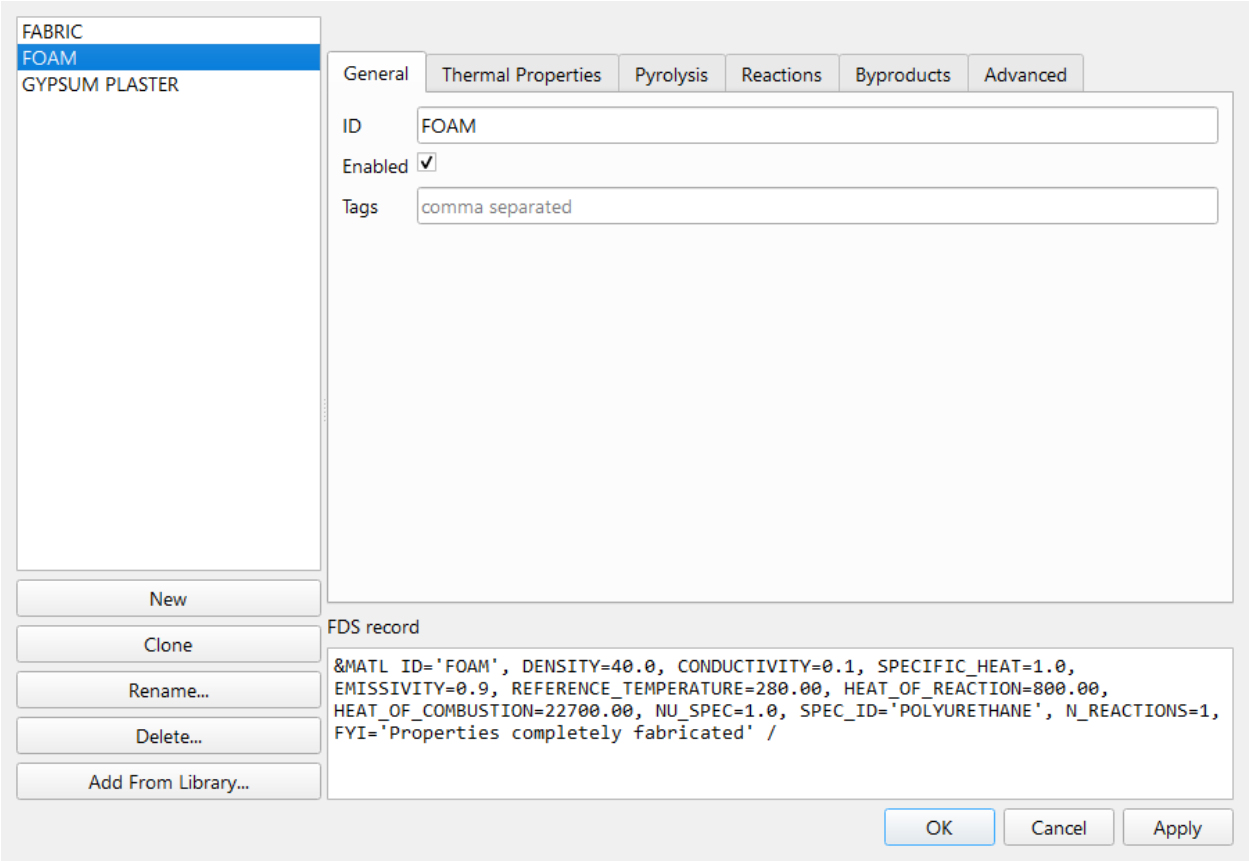


Figure 59. Model > Edit Materials, FOAM: density, conductivity, specific heat and the pyrolysis reaction that produces POLYURETHANE fuel gas.

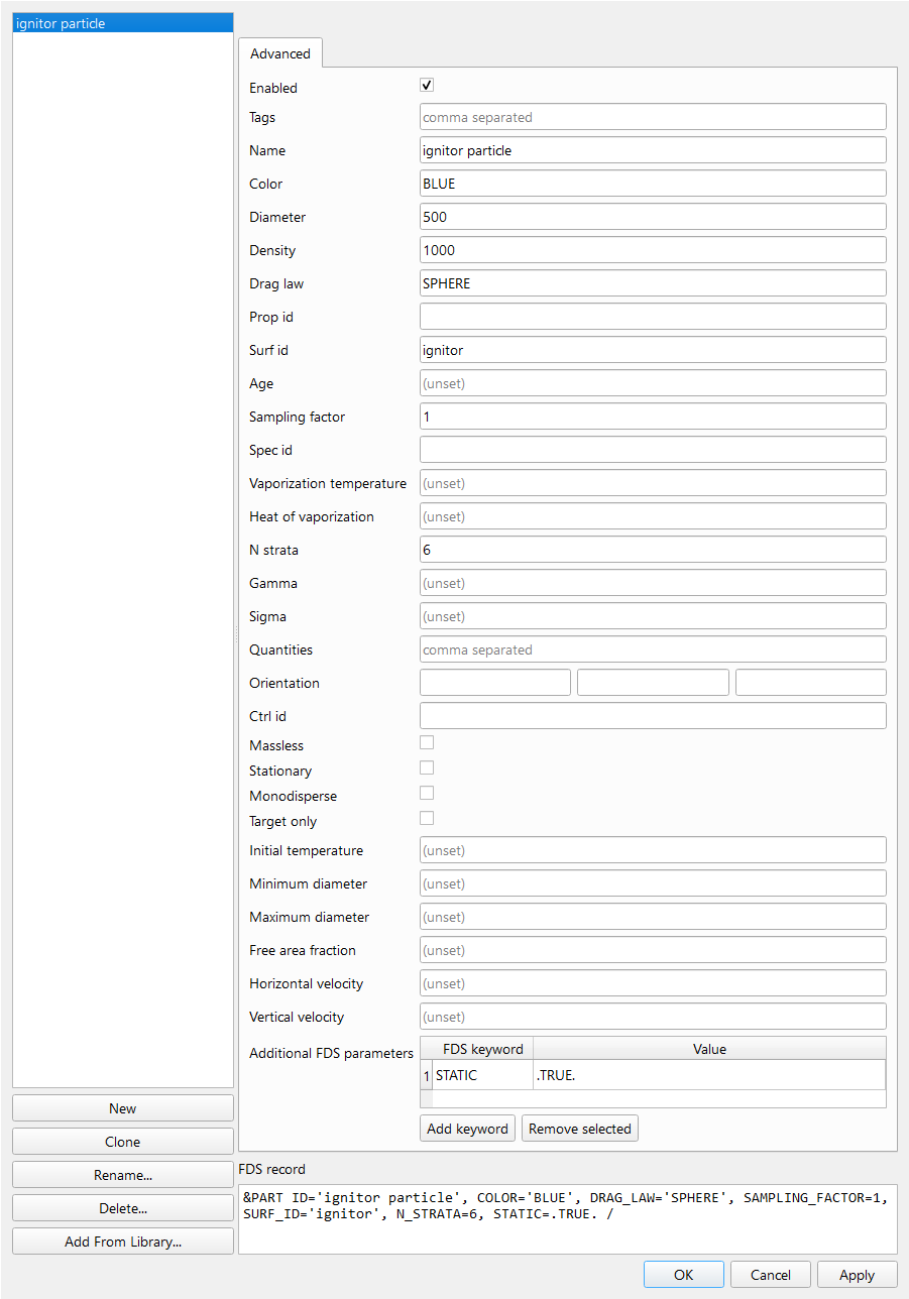


Figure 60. Model > Edit Particles: the static 'ignitor particle', a 1000 C cylinder placed on the seat cushion that starts the fire.

The only change: Model > Simulation Parameters, T_END from 600 s to 25 s.

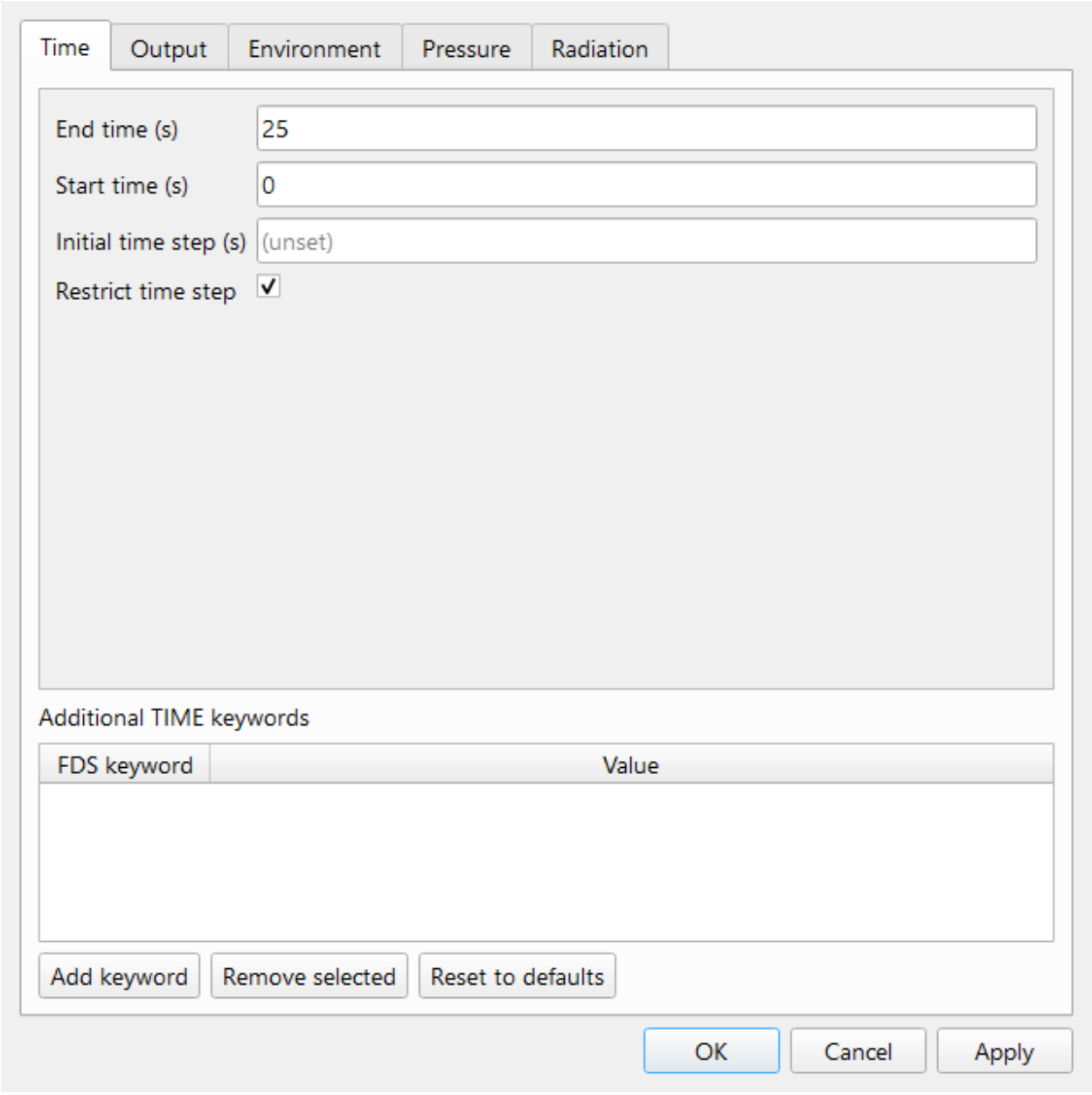


Figure 61. Model > Simulation Parameters: T_END reduced from 600 s to 25 s so the demo runs in about half an hour; nothing else in the case is changed.

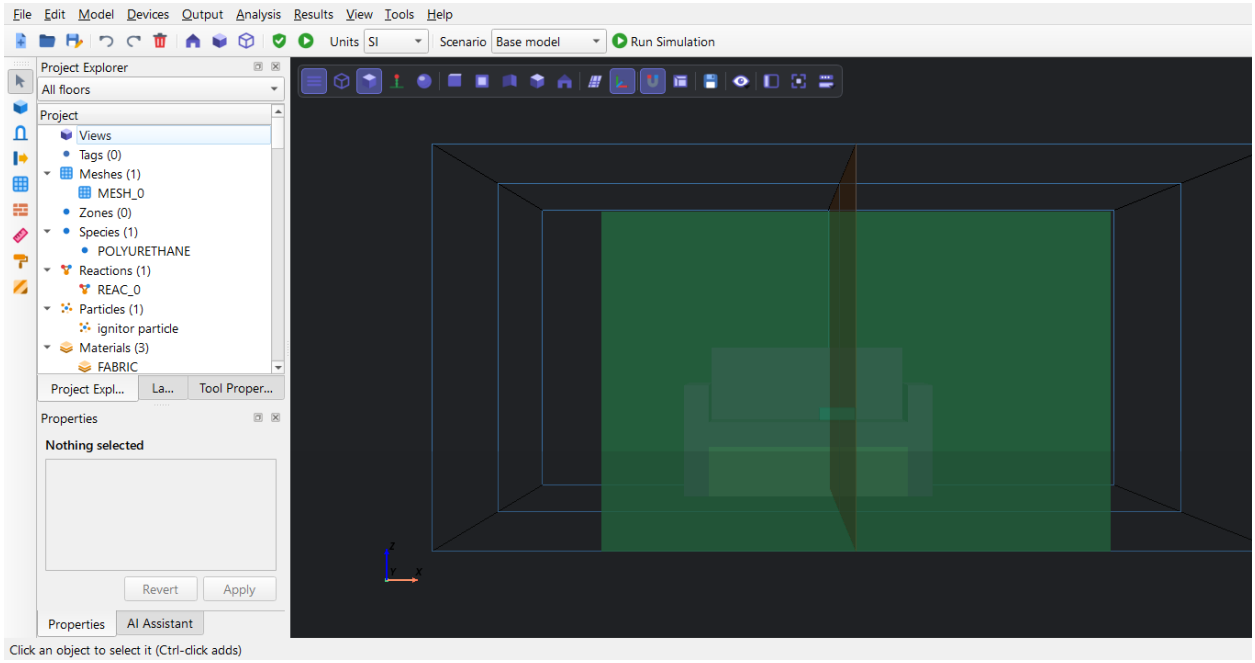


Figure 62. Front view: the open vent in the south wall (green) is the only opening; the other boundaries are the default solid wall surface.

Note. When the project is saved and written back to FDS, PyFDS adds its standard output intervals to &DUMP (slices, boundary files and devices every 1 s, Plot3D every 10 s) in place of the original NFRAMES=3000, and writes the &MISC, &PRES and &RADI defaults explicitly (they equal FDS's defaults). These change how often results are saved, not the physics.

D Validation and running the simulation

Model > Validate: 0 errors, 0 warning(s).

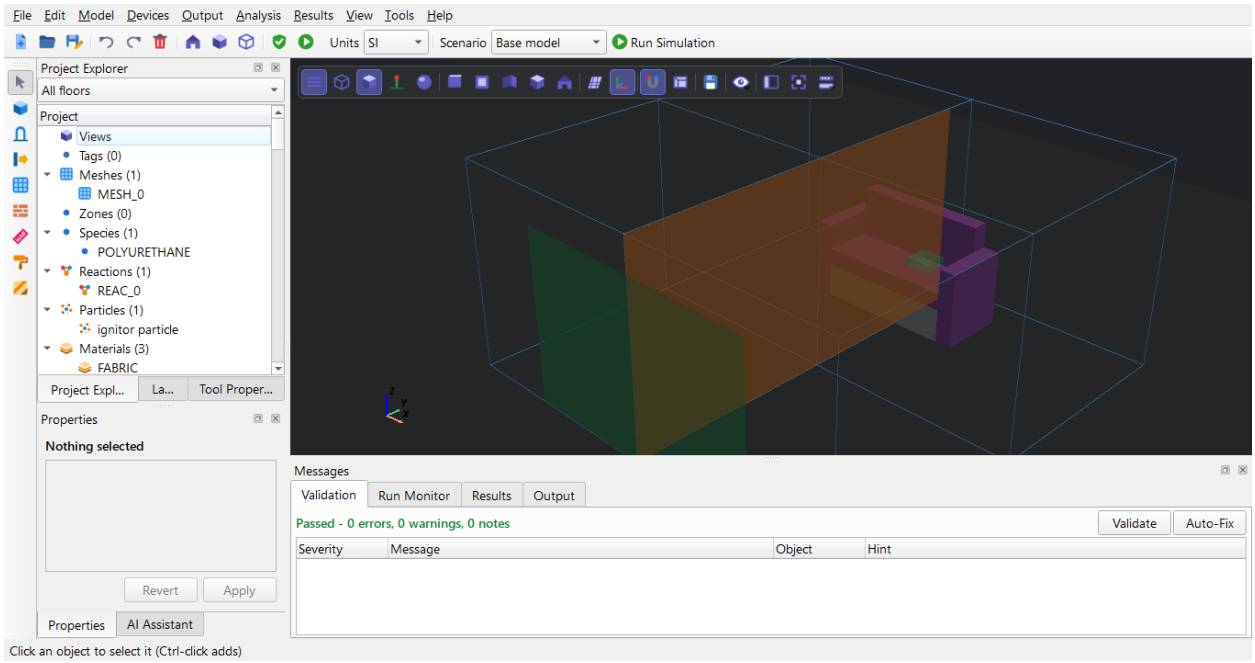


Figure 63. Model > Validate on the imported case.

Run > Run Simulation. Because the model has 4 meshes, the dialog proposes 'This computer, in parallel (MPI)' with 4 processes - FDS needs one process per mesh - and shows which mesh each process computes.

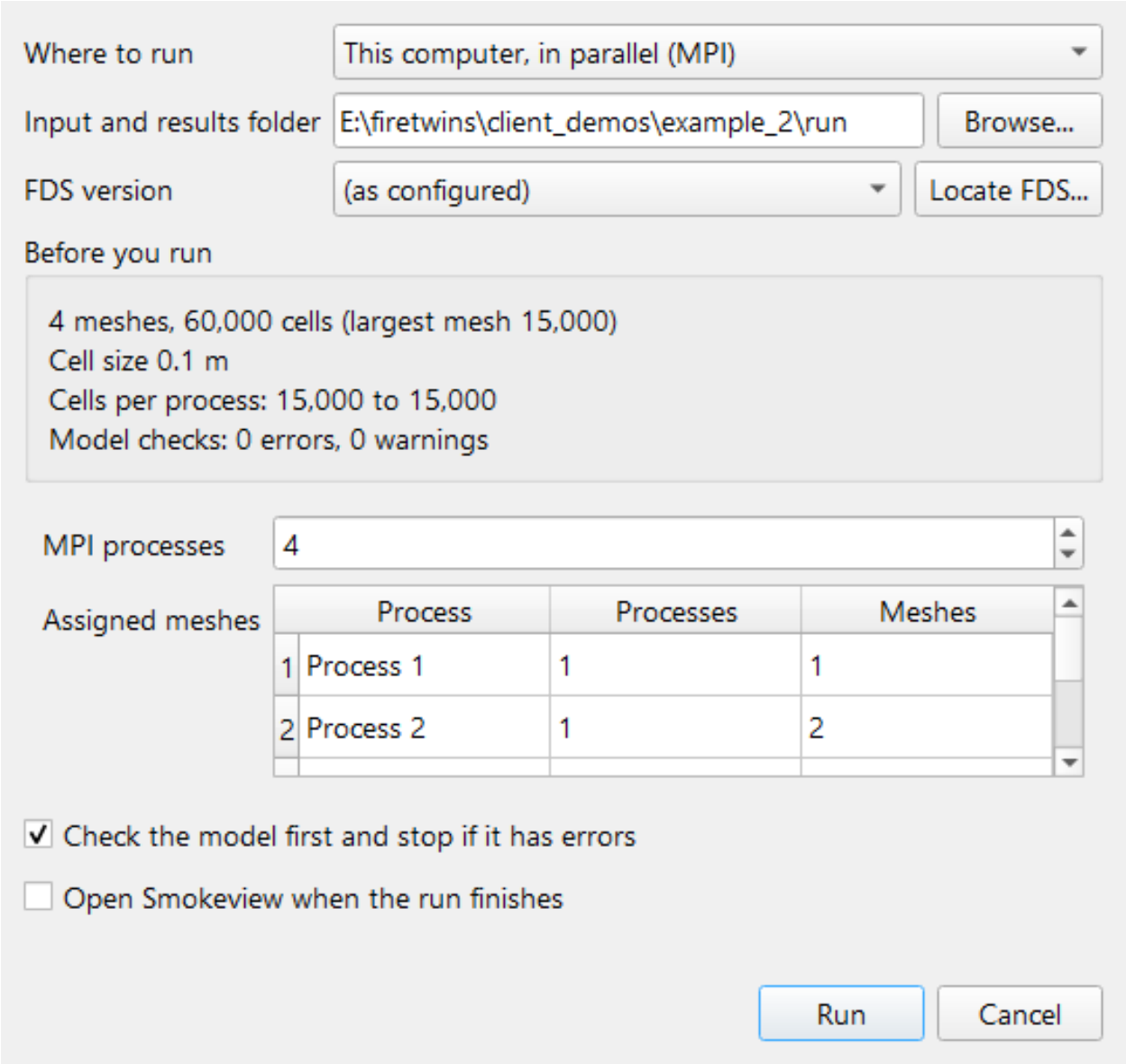


Figure 64. Run > Run Simulation with four meshes: 'This computer, in parallel (MPI)' is preselected with one process per mesh; the table shows which mesh each process computes and the summary shows the cells per process.

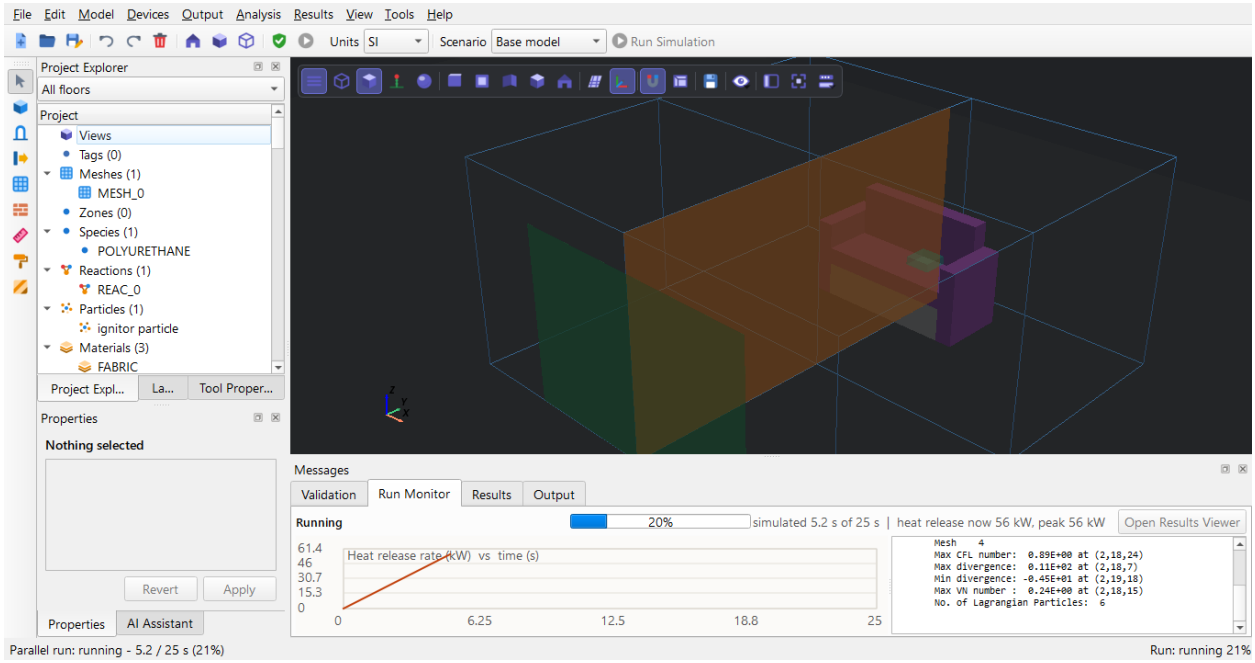


Figure 65. Run Monitor at 20 %.

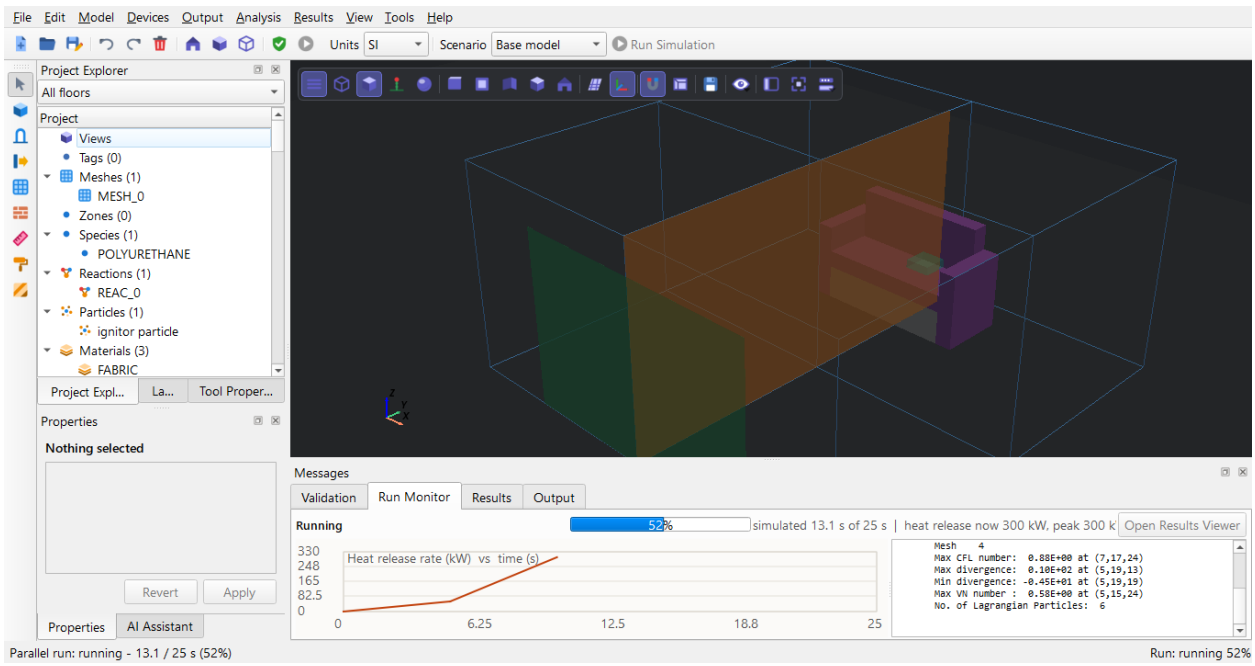


Figure 66. Run Monitor at 50 %.

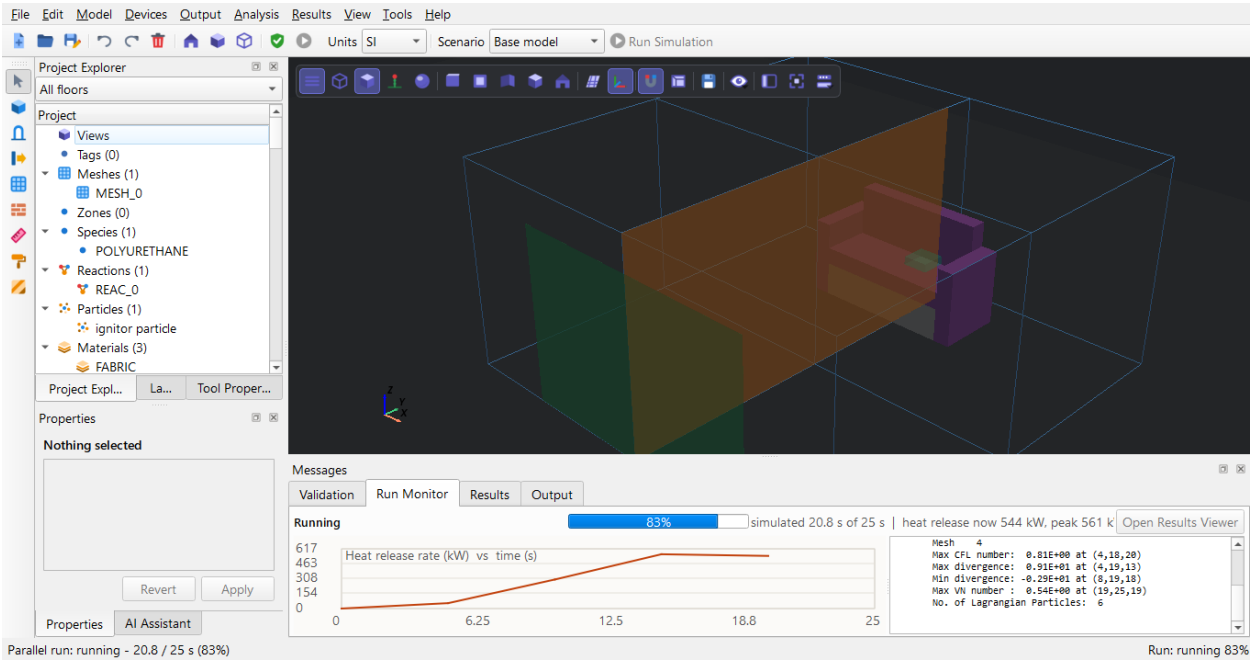


Figure 67. Run Monitor at 80 %.

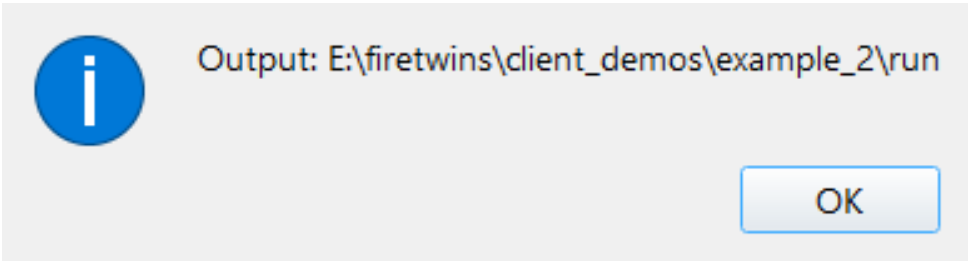


Figure 68. Message: Simulation complete

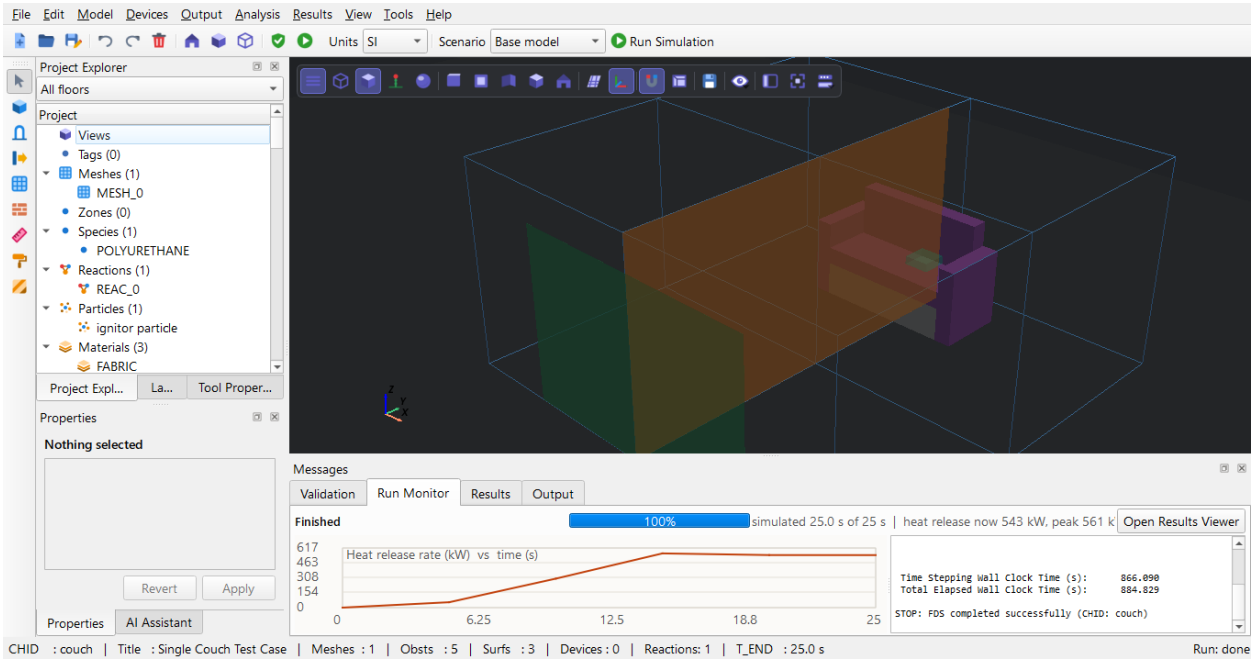


Figure 69. Run finished on 4 MPI processes.

Outcome on the test computer: FDS completed successfully; wall clock time 884.829 s for 25 s of simulated time on 4 MPI processes.

E Post-processing

E.1 PyFDS Results

File > Load Results with client_demos\example_2\run\couch.smv. Results of the four meshes are listed per mesh; tick all four to see the whole plane or surface.

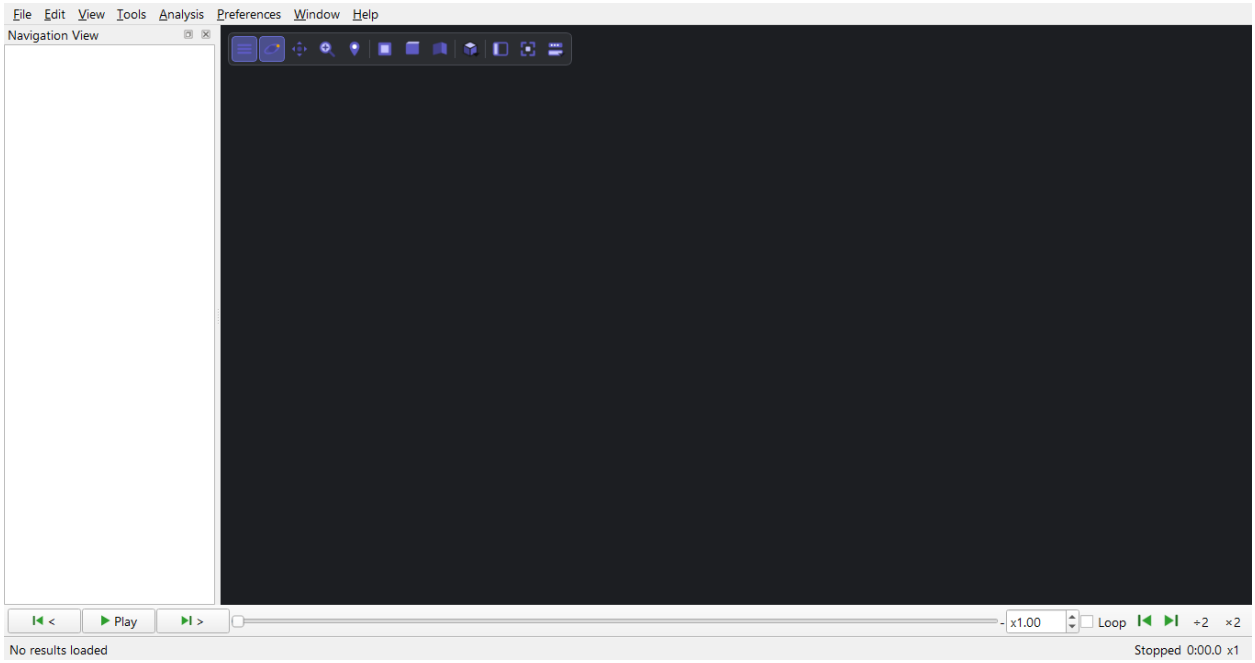


Figure 70. PyFDS Results (Results > Open PyFDS Results in the editor, or the PyFDS Results shortcut) before a run is loaded.

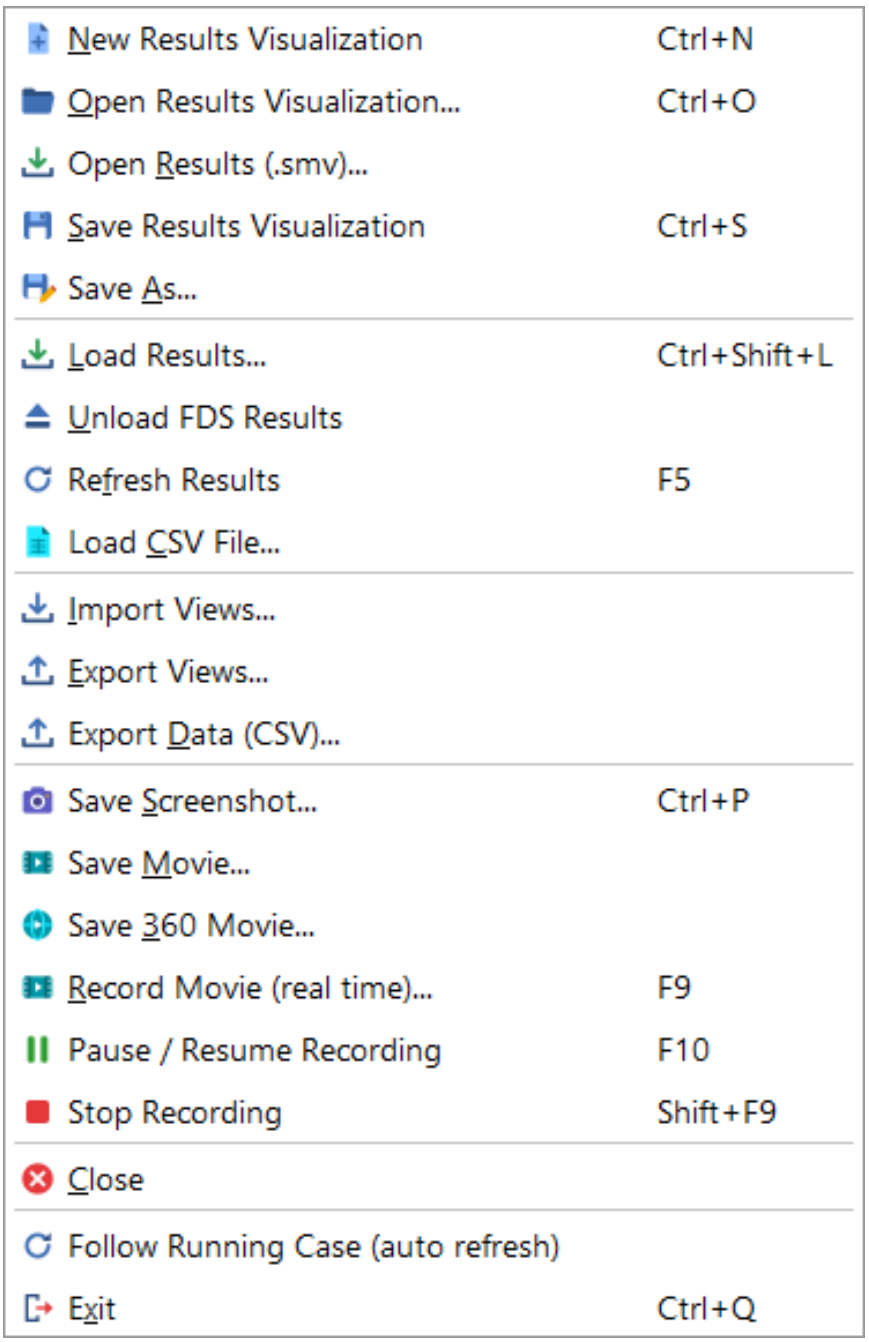


Figure 71. PyFDS Results, File menu: Load Results opens the .smv file of a run.

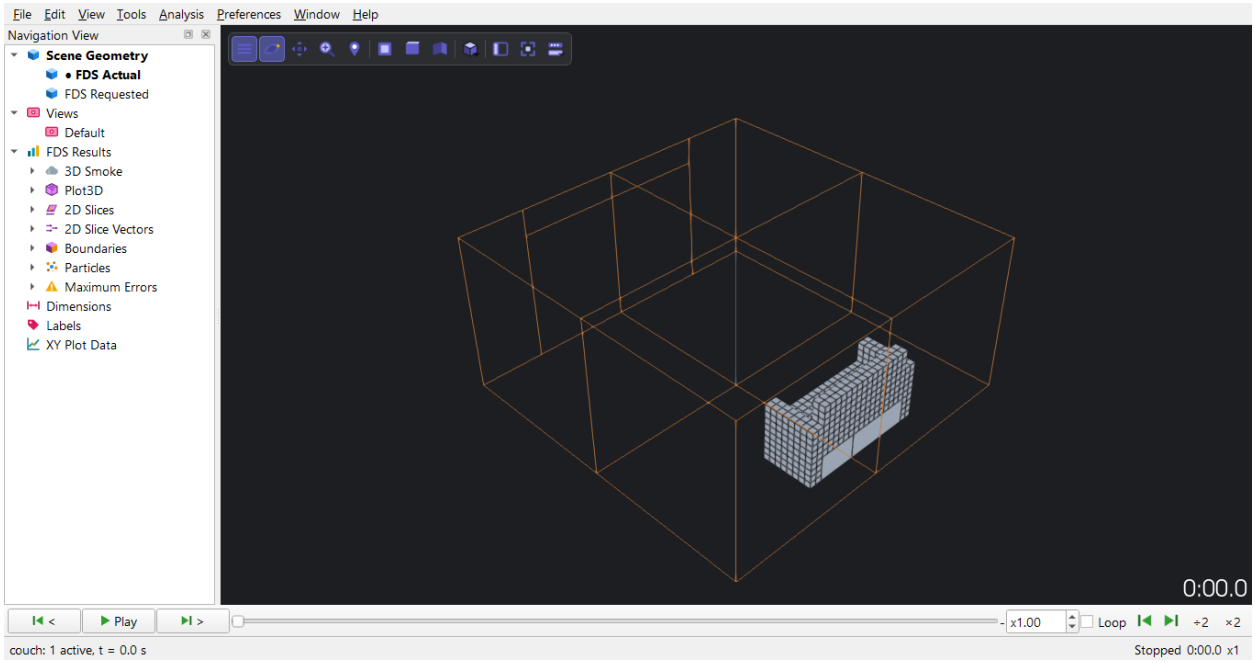


Figure 72. File > Load Results with couch.smv: the geometry is shown and the Navigation View on the left lists every result of the run.

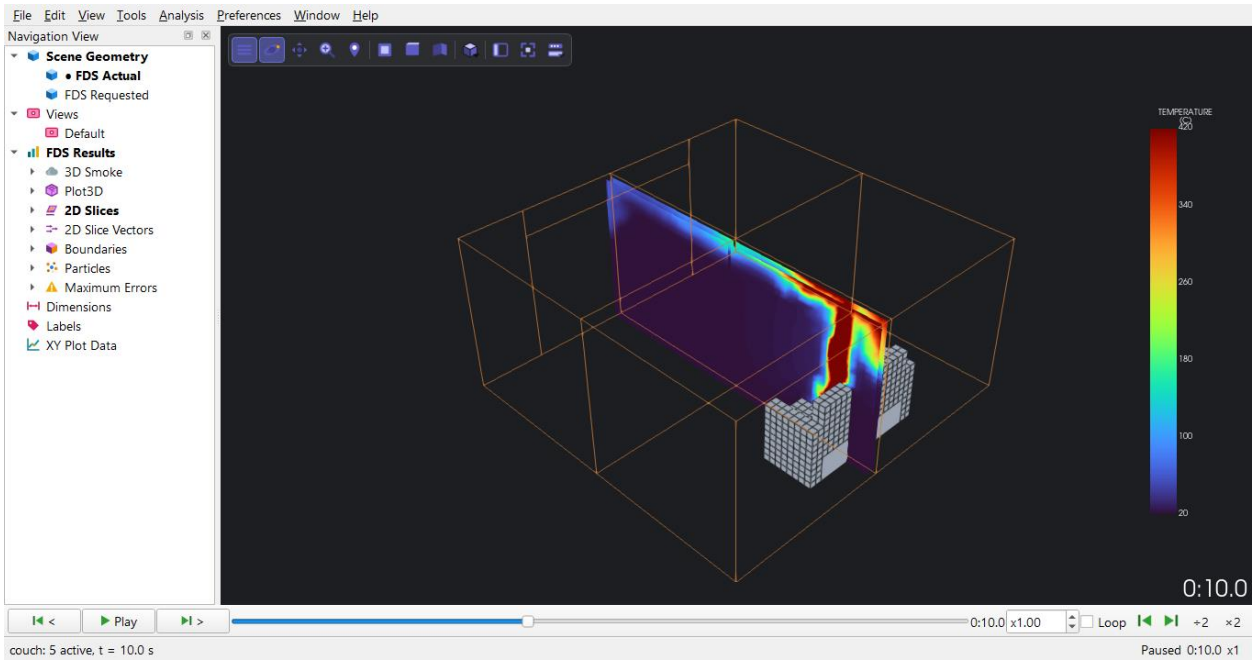


Figure 73. Temperature slice through the couch (X = 2.5 m) at 10 s.

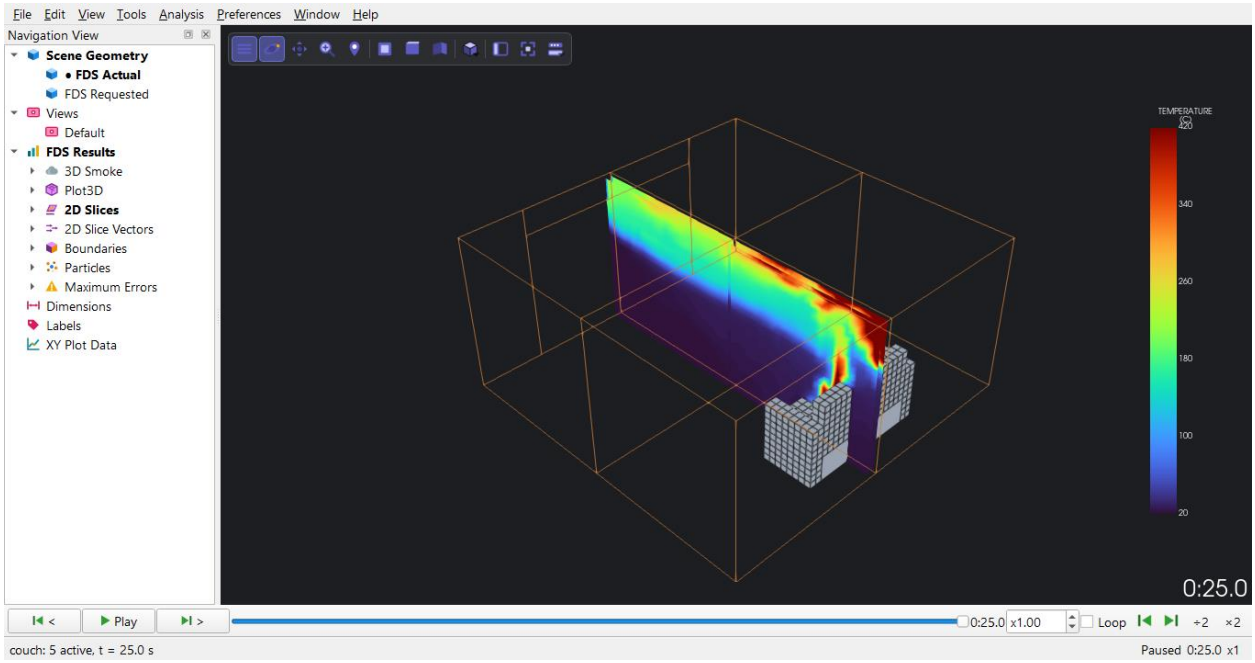


Figure 74. The same slice at the end of the run (25 s): the flame plume over the seat cushion.

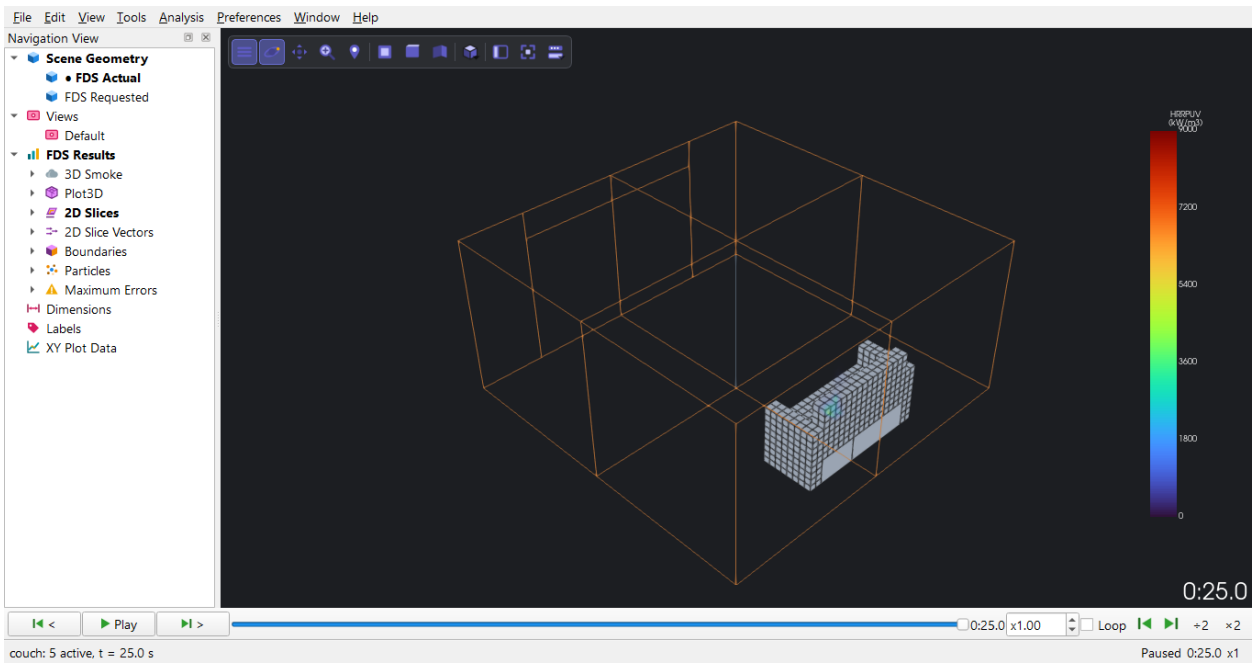


Figure 75. Heat release rate per unit volume (the flame) on the same plane at 25 s.

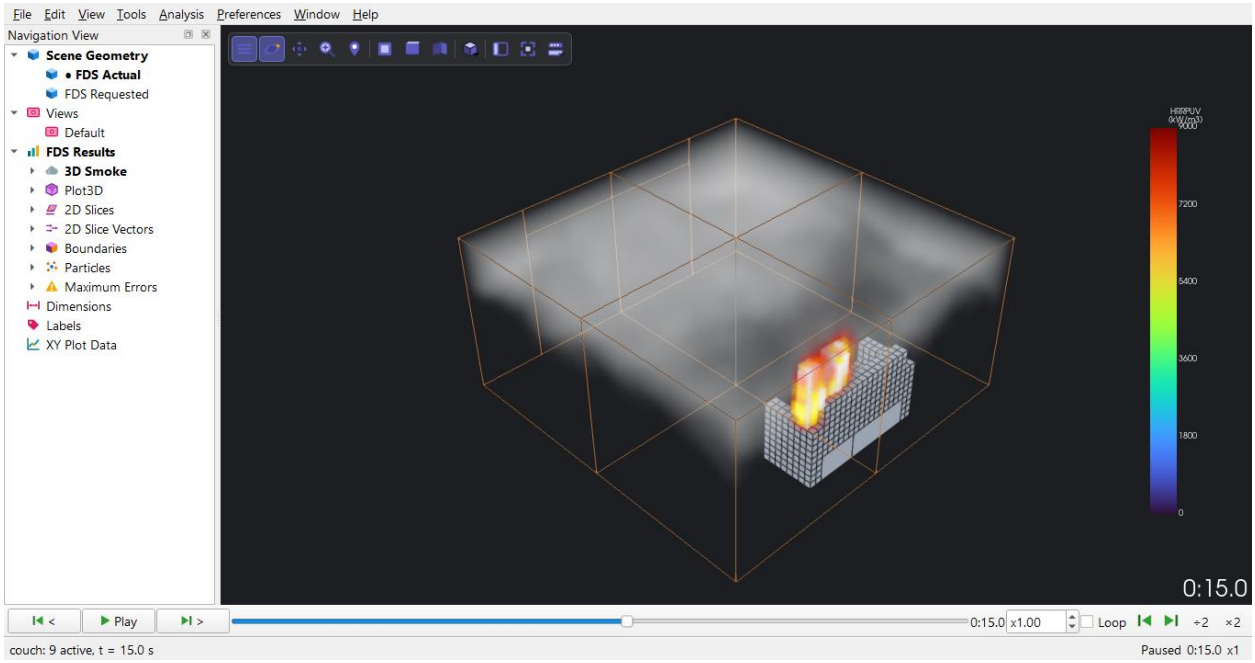


Figure 76. 3D smoke (soot density) and flame at 15 s.

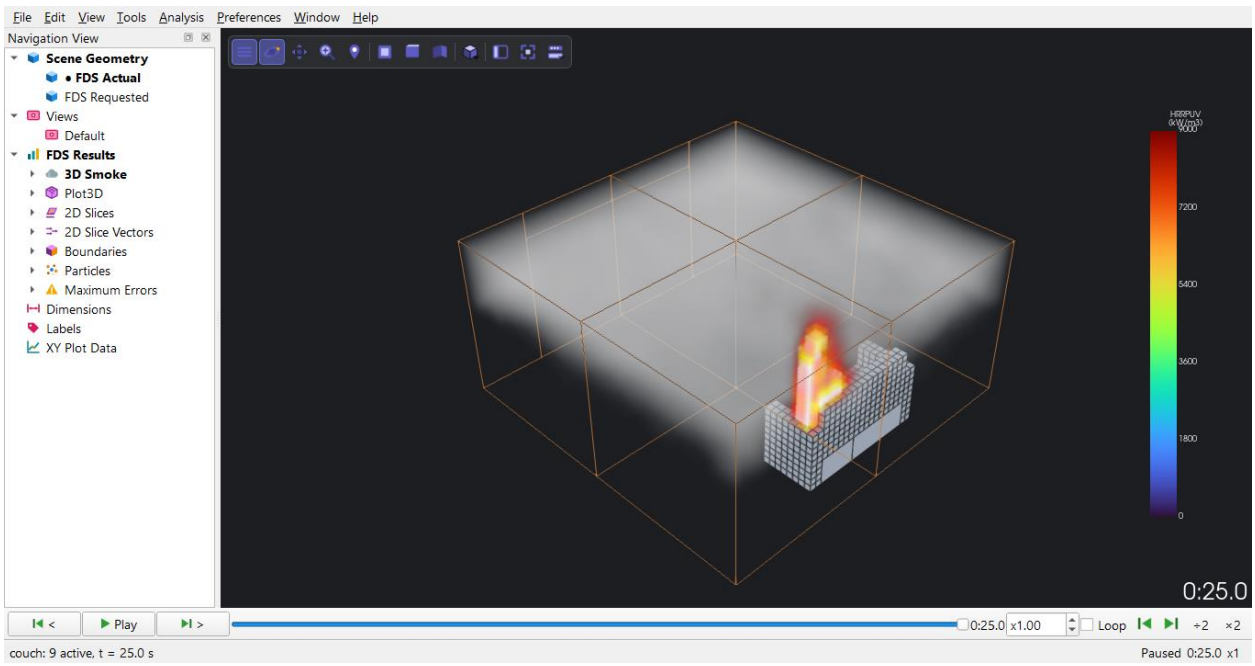


Figure 77. 3D smoke and flame at 25 s.

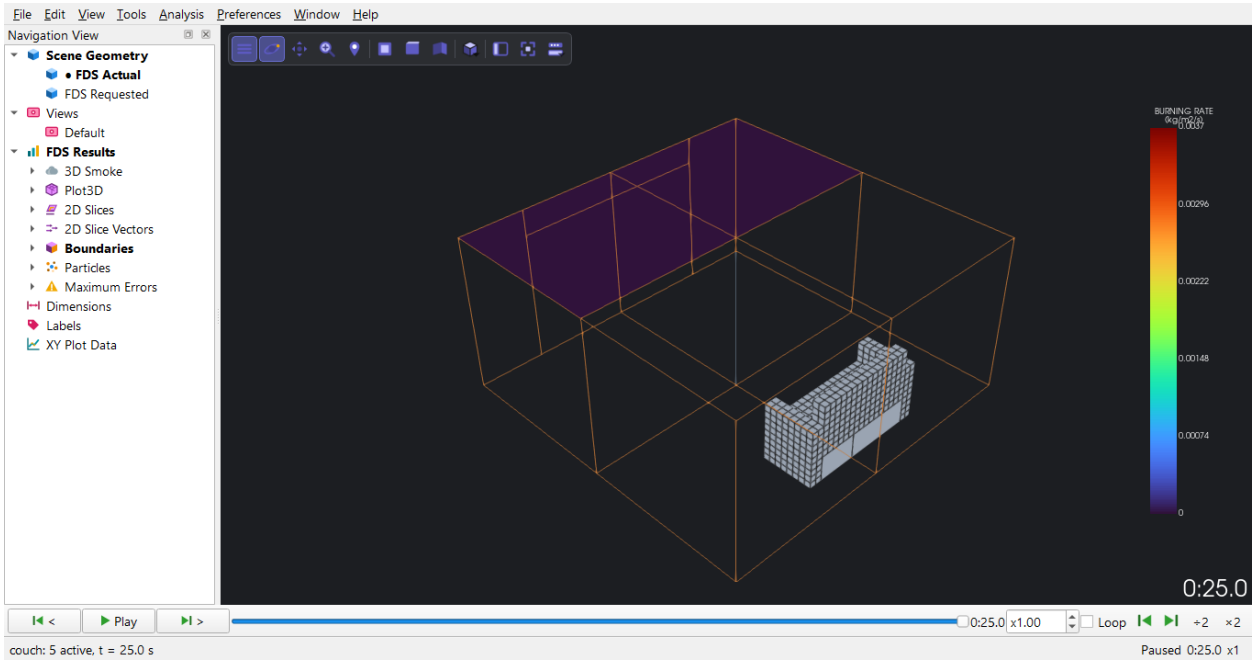


Figure 78. Boundary file, burning rate on the couch surfaces at 25 s (kg/m2/s).

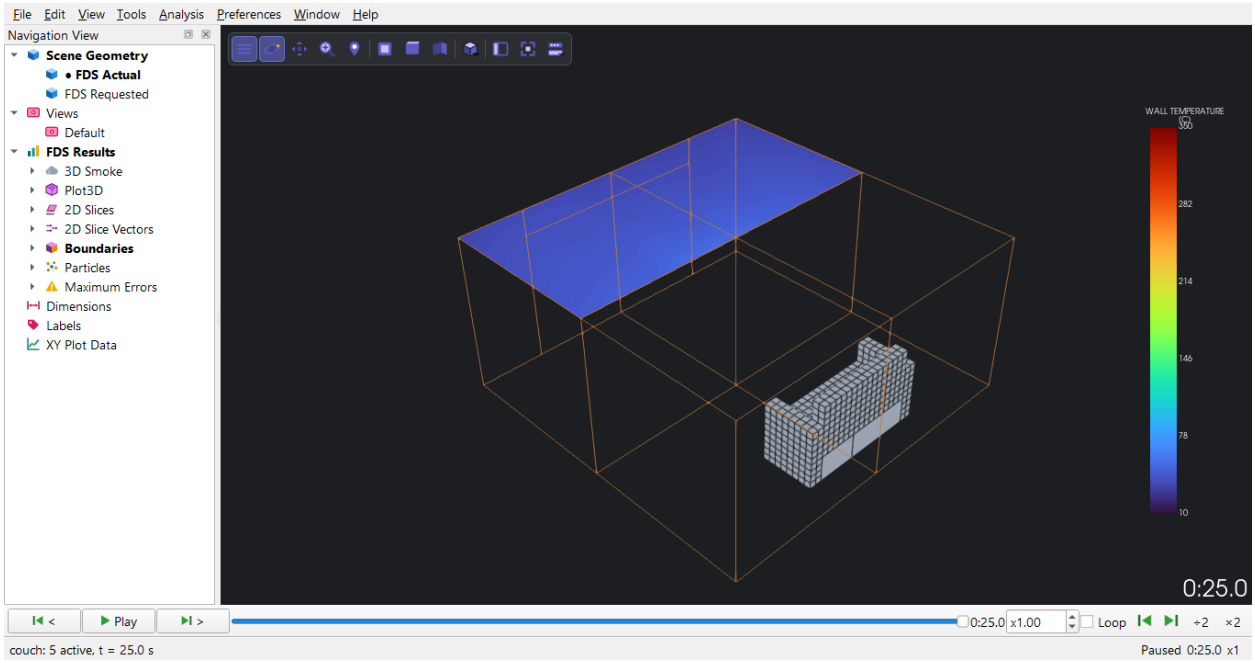


Figure 79. Boundary file, wall temperature at 25 s.

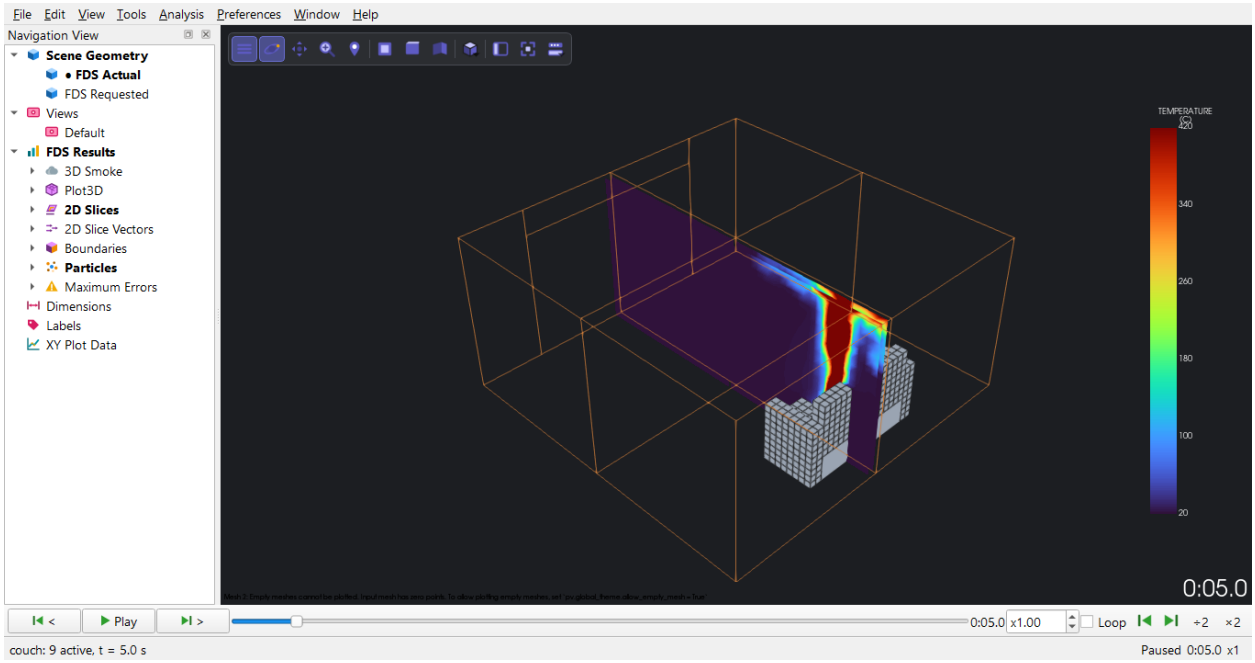


Figure 80. Static ignitor particle and the temperature slice at 5 s.

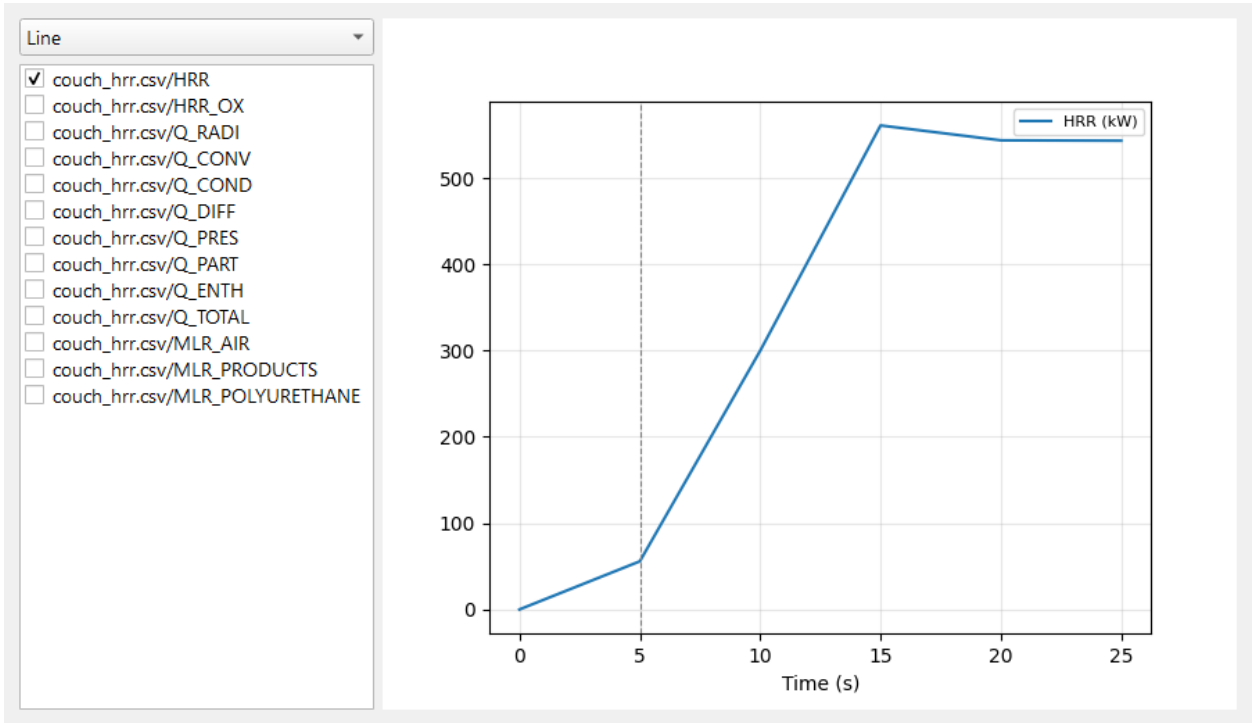


Figure 81. Analysis > Time History Plot after File > Load CSV with the heat release (_hrr.csv) and device (_devc.csv) files: tick the series to plot.

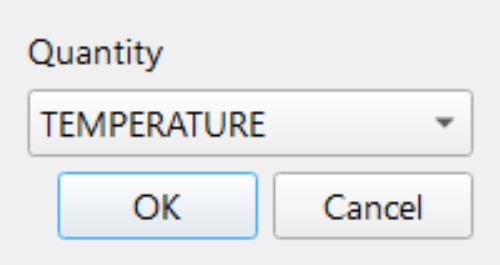


Figure 82. Preferences > Colorbar: range mode, limits, colour map and labels of a quantity's colour bar.

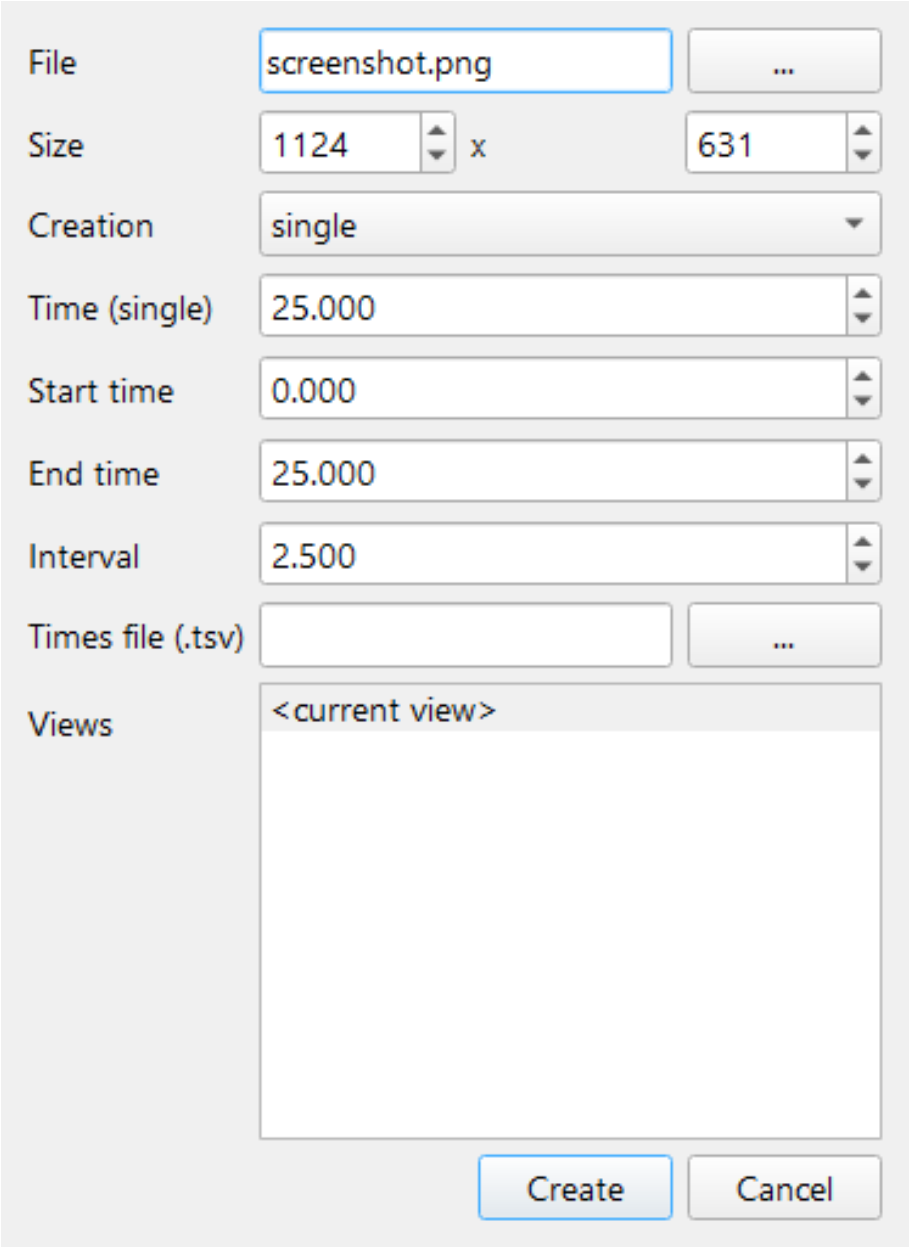


Figure 83. File > Save Screenshot: one picture, a series over time, or one per saved view, at a chosen size.

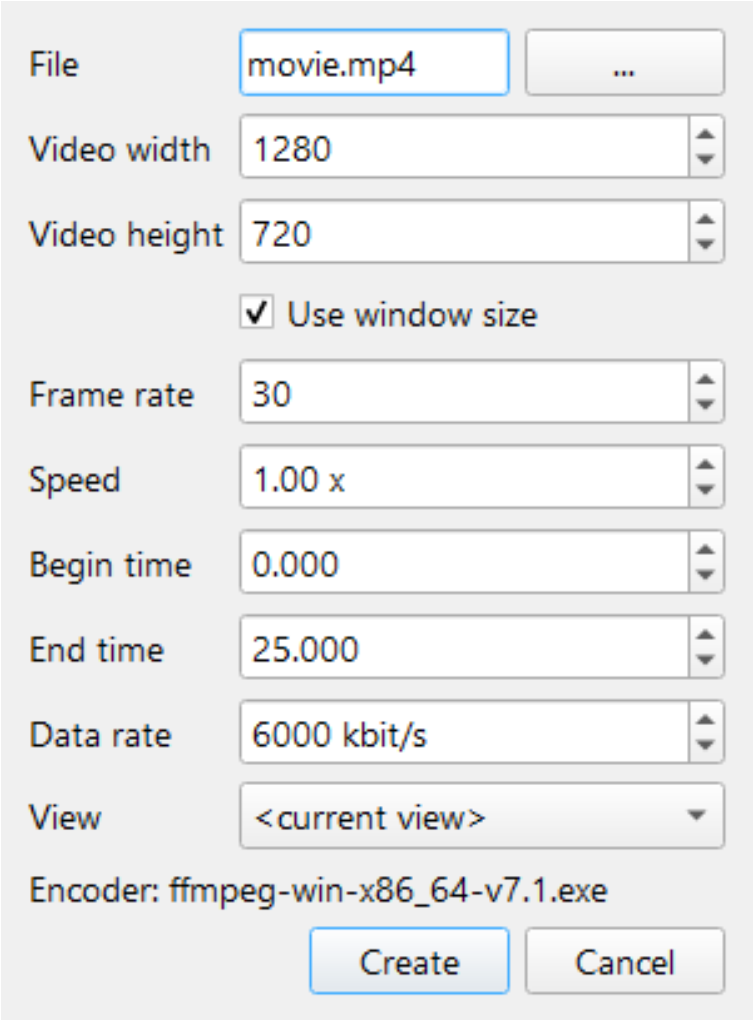


Figure 84. File > Save Movie: an MP4 of the time range at a chosen frame rate and speed.

E.2 Smokeview

Open couch.smv in Smokeview (double-click). Right-click > Load/Unload for slices, boundary files and 3D smoke; the pictures below come from a Smokeview script.



Figure 85. Smokeview 6.10.1 (Run > Open Smokeview, or double-click the .smv file) while the script runs: 3D smoke and the flame over the couch at t = 15 s. Right-click in the window for the menus.

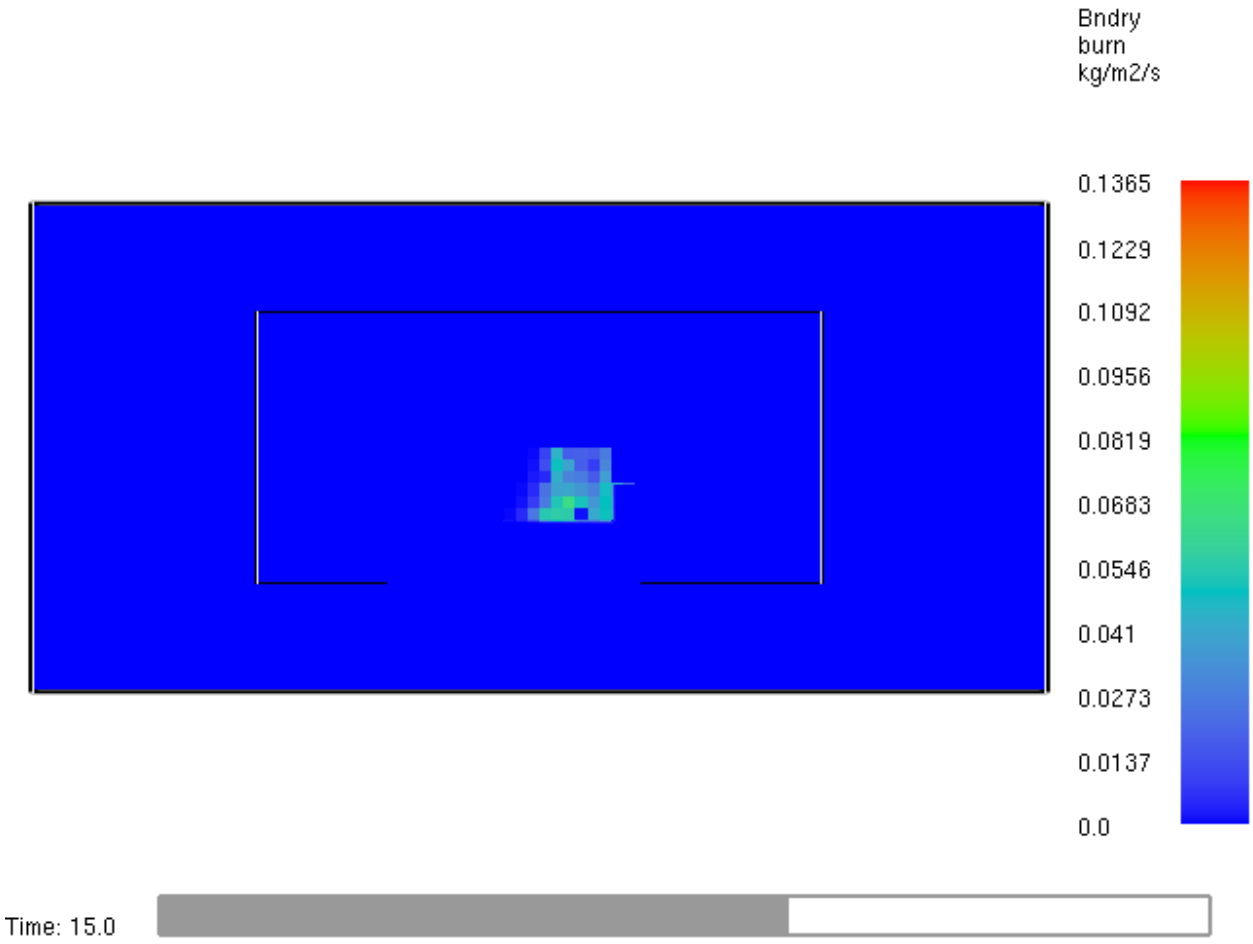


Figure 86. Rendered by Smokeview from the script: smv_burning_rate_t15.png

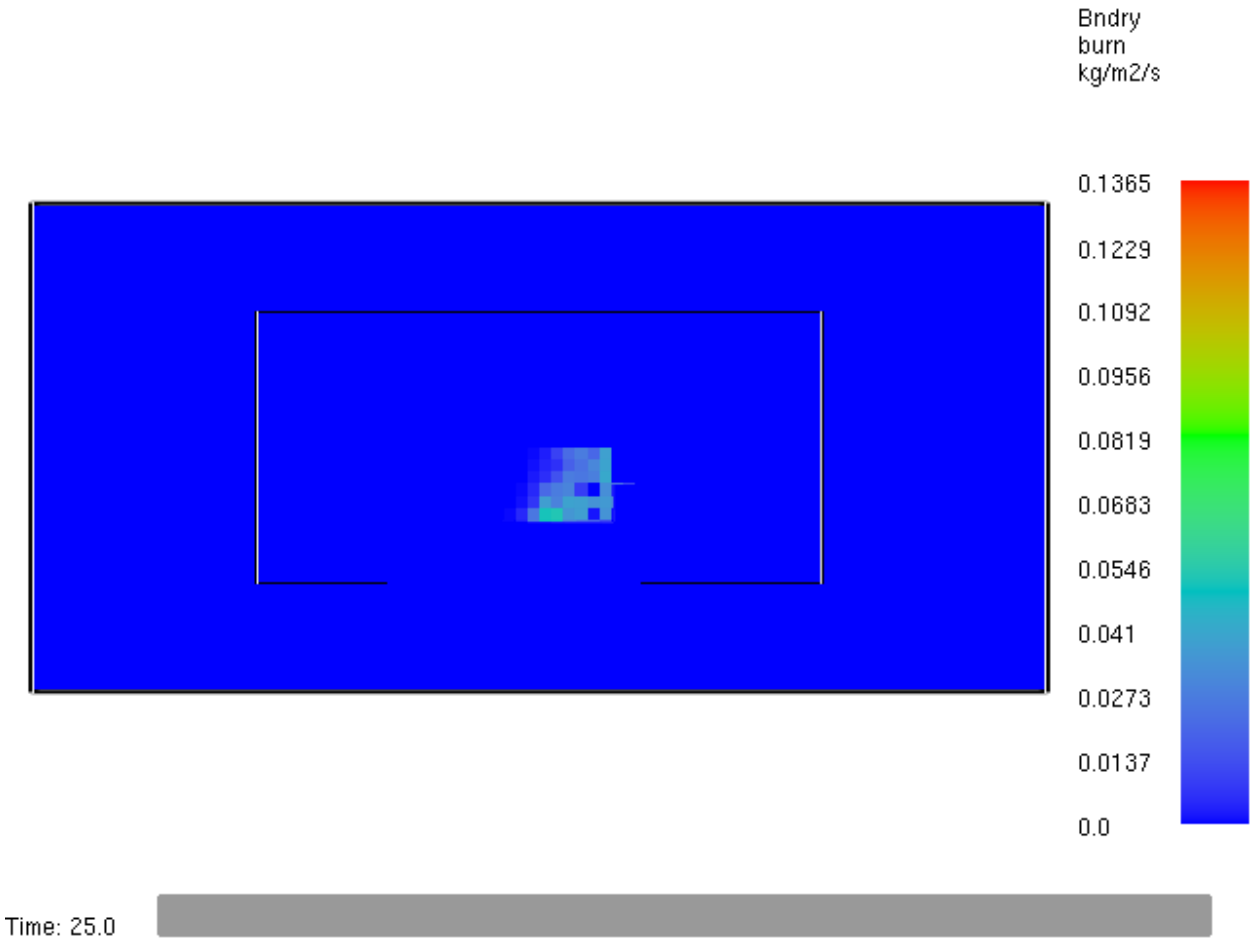


Figure 87. Rendered by Smokeview from the script: smv_burning_rate_t25.png

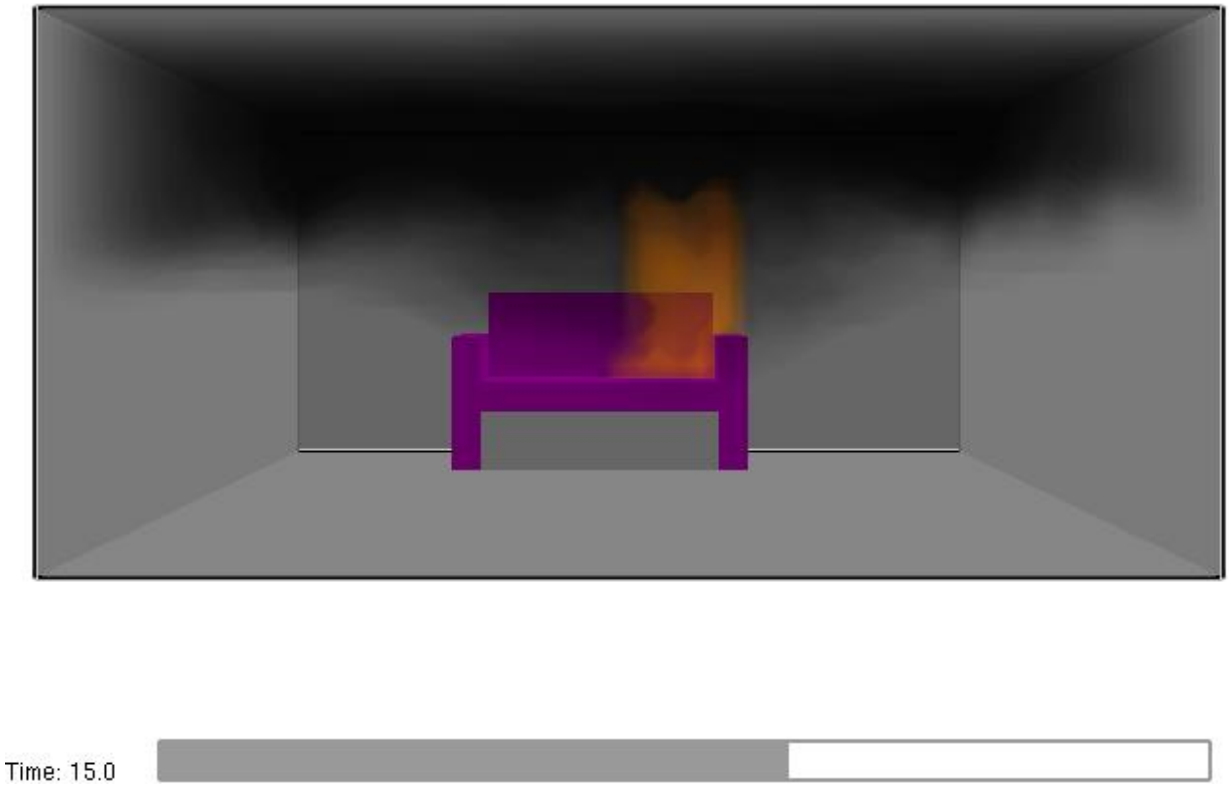


Figure 88. Rendered by Smokeview from the script: smv_smoke_t15.png

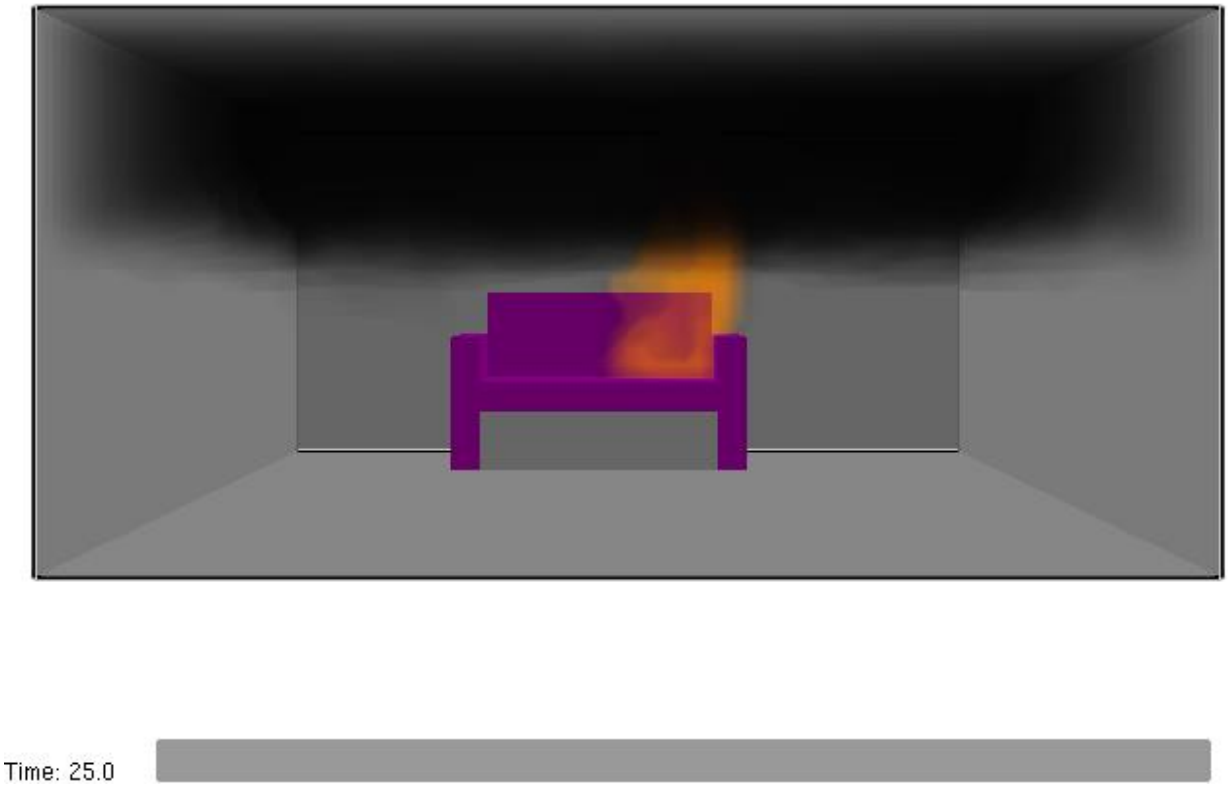


Figure 89. Rendered by Smokeview from the script: smv_smoke_t25.png

F Results and interpretation

Time (s)	5	10	15	20	25
Heat release rate (kW)	56	300	561	544	543

- The ignitor heats the seat fabric; flames spread over the seat and up the back - the burning-rate boundary file shows where material is pyrolysing.
- The heat release rate rises as more surface burns, to about 560 kW at 15 s, and then stays near 545 kW to 25 s while the couch keeps burning - the run is stopped long before the original 600 s.
- Wall temperatures rise only near the couch in this short time.

Limitations. The material properties are the FDS example's invented values, the run covers the first 25 s only, and the 0.1 m mesh is the example's. The case demonstrates the workflow; its heat release rate is not a prediction for a real couch.

G Troubleshooting

Symptom	Cause and remedy
'4 processes for N meshes: FDS needs one process per mesh'	Set MPI processes to the number of meshes (4) or use 'This computer' with one process.
MPI run does not start	FDS's MPI (mpiexec) must be installed with FDS; Locate FDS... picks an install that has it.
Run takes far longer than expected	Each of the 4 processes needs a free CPU core; close other heavy programs or reduce T_END.
Only a quarter of a slice is shown	Results are per mesh: tick the slice for all four meshes.
A Smokeview script stops with 'Slice files of type TEMPERATURE failed to load'	This case's slices are cell-centred; Smokeview 6.10.1 scripts do not load them. Load the slice interactively (right-click > Load/Unload > Slice file) or view it in PyFDS Results.

H Quick reference

12. Start PyFDS; File > Open: couch_original.fds
13. Inspect: Model > Edit Meshes / Surfaces / Materials / Particles
14. Model > Simulation Parameters: T_END 25 s
15. Model > Validate; File > Save As couch.pyfds
16. Run > Run Simulation: parallel (MPI), 4 processes, folder, Run
17. PyFDS Results: File > Load Results (couch.smv); tick all 4 meshes of a result
18. Smokeview: open couch.smv